How In-Kind Transfer Design Shapes Supply Incentives: Evidence from U.S. Housing Vouchers*

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November 2, 2026

Abstract

Many safety-net programs deliver in-kind benefits through private providers, and program design determines who participates and at what price. We study landlord participation and pricing decisions in the U.S. Housing Choice Voucher program, linking voucher administrative data to national rental listings and to a new panel of the program's price ceilings. We show that landlords are more likely to participate and to explicitly target voucher holders in high-poverty neighborhoods and that they adjust rents substantially toward the program's ceiling between listing and contract. We estimate a model of landlord participation and pricing. Counterfactuals show that targeting higher price ceilings to low-poverty neighborhoods shifts voucher holders toward them more cost-effectively than uniform increases or sign-up bonuses, while stricter rent review can finance such increases without raising program spending.

Keywords: in-kind transfers, housing vouchers, landlord participation, rent ceilings, bunching, neighborhood access.
JEL codes: H42, H53, I38, R21, R31, R38.

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1 Introduction

Several of the largest U.S. safety-net programs, including the Housing Choice Voucher (HCV) program, the Supplemental Nutrition Assistance Program (SNAP), Medicaid managed care, and the Women, Infants, and Children (WIC) program, share a common structure. The government finances these in-kind benefits, often up to an administratively set price ceiling, while private providers, such as landlords, grocers, and insurers, deliver them. Program outcomes therefore depend critically on whether providers choose to serve program participants and at what price. These provider decisions in turn determine what goods and services participants can access, their quality, and what the program pays.

In the HCV program, provider incentives are particularly sensitive to program design because housing is a highly heterogeneous good while the program applies a relatively uniform price ceiling. Voucher holders lease units from private landlords and pay 30 percent of their income toward rent, and the government covers the remainder up to the *payment standard*, a price ceiling that only varies by market (metropolitan area or ZIP code) and bedroom count.[1] This program feature ties landlords' pricing directly to their decision to participate, since voucher holders bear none of a rent increase below the standard but all of any rent above it. A landlord listing below the payment standard therefore can raise the rent toward it at no cost to the voucher holder, which we term *upcharging*, and one listing above it can attract voucher holders by cutting the rent, which we term *downcharging*. Because rental units are heterogeneous while the payment standard is common within a market and bedroom count, nearly every listing faces one of these two incentives.

Understanding these incentives is especially important in the HCV program, the largest rental assistance program in the United States, which has struggled with two of its main goals, successful lease-up and expanding neighborhood access for low-income families.[2] Only about 60 percent of households issued a voucher succeed in leasing a unit (Ellen et al., 2024), and those who do remain concentrated in high-poverty neighborhoods (Galvez, 2010; Lens, 2013; Horn et al., 2014; Collinson et al., 2019). Landlords are a likely part of the explanation. While the program offers guaranteed government payments and a broader tenant pool, benefits that are especially valuable where nonpayment and vacancy risks are high, it also imposes administrative costs such as unit inspections and paperwork. Consistent with this, voucher holders face widespread discrimination in rental markets (Phillips, 2017; Cunningham et al., 2018; Aliprantis et al., 2022; Faber and Mercier, 2022), particularly in low-poverty areas (Katz, 2026). Relatively little is known, however, about how program design shapes landlords' incentives to participate and set rents, or how these supply

[1] This distinguishes HCV from programs such as SNAP, where prices vary across stores and promotions but the program imposes no administratively set, beneficiary-specific price schedule.

[2] In 2025, it served about 2.4 million households at an annual cost of more than $30 billion.

responses affect where voucher holders live and what the program costs.

In this paper, we address this gap by studying how landlord participation and pricing respond to HCV program design and shape program outcomes. Combining online rental listings with voucher contract data, we document that landlords in high-poverty neighborhoods are far more likely to target voucher holders and ultimately participate in the program, and that their rents adjust toward the payment standard between listing and contract. Guided by these facts, we build and estimate a model of landlord participation and pricing and use it to evaluate how alternative program designs affect where voucher holders lease, what the program costs, and who captures the subsidy.

A central contribution of this paper is a new national linkage of rental listings to administrative voucher contracts, which allows us to follow the same rental unit from its listing stage through the eventual voucher contract and observe how its rent changes along the way. To do so, we use online rental listings from RentHub, which aggregates postings across property management and rental platforms and allows us to observe whether landlords explicitly target voucher holders at the listing stage. We classify units posted on affordable housing platforms specialized in HCV as *hard-targeted* and units posted elsewhere but explicitly soliciting voucher holders in their descriptions as *soft-targeted*. We then link these listings at the address level to restricted-access HUD administrative data on all HCV contracts, which report voucher holder and program characteristics and the applicable payment standard. The linkage allows us to measure participation by targeting status and the degree of upcharging and downcharging between listing and contract. We also construct the first nationwide panel of payment standards, combining inference from HUD administrative records with manual collection. Together, these data cover more than 900 CBSAs and roughly 2,000 local public housing agencies administering vouchers between 2015 and 2025, with payment standard information for nearly 500 of the largest agencies.

We document several facts about landlord participation and pricing in the HCV program. First, landlord targeting and voucher participation are concentrated in high-poverty neighborhoods. On average, 1.5 percent of listings hard- or soft-target voucher holders, rising to 3.3 percent in the highest-poverty quartile. Targeting also rises with the shares of Black and Hispanic residents, with lower education levels, and with higher vacancy rates in the neighborhood. Targeting strongly predicts participation, as hard-targeted units are roughly twenty times as likely as non-targeted units to lease under the program, and soft-targeted units are nearly eight times as likely.

Second, the payment standard plays a central role in landlords' pricing decisions. Contract rents for new voucher contracts bunch sharply at the payment standard. Listed rents also bunch at the payment standard, though more modestly and primarily among targeted listings, indicating that landlords respond to the price ceiling before any voucher holder makes contact. Contract rent bunching alone, however, cannot separate landlord pricing from tenant search, since voucher hold-

ers have their own incentive to lease units priced at the payment standard, where their subsidy is maximized. By observing listed and contract rents for the same unit, our HCV–listings linked sample allows us to attribute part of the bunching to listing-to-contract rent adjustments.

In the linked sample, bunching is far more pronounced in contract rents than in listed rents, and listed rents are more dispersed than contract rents, consistent with downcharging among listings above the standard and upcharging among those below it. To quantify these adjustments, we implement a differences-in-bunching design that exploits variation over time in the real value of rents relative to nominal payment standards to recover counterfactual rents absent the program. We estimate that 3.8 percent of linked voucher units bunch at the payment standard. Only about a quarter of this bunching arises at the listing stage, with three-quarters emerging between listing and contract. Excess mass extends to units priced up to 20 percent below it, indicating that upcharging also includes smoother, partial adjustment. The offsetting missing mass appears on both sides, among units 20 to 50 percent below the payment standard and among units up to 40 percent above it, reflecting upcharging and downcharging, respectively.

These listing-to-contract rent adjustments are large and absent in comparable market leases. Among linked units, 37.5 percent of landlords raise rents from listing to contract and 34.3 percent lower them, by about 8 and 9 percent on average, respectively. The adjustments grow with the distance between the listed rent and the payment standard in both upcharging and downcharging. In Multiple Listing Service transactions outside the program, the same listing-to-contract comparison yields changes near zero over a comparable range of listed rents around the payment standard.

Finally, we exploit the staggered adoption of ZIP code-level payment standards in designated metropolitan areas to identify how changes in the price ceiling affect landlord targeting and pricing. The reform replaces the metro-level payment standards with ZIP-level ones, raising payment standards in higher-rent ZIP codes and lowering them in lower-rent ones within the same metro. Landlords respond on both margins. Targeting of voucher holders rises, most in ZIP codes where the payment standard falls, where landlords appear to compete more actively for voucher tenants even as the subsidy shrinks. Rent adjustments also follow the payment standard. Where it falls, contract rents end up about 3 percentage points lower relative to listed rents, as landlords have less room to upcharge. Consistent with prior work (Collinson and Ganong, 2018; Ellen et al., 2025; Eriksen et al., 2025), voucher holders in adopting areas end up in somewhat lower-poverty neighborhoods, by about 2 percentiles of the within-metro poverty distribution on average.

Guided by these facts, we develop and estimate a three-stage equilibrium model of landlord behavior in the HCV program. In the first stage, landlords set a listed rent and choose whether, and how intensively, to target voucher holders, at a cost. In the second stage, a fixed flow of new vouchers in each market is allocated across listings according to their rents and targeting, so a listing's

chance of a voucher match reflects both its own choices and those of other landlords. The third stage allows landlords matched to a voucher holder to move the contract rent toward the payment standard, upward from below the standard and downward from above it. Using the model, we estimate the net gain of participating in the program, defined as the difference between the value of its voucher-tenancy benefits relative to an uncertain market tenancy minus the program's administrative costs, as well as the cost of targeting voucher holders. Intuitively, because the probability of a voucher match drops discretely just above the payment standard, the excess mass of listed rents at the standard and the missing mass just above it identify the net gain from a voucher, while targeting shares given voucher net gains identify the cost of targeting. These are the two key objects that determine which landlords participate, where they are located, and how participation and prices respond to program design.

The estimates imply that fewer than 2 percent of landlords find it worthwhile to actively seek out voucher holders. For 71 percent of landlords, a voucher tenancy would be more valuable than a market tenancy if a match occurred, but the probability of a match is too low to justify the cost of targeting. Relative to a voucher tenancy, a market tenancy yields about 79 percent of market value, so the voucher benefits to landlords are worth roughly one-fifth of rent. At the median rent, the net gain from a voucher match, accounting for participation costs, is $516 per month in the lowest-poverty quintile of tracts but only $82 in the highest. Participation nevertheless concentrates in high-poverty tracts for two reasons. Targeting voucher holders is cheaper there, which may reflect landlords' familiarity with the program and willingness to participate, and voucher holders are more likely to match with listings there at a given rent, partly reflecting their search behavior. The model reproduces rent adjustments between listing and contracting and variation in voucher matches and targeting across rents and neighborhoods. It also matches the larger targeting response to ZIP-level payment standards in areas where the standard falls, an outcome not targeted in estimation.

We then use the model to evaluate counterfactual program designs, focusing on where voucher holders lease and what the program costs. First, targeting higher payment standards to low-poverty neighborhoods is far more cost effective than raising them broadly. Raising payment standards only in the lowest-poverty quartile achieves a comparable shift toward lower-poverty neighborhoods at less than one-twentieth of the cost of a uniform increase, most of whose spending goes to listings that would have leased to vouchers anyway. Rent-indexed payment standards, a stylized Census tract version of ZIP code-level payment standards, likewise shift voucher holders toward lower-poverty areas while reducing spending. Second, sign-up bonuses raise landlords' gains from voucher matches but create additional matches mainly into high-poverty tracts, and even when targeted to low-poverty tracts they shift few voucher holders. Third, tighter rent-reasonableness enforcement reduces upcharging and saves spending. Using these savings to fund higher payment standards in lower-poverty tracts or rent-indexed standards shifts voucher holders toward lower-

poverty neighborhoods at no net budget cost.

Related literature. We contribute to several strands of literature. We first add to the economics of in-kind transfers and the private provision of public programs. A large literature shows that private providers respond to contract design on margins that are difficult to contract on (Hart et al., 1997; Galiani et al., 2005; Knutsson and Tyrefors, 2022; Duggan et al., 2023; Agafiev Macambira et al., 2026; Coube et al., 2026), and studies of administered payment schedules find that providers adjust participation and treatment when fees change (Clemens and Gottlieb, 2014; Alexander and Schnell, 2024). Closest to our setting, providers may price-discriminate between program and non-program customers, as documented for WIC (Meckel, 2020) though not for SNAP (Goldin et al., 2022). A complementary public finance literature attributes incomplete take-up to informational barriers and transaction costs on the recipient side (Bettinger et al., 2012; Bhargava and Manoli, 2015; Alatas et al., 2016; Deshpande and Li, 2019; Finkelstein and Notowidigdo, 2019). We expand this literature in the housing voucher context, one of the primary programs relying on private providers where the supply side remains relatively underexplored. We show how the program's design, primarily through the rent ceiling, affects landlord incentives at the listing and contracting stages and shapes both the quality of the housing the program delivers, in terms of neighborhood access, and its price.

We also contribute to a growing body of work on landlord incentives in low-rent markets. Recent studies provide evidence of high investment returns among low-rent properties (Damen et al., 2025), learning about tenants' nonpayment risk as an important driver of eviction decisions (Humphries et al., 2025), and landlords' demand for rent insurance (Bézy et al., 2025; Abramson and Van Nieuwerburgh, 2026), together indicating that insuring rental income is valuable, especially in the low-rent market segment. We connect this work to the housing vouchers context by showing that some landlords use vouchers themselves as insurance, prioritizing voucher holders at the listing stage and, at times, forgoing part of the rent to attract them.

Finally, we build on prior work on housing vouchers. One strand examines their effect on private market rents, generally finding that vouchers raise rents through demand (Susin, 2002; Fack, 2006; Eriksen and Ross, 2015; Park, 2026), with case study evidence that landlords charge voucher holders more than comparable tenants (Desmond and Perkins, 2016). Another studies how program design shapes outcomes through the demand side of the market, documenting bunching at payment standards in particular metropolitan areas (McMillen and Singh, 2020; Aliprantis et al., 2022, for LA and DC, respectively), showing that raising payment standards in low-poverty areas improves neighborhood outcomes for voucher holders without reducing lease-up (Collinson and Ganong, 2018; Ellen et al., 2025), and that customized support for voucher holders can substantially improve these outcomes as well (Bergman et al., 2024). Galiani et al. (2015) and Davis et al. (2021) estimate demand models of neighborhood choice to evaluate restricting voucher use to low-

poverty neighborhoods. We complement this demand-side evidence by modeling the supply side, studying participation and pricing jointly from the listing stage onward, and moving the analysis of landlord behavior from localized settings to national scale.

2 Institutional Background

2.1 The Housing Choice Voucher Program

The Housing Choice Voucher (HCV) program, established in 1974 under the Housing and Community Development Act and formerly known as Section 8, is the largest rental assistance program in the United States, serving 2.4 million households as of 2025.[3] The program is federally funded by the U.S. Department of Housing and Urban Development (HUD) and administered locally by approximately 2,200 Public Housing Authorities (PHAs). Unlike project-based subsidies, which tie assistance to specific developments, most HCV assistance is tenant-based, meaning that eligible households may lease any unit in the private market that meets the program's quality and pricing rules.

A defining feature of the program is that voucher holders must themselves find a willing landlord in the private rental market. Once a voucher is issued, the household has a fixed window, typically 60 to 120 days, depending on the PHA, to find a unit, have it pass inspection, and sign a lease. PHAs may grant extensions at their discretion, but these are not guaranteed and vary widely in practice. If the household fails to sign a lease within the allotted window, the voucher expires and reverts to the PHA, and the household returns to the waiting list.[4] This search-on-deadline structure places voucher holders under substantial time pressure. Approximately 40 percent of households who received a voucher fail to lease up within the search window, and failure rates are higher in tight rental markets, among larger households, and among Black and Hispanic households (Ellen et al., 2024). These search frictions imply that the effectiveness of the program depends directly on landlord participation, since a voucher that no landlord will accept provides no assistance. Participation is voluntary in most of the country. Outside jurisdictions with source-of-income (SOI) anti-discrimination laws, landlords may refuse to accept vouchers for any reason, and even where SOI protections exist, enforcement tends to be weak.[5] This dependence on voluntary landlord par-

[3] Source: HUD Picture of Subsidized Households (PSH).

[4] Average waiting times for HCV vouchers exceeded two years in most large metropolitan areas as of the late 2010s, and several major PHAs closed their waitlists entirely during this period (Collinson et al., 2019).

[5] As of 2023, roughly half of U.S. voucher holders live in jurisdictions with some form of source-of-income protection. Source: Poverty & Race Research Action Council. (January 2023). *Appendix B: State, local, and federal laws barring source-of-income discrimination. Expanding Choice: Practical Strategies for Building a Successful Housing Mobility Program.* See Cunningham et al. (2018) and Aliprantis et al. (2022) for evidence on the prevalence of voucher

ticipation motivates our focus on the supply side of the market.

Even a willing landlord must clear several further requirements before a lease can be signed. The unit must pass an inspection upon lease-up and periodically thereafter, and the contract rent must pass a rent reasonableness test. The landlord then signs a Housing Assistance Payments (HAP) contract with the PHA and the tenant signs a standard lease with the landlord. Once the contract is executed, the tenant generally pays 30 percent of their income toward rent, and the PHA pays the remainder directly to the landlord each month. The PHA's payment is capped by a price ceiling, the payment standard. We now turn to several of these features, which are central to our setting.

2.2 Key Program Features

Payment standards. The payment standard, set by each PHA, determines both the maximum subsidy a voucher holder can receive in the PHA's jurisdiction and the maximum gross rent, the sum of contract rent and a utility allowance, that the program will subsidize in full. For a voucher holder with monthly income I_i, their out-of-pocket payment stays at $\tau I_i = 0.3\, I_i$, 30 percent of their income, as long as gross rent R^g does not exceed the payment standard S. Gross rent above S is passed on to the tenant dollar-for-dollar, $\tau I_i + (R^g - S)$, subject to an overall out-of-pocket cap of $\bar{\tau} I_i = 0.40\, I_i$. On the flip side, the government's housing assistance payment (HAP) rises one-for-one with rent up to $R^g = S$ and is flat beyond, because each additional dollar above the ceiling must come from the tenant. Appendix Figure D.1 illustrates the government's and the tenant's shares of the rent as a function of the rent relative to the payment standard.

Each PHA may set its payment standards within a permitted range, usually between 90 and 110 percent of the metro-by-bedroom count-level Fair Market Rent (FMR), published by HUD, which is based on the 40th percentile of recent-mover gross rents. Since 2016, HUD has required a subset of metropolitan areas to set payment standards at the ZIP code level using Small Area FMRs (SAFMRs), which track sub-market rent differences more finely than metro-wide FMRs.[6] We return to this policy variation in our quasi-experimental analysis in Section 5.

Unit inspection and paperwork. Participation comes with administrative requirements. Before a lease can begin, the unit must pass a Housing Quality Standards (HQS) inspection and be reinspected periodically thereafter. A failed inspection requires repairs and a follow-up visit, and the PHA pays no subsidy until the unit passes, so inspection lags translate into forgone rent on a unit

denial under both regimes.

[6] SAFMRs were introduced in 24 metros in 2018 and in 41 metros in 2025. Some PHAs adopted ZIP code payment standards voluntarily before the mandatory rule took effect. See Collinson and Ganong (2018) and Ellen et al. (2025) for the policy history and evaluations of uniform and SAFMR-based ceiling changes.

the landlord holds off the market. Beyond inspections, the landlord must execute the HAP contract and the program's lease addendum, document ownership and tax information, and obtain PHA approval for subsequent rent increases. Interview evidence suggests that scheduling, paperwork, and payment setup often add weeks between tenant selection and lease-up (Greenlee, 2014; Garboden et al., 2018; Aliprantis et al., 2022).

Rent reasonableness. Another friction facing landlords is the rent reasonableness test. Before executing a HAP contract, the PHA must document that the proposed rent is reasonable relative to comparable unassisted units and the unit's own characteristics. In principle, the test can also bind below the payment standard, since a rent well under the ceiling may still be deemed unreasonable if comparable units rent for less. In practice, implementation varies across PHAs, and the test's effective stringency has been the subject of debate. Qualitative work suggests landlords retain scope to secure higher approved rents by upgrading or reclassifying unit features, for example, counting extra rooms or making cosmetic renovations at little to no cost to justify higher rent requests under the program (Rosen, 2014; Garboden et al., 2018). A HUD Office of Inspector General audit of the New York City Housing Authority found no evidence of a rent-reasonableness determination in 49 of 115 sampled files and estimated that $1.16 billion in housing assistance payments may have lacked such support (U.S. Department of Housing and Urban Development, Office of Inspector General, 2014). We return to this channel in our counterfactual analysis in Section 8, where we model tightened enforcement as a proportional scaling down of the expected upcharging.

2.3 Landlord Participation and Pricing Incentives

Landlords face a trade-off when leasing to voucher holders, well documented in prior qualitative studies. On the benefit side, the program offers guaranteed rent payments, since the government directly covers a portion of the rent, along with reduced nonpayment and vacancy risk and access to a broader tenant pool (Rosen, 2014; Greenlee, 2014). On the cost side, participating landlords face unit inspections, administrative paperwork, and the soft rent ceilings imposed by payment standards (Greenlee, 2014; Garboden et al., 2018; Aliprantis et al., 2022). The balance between these benefits and costs varies across landlords and markets. Participation is more attractive where rents are low, landlords' outside options are thin, and the guaranteed payment is more valuable, and less attractive where market rents substantially exceed the payment standard.

On the pricing margin, two landlord behaviors motivate our analysis. First, landlords whose market rents would exceed the payment standard may lower their rents to the payment standard, where the tenant pays only the base income share. In exchange for the forgone rent, these landlords secure the guaranteed government payments and the program's insurance against nonpayment and

vacancy, protection worth most in low-rent markets (Humphries et al., 2025). We call this *downcharging* behavior. Second, landlords whose market rents would fall below the payment standard may instead raise them toward it to capture a larger government transfer. They can do so because voucher holders' demand is inelastic below the payment standard, where the tenant's payment is a fixed share of income no matter the rent. We call this *upcharging*, consistent with qualitative accounts of such behavior under the program (Rosen, 2014; Desmond and Perkins, 2016; Garboden et al., 2018). Bunching exactly at the standard maximizes the landlord's subsidy, but frictions such as the rent reasonableness test may stop some landlords partway.

Downcharging and upcharging have opposite budgetary implications. Downcharging transfers rent from landlords to voucher holders and can open markets that would otherwise be out of reach, while upcharging transfers taxpayer dollars to landlords. They are also inseparable from the participation decision itself, since the rent a landlord expects to collect under the program, whether above the listed rent through upcharging or below it through downcharging, enters the return to leasing to voucher holders in the first place. For example, some landlords who would upcharge may participate only because the higher program rent offsets the administrative costs of HCV participation. A program design that curbs upcharging therefore saves money but may discourage participation, while one that accommodates downcharging expands access at landlords' expense. Distinguishing the two behaviors, and quantifying their relative importance, is a central task of this paper.

3 Data

Our analysis combines three data sources: restricted-access administrative data on all HCV participants from HUD, rental listings from RentHub, and a nationwide panel of payment standards that we construct by combining information inferred from HUD data with manual collection.

3.1 HUD voucher administrative data

We use HUD's restricted-access contract-level administrative data on the HCV program, covering all voucher holders from 2015 through 2025. The data provide individual-level records on household characteristics (composition, race, ethnicity, age, disability status, and income), unit characteristics (precise address, bedroom count, contract rent, gross rent, tenant out-of-pocket payment, HAP, and utility allowance), and program parameters (the payment standard applied to each contract, and indicators for new admissions).

Our main analyses focus on the 1.5 million new admissions observed between 2015 and 2025. New admissions are the natural sample for studying landlord pricing because the contract rent is

freshly negotiated under the payment standard in force at the time of lease-up, while continuing contracts may inherit the rent amount from prior periods. We impose two restrictions in our analyses. First, we exclude project-based voucher participants, since our focus is on landlords who are not bound by development-level contracts. Second, we restrict to units with 0 to 4 bedrooms, the range for which payment standards are consistently published. Column (1) of Table 1 reports summary statistics for this new admissions sample. The median contract rent is $850, the median payment standard is $990, and it spans 924 CBSAs.

3.2 RentHub rental listings data

RentHub (2026) compiles rental listings from property management websites, rental platform aggregators, and other public sources and re-scrapes active listings on a weekly to biweekly cadence. A given unit therefore appears in multiple scrapes while it remains on the market. Each scrape records the asking rent, bedroom count, bathroom count, square footage, property type, amenities, text description, and street address, which we geocode to a standardized address, Census tract, and county. Each scrape also includes platform and unit identifiers assigned by RentHub. We use the platform identifier and description text to classify voucher targeting and use the unit identifier together with the geocoded address to chain scrapes into listing spells, as described below.

We organize the raw scrapes into listing spells intended to capture a single unit's continuous time on the market. We identify a unit using the RentHub unit identifier when available, which maps to a single geocoded address for the large majority of listings, and otherwise use the geocoded address and any extracted unit number. Within each unit, we order scrapes by date and begin a new spell whenever the gap between consecutive scrapes exceeds 60 days. We treat a longer gap as evidence that the unit left the market between spells, typically because it was leased.[7] For each spell, we keep the first and last listed rents, the minimum and maximum listed rents, the first and last scrape dates, spell duration, number of scrapes, bedroom count, bathroom count, square footage, building type, and the voucher-targeting classification described below.

The listing spell is our primary unit of observation in the listing-to-contract analysis in Section 4 and the model estimation in Section 7. We use scrapes from 2015 through May 2026 to construct listing spells first observed between 2015 and 2025. We restrict the sample to studio and one- to four-bedroom units with a text description and a valid CBSA location. We also drop observations with listed rents below $100 or above the 99th percentile within each state.

We measure landlords' voucher-targeting behavior in listings by classifying each listing spell

[7]The 60-day threshold sits in a valley of the within-unit gap distribution between the weekly to biweekly scrape cadence of an active listing and the much longer gaps that precede genuine re-listings, so the exact cutoff is not consequential. See Appendix B.1.

into one of three mutually exclusive categories based on platform and listing text, with hard targeting taking precedence over soft and soft over non-targeted (see Appendix B.2 for further details).

- **Hard-targeted.** Listings on an affordable housing platform or with an affordable housing management organization, channels that exist specifically to serve voucher holders. Within our data, these are dominated by AffordableHousing.com (formerly GoSection8.com), the most widely used voucher-specific platform.[8]

- **Soft-targeted.** Listings that are not on an affordable housing platform but whose description explicitly welcomes vouchers, for example "Section 8 welcome," "vouchers accepted," or "all vouchers welcome." Acceptance language is screened against a refusal list, so phrases such as "no section 8" do not qualify.

- **Non-targeted.** Listings that are not on an affordable housing platform and whose description does not explicitly welcome vouchers. This includes listings on mainstream platforms (e.g., Zillow, Realtor.com, Apartments.com) and listings whose only voucher language is a refusal.

We use this classification in Fact 1 (Section 4) to document how targeting intensity correlates with voucher match rates and neighborhood characteristics, and as the landlord's choice variable in the model (Section 6), where the share of listings in each tier identifies both the cost and the match rate benefit of active voucher targeting.

We report summary statistics at two levels of observation in Table 1, raw scrapes (column 3) and sample listing spells (column 4). The RentHub sample consists of 421 million scrapes across 928 CBSAs, which collapse to 39 million listing spells, an average of about 11 scrapes per spell. Rental units in the RentHub data are considerably more expensive than the voucher stock. The median listed rent is $1,800 at the scrape level and $1,900 at the spell level, more than twice the $850 median contract rent among all new voucher admissions (column 1) and well above the $1,100 median among the admissions we link to a listing (column 2). Part of this gap reflects size, since the median listed unit has 1,094 to 1,174 square feet against 861 for linked voucher units, and part reflects that listings concentrate in higher-rent areas, where the median payment standard is $1,538 to $1,776 against $990 for all new admissions. Voucher targeting is rare among listings as a whole. Only 0.39 percent of listing spells are hard-targeted and 1.12 percent soft-targeted. It is an order of magnitude more common among listings we link to a voucher contract, at 6.54 and 6.92 percent, which we describe in Section 3.4.

[8]Over 97 percent of our hard-targeted listings originate from AffordableHousing.com. Because the platform and company metadata that flag hard-targeted listings fill in sharply from 2019, all specifications include year fixed effects. We also re-estimate the model parameters using 2023–2025 data, when the RentHub coverage is more complete overall, as a robustness check.

3.3 Payment standards

Payment standards are the main program feature we focus on to study landlords' participation and pricing decisions, so measuring them accurately is essential. Yet no centralized or consistently maintained nationwide dataset of payment standards exists. PHAs vary widely in how they publish this information, with some posting schedules in administrative plans or on their websites and others providing little accessible documentation. HUD administrative data record the payment standard applied to each contract, but these records do not indicate whether it is set at the metro or ZIP code level and, in raw form, cannot be used to build a panel of payment standards applicable to units that did not actually enter the HCV program. In addition, HUD administrative data report payment standards only for voucher contracts in PHAs that do not participate in the Moving to Work (MTW) Demonstration program.[9]

We construct a harmonized monthly panel of payment standards by PHA, geographic unit, and bedroom size, combining three complementary sources. First, we infer each PHA's schedule from HUD administrative data using the modal payment standard in each PHA-period-bedroom cell. This procedure yields valid metro-level payment standard inferences for 454 PHAs over all or part of the 2015–2025 sample period, including PHAs that adopted ZIP-level payment standards partway through the sample (for which the metro-level payment standard applies in pre-adoption years). Second, we exploit the regularity that PHAs typically set payment standards as a fixed multiplier of the FMR or SAFMR (e.g., 100 or 110 percent) and recover the applicable multipliers from the same administrative records to extend coverage to finer geographic levels, particularly for PHAs applying ZIP-level payment standards. We obtain data for 404 PHAs using this method. Third, for 76 of the largest PHAs considered for the SAFMR mandate in 2016, we manually collect payment standards from administrative plans, websites, and direct correspondence. When multiple sources cover the same PHA-period-bedroom cell, we apply a priority hierarchy that favors manually collected data and, within the two inference methods, prioritizes the modal method over the multiplier-based one. In total, our dataset contains payment standard information for at least some period in 2015–2025 for 474 PHAs. Appendix A details the construction of this dataset.

Utility allowances, which enter the tenant's gross rent calculation but not the landlord's contract rent, vary across PHAs and bedroom sizes, with some PHAs reporting zero allowances for all contracts and others consistently reporting positive values. Although the payment standard formally applies to gross rent, our analyses focus on contract rent, which is the object landlords choose and the one that maps directly to listed rents.

[9]The Moving-To-Work Demonstration program grants participating PHAs flexibility to waive or adapt standard public housing and HCV rules, including how payment standards are set. MTW began with 39 initial agencies, which retain MTW status throughout most of our sample period. Under the expansion authorized by the 2016 Consolidated Appropriations Act, HUD began designating up to 100 additional agencies starting in 2021.

3.4 Linking rental listings to HUD contracts

A contribution of our data construction is the linkage between HUD administrative records and RentHub rental listings, which lets us observe both the listed rent and the signed contract rent for the same voucher unit. This linkage identifies the listing-to-contract rent change studied in Fact 3 (Section 4) and disciplines the Stage 3 contracting parameters in the structural model.

We link each HCV new admission to any RentHub listing spell posted within the six months prior to the admission date, matching on cleaned street address, city, state, and bedroom count. Address cleaning proceeds in several steps. We first extract unit identifiers (e.g., "apt," "unit," "suite," "#") from the full address field, using a dictionary of prefixes followed by alphanumeric sequences, and strip these identifiers from the address string. We geocode the cleaned addresses with the Census Geocoder to obtain standardized street numbers, street names, and street types; for addresses without a successful Census match, we fall back to the ArcGIS World Geocoder and retain the original address when both attempts fail. We then standardize all key address components in both the HUD and RentHub files.

Because unit identifiers (apartment or suite numbers) are missing or unreliable in RentHub for most multi-unit properties, our main linkage is at the address level rather than the unit level. The correspondence between admissions and listing spells is therefore many-to-many rather than one-to-one. A single admission may be linked to several spells, either because the address was posted more than once or because a multi-family building contains units beyond the one the admission refers to, and a single spell may in turn link to several admissions in the same building. In multi-family housing, then, a linked spell may correspond to a different unit than the voucher holder's. This procedure links 14 percent of new admissions over 2015–2025.

For analyses conducted at the voucher-contract level, we address this concern in two ways. First, when multiple listings are linked to an address, we retain the single listing whose posting date is closest to the admission date and use it as our main sample. Second, we repeat all analyses on a robustness subsample restricted to single-unit addresses, for which the concern is absent by construction.[10]

For analyses conducted at the listing spell level, where the unit of observation is the spell itself, we retain every linked spell and attach a linkage weight in place of a binary linked indicator. Each linked admission receives a total weight of one, divided equally among the spells linked to it, so an admission linked to k spells contributes $1/k$ to each. A spell's linkage weight is the sum of these fractions over the admissions linked to it, and is zero for an unlinked spell. Because each

[10] We classify an address as single-unit if none of the following hold in either the HUD or the RentHub data: the unit is recorded as being in a multi-family building; the record carries an apartment number; or the same address appears elsewhere in that source with an apartment number.

admission's weight sums to one regardless of how many spells it touches, the linkage weights sum across spells to the number of distinct linked admissions, so no admission is counted more than once.

The HCV units we link to listing spells are not a random sample of HCV admissions. Table 1, Column (2), reports summary statistics for the linked sample. Relative to the full sample of new admissions, linked units have higher contract rents and payment standards, though they have lower rents and are smaller than the average RentHub listing. Appendix Figure D.2 further plots coefficients from linear probability models of linkage on tenant and unit observables (standardized within PHA). Linkage is more likely not only for units with higher contract rents but also for units in neighborhoods with higher renter shares and in multi-family structures, consistent with online listing platforms over-representing rental housing in denser, more heavily rented submarkets.

4 Stylized Facts

This section documents four facts about landlord participation and pricing under the HCV program. The first fact shows that voucher targeting is concentrated in high-poverty neighborhoods and strongly predicts participation. The remaining facts turn to pricing. We find that rents bunch at the payment standard, most sharply in voucher contracts and, among listings, in those targeting voucher holders. Using linked listings and contracts and a differences-in-bunching design, we find that landlords move rents toward the payment standard from both sides. Most of these adjustments occur between listing and contracting stages, are far larger than what we observe in a market-rate benchmark, and vary systematically with neighborhood characteristics. Together, these facts show that participating supply is geographically selected, likely affecting where vouchers can lease up, and that landlords' pricing responds to the payment rule at both stages, two features the model in Section 6 is built to capture.

Fact 1: Voucher targeting is concentrated in high-poverty neighborhoods and strongly predicts participation

Landlords are more likely to seek voucher tenants in high-poverty neighborhoods within a metropolitan area. Figure 1a plots the share of listing spells that are hard- and soft-targeted, defined as in Section 3.2, by the poverty quartile of the unit's Census tract, computed within the CBSA. The share of hard-targeted listings, those advertised on affordable housing platforms, rises from less than 0.1 percent in the lowest-poverty quartile to roughly 1.0 percent in the highest. The share of soft-targeted listings, those whose text explicitly seeks out voucher holders, rises from

0.42 percent to almost 2.3 percent across the same quartiles, more than a five-fold increase. Soft targeting exceeds hard targeting throughout the poverty distribution, indicating that landlords more often signal openness to vouchers within standard listings than post on voucher-specific platforms. A plausible explanation is that landlords seek to limit the cost of posting across multiple platforms while preserving the appeal to a broader tenant pool that includes voucher holders.

Several other tract characteristics predict targeting. Figure 1b reports coefficients from separate listing-level regressions of a hard- or soft-targeted indicator on standardized Census tract characteristics, each including CBSA-by-year and CBSA-by-bedroom fixed effects, so the coefficients measure how each characteristic relates to the probability of targeting across neighborhoods within the same metropolitan area. Both forms of targeting are more likely in tracts with larger Black and Hispanic populations, lower education levels, higher unemployment, and lower median household income. Among housing market characteristics, targeting is more common in higher-vacancy markets with larger renter shares and lower median rents.

Landlords who target vouchers are also more likely to participate in the program. Figure 2a shows, by targeting status and within-CBSA poverty quartile, the share of listing spells linked to a new voucher contract. Among hard-targeted listings, the linkage rate rises from about 4.5 percent in the lowest-poverty quartile to 7.5 percent in the highest. Among soft-targeted listings, it rises from 1.8 to 3.4 percent. Non-targeted listings link at rates near zero throughout, from 0.2 percent in the lowest-poverty quartile to about 0.7 percent in the highest. Within CBSA, hard-targeted listings are roughly twenty times as likely as non-targeted listings to enter the voucher program, and soft-targeted listings nearly eight times as likely.

These differences in program participation also appear across other tract characteristics. Figure 2b reproduces the pairwise regressions of Figure 1b using an HCV-linked indicator as the outcome, a proxy for program participation, estimated separately for hard-, soft-, and non-targeted listings. The pattern mirrors that of targeting. Listed units are more likely to enter the program in tracts with larger Black populations, lower education levels, lower median household income, larger renter shares, and lower median rents. Most associations hold across the three samples, with a few exceptions. Among hard-targeted listings, we cannot reject that linkage rates are unrelated to tract Black share, education, and median rent. That is, once hard targeting, which is already concentrated in these areas, is accounted for, these characteristics carry no additional weight for program entry. Likewise, landlords in higher-vacancy areas are no more likely to enter the program in any targeting category, once their greater propensity to seek voucher holders is taken into account.

Together, these patterns are consistent with landlords seeking voucher tenants where the benefits of participation presumably outweigh the costs. That targeting is concentrated in lower-income, less-educated areas suggests landlords there value the reduced nonpayment risk vouchers provide,

since part of the rent is guaranteed by the government. Higher targeting in higher-vacancy areas is consistent with landlords using vouchers to reduce vacancy risk by expanding their tenant pool. That several of these characteristics lose predictive power once targeting is held fixed indicates that they operate through the decision to seek voucher holders in the first place.

Fact 2: Rents bunch at the payment standard, most sharply in voucher contracts and, among listings, where landlords target vouchers

We next examine how rents are distributed around payment standards for landlords participating in, or seeking to participate in, the HCV program. Figure 3 plots the distribution of rents around payment standards in 0.5-percentage-point bins for four samples: contract rents for new admissions in the administrative data (Panel a), and listed rents for hard-, soft-, and non-targeted listing spells (Panels b–d). Voucher contract rents bunch sharply at payment standards. Listed rents also bunch at the payment standard, though less sharply than contract rents, and the bunching increases with targeting intensity. It is strongest among hard-targeted listings, weaker among soft-targeted listings, and smallest among non-targeted listings. Because listed rents are set before a voucher holder is involved, this pattern indicates that landlords themselves drive part of the adjustment toward the payment standard at the listing stage, especially when they actively seek voucher holders.

The extent of contract rent bunching also varies across neighborhoods. Appendix Figure D.3 reports pairwise regressions of an indicator for the contract rent equaling the applicable payment standard on standardized tract characteristics, with PHA-by-year and PHA-by-bedroom fixed effects. Bunching is more common in high-poverty, high-vacancy tracts with more renter households, where listed rents tend to sit below the standard and upcharging can carry rents to it, but also in tracts with lower Black population shares and more educated residents, where rents more often start above the standard and bunching is more plausibly reached by downcharging. Because these regressions condition on a signed contract, they cannot separate the two forces, a distinction we take up directly in Fact 3.

Fact 3: Both upcharging and downcharging contribute to the bunching, mostly at the contracting stage

To decompose the excess mass at the payment standard by direction and timing, we restrict attention to newly admitted HCV units linked to our listings sample, constructed as in Section 3.4. This sample follows rents from the listing stage to the contracting stage while holding unit composition fixed. Figure 4 overlays histograms of contract and listed rents for these linked units.

Bunching is far more prominent in contract rents, indicating substantial rent changes between listing and contracting, but it is present in listed rents as well. This listing-stage bunching may reflect voucher holders seeking out units priced at the payment standard, so that rents at the standard are overrepresented among linked listings, or it may reflect landlords already manipulating rents when they list, as the targeting patterns of Fact 2 suggest. Either way, the listed rent distribution cannot serve directly as the no-manipulation benchmark, and quantifying the full extent of rent adjustment under the program requires estimating one. Intuitively, once that benchmark is in hand, missing mass above the payment standard points to downcharging, while missing mass below it points to upcharging.

We recover the counterfactual distribution of listed rents under no manipulation with a differences-in-bunching design that exploits the interaction of nominal payment standards with rent inflation (Dube et al., 2025).[11] Because payment standards are set in nominal terms and do not generally adjust with inflation, a focal year's payment standard falls at arbitrary, non-salient points of other years' rent distributions once those rents are expressed in focal-year dollars. The deflated distributions from other years are therefore free of bunching at the focal year's payment standard and can be used to infer its counterfactual.

We implement the design by stacking the data by focal year, pairing each focal year's observations with those from years more than two apart, since standards can be sticky across adjacent years, deflating rents into focal-year dollars, and binning them in 0.5-percentage-point intervals.[12] Let $\tilde{r}_{i,t,\tau}$ denote the deviation of unit i's year-t rent, deflated into year-τ dollars, from the year-τ payment standard, in percent of the standard, and let $p_{\tilde{r},t,\tau}$ denote the share of stack-τ observations from year t in bin $\tilde{r}$. Within each stack, we estimate

$$p_{\tilde{r},t,\tau} = \sum_{\tau'} \sum_{i=0}^{k=20} \alpha_{\tau,i} \, \mathbb{1}(\tau = \tau') \times \tilde{r}^i + \sum_{\tau'} \sum_{j=-w}^{w} \beta_{\tau,j} \, \mathbb{1}(\tau = \tau') \times \mathbb{1}(\tilde{r} = j) \times \mathbb{1}(t = \tau) + \varepsilon_{\tilde{r},t,\tau} \quad (1)$$

where the first term fits a 20th-order polynomial as the counterfactual from the non-focal years and the second term recovers the focal year's deviation from it in every bin, so excess and missing mass can be estimated anywhere in the distribution rather than only near the threshold. We estimate the model on a sample of 441 PHAs with payment standard data available consistently over 2015 to 2025, computing standard errors with 1,000 iterations of a bootstrap procedure following the

[11] The traditional bunching estimator (Saez, 2010; Chetty et al., 2011; Kleven, 2016) fits a high-order polynomial to the focal distribution outside an excluded manipulation range around the threshold. It relies on strong parametric assumptions to identify the counterfactual (Blomquist et al., 2021) and detects only sharp adjustments near the threshold, whereas the differences-in-bunching design also captures smoother manipulation far from it.

[12] We deflate rents using the smoothed monthly Zillow Observed Rent Index for the United States, applying the December value of each year.

literature (Kleven and Waseem, 2013; Dube et al., 2025).[13][14]

Figure 5 shows the results, where Panel (a) overlays the contract rent histogram and the estimated counterfactual listed rent distribution, and Panel (b) plots their difference. We estimate that 3.8 percent of linked new admissions bunch at the payment standard, with excess mass extending to units priced up to 20 percent below it rather than confined to the threshold. The excess mass alone cannot reveal how much comes from units priced above versus below their counterfactual rent, but the offsetting missing mass can. The difference plot in Panel (b) reveals significant missing mass among units with contract rents 50 to 20 percent below the payment standard, reflecting upcharging landlords, and significant missing mass among units 0 to 40 percent above the payment standard, reflecting downcharging landlords.

This approach also characterizes the timing of manipulation. Comparing the listed rent distribution with its counterfactual, we find that only about 1 percent of participating landlords bunch at the payment standard at the listing stage (see Appendix Figure D.6). That is, roughly one-quarter of voucher contract bunching occurs at the listing stage and the remaining three-quarters at the contracting stage.

These results survive a range of robustness checks, including alternative polynomial orders and bin widths, excluding the post-COVID period of unusually fast rent growth, and using absolute rather than normalized deviations from the payment standard (see Appendix Table D.2). One exception is the sample of single-unit addresses, where we estimate smaller though still statistically significant bunching of 1.4 percent (see Appendix Figures D.4(a) and D.7, and column 8 of Appendix Table D.2). A plausible explanation is that single-unit addresses over-represent smaller landlords, who may be less attentive to the rent ceiling, hold less bargaining power, or be more willing to accommodate utility allowances so that the payment standard binds on gross rather than contract rent. At the same time, the missing mass from above the payment standard is larger in this sample, rising from 21 percent in the main sample to 30 percent. This pattern suggests that smaller landlords value the program's payment guarantee, and its access to a broader tenant pool, more than larger landlords.

[13]The sample restriction is necessary because the design applies a given year's payment standards to contract rents from other years. We therefore exclude PHAs whose payment standards we could neither infer from HUD administrative records (mostly MTW and small PHAs) nor collect directly.

[14]In each bootstrap iteration b, we resample residuals from Equation (1) with replacement, add them to the fitted values, and re-estimate the model; we repeat this 1,000 times and compute standard errors from the resulting empirical distribution.

Fact 4: Rent adjustments far exceed the market-rate benchmark and are larger for units listed farther from the payment standard

We now evaluate the size of listing-to-contract rent adjustments. Figure 6a shows a binned scatter plot of the relative change from listed to contract rent against the listed rent among linked units. Units with listed rents at least roughly 10 percent below the payment standard end up with higher contract rents, and the gap grows with distance from the ceiling, reaching increases of up to 50 percent, though few units listed that far below. This is consistent with the upcharging incentive being stronger well below the payment standard where there is ample room to raise rents while the tenant's payment stays fixed and demand from voucher holders is perfectly price-inelastic. Above the payment standard the pattern reverses as the downcharging incentive takes over. Rents fall from listing to contract by up to 30 percent, indicating that some landlords find the program valuable enough to accept lower rents than they listed in the private market. These patterns hold for single-unit addresses (Appendix Figure D.4(b)) and regardless of whether the landlord targets voucher holders (Appendix Figure D.5).

Overall, we estimate that 37.5 percent of landlords adjust rents upward by 7.9 percent, and 34.3 percent adjust rents downward by 8.7 percent. Scaled by baseline rents, the first group ends up with roughly $48 more per month under the program, and the second forgoes roughly $69 per month on average in 2015 dollars.

The magnitude of these listing-to-contract rent changes is far larger than those in private rental markets. Figure 6b repeats the exercise for market-rate MLS rental transactions outside the program, which record both the listed and the signed contract rent. Over the same range of listed rents relative to the payment standard, listing-to-contract changes are close to zero, with only a small discount that grows modestly with the rent level. Rent adjustments toward the ceiling, from both sides and increasingly with distance, are therefore specific to the voucher program rather than a general feature of rental transactions.

We also relate the magnitude of these rent changes to voucher holder and neighborhood characteristics. Figure 7 plots coefficients from pairwise regressions of the rent change on standardized voucher-holder and Census tract characteristics, estimated separately for linked units listed above the payment standard (left panel), which on average lower their rents under the program, and units listed below it (right panel), which on average raise them. To ease interpretation, the left panel uses the negative of the rent change as the outcome, so that coefficients in both panels measure the size of the adjustment toward the payment standard.

For units initially listed above the payment standard, one pattern stands out. Discounts at contracting are largest when the program's guarantee is presumably worth the most, for elderly tenants,

whose Social Security income is stable, and for lower-income tenants, for whom the government covers a larger share of the rent, as well as in lower-income, less-educated areas, where market-rate tenants tend to carry larger nonpayment risks and the outside market tends to be thin. The pattern reverses for units initially listed below the payment standard. These units raise rents more when the voucher holder's income is higher and in higher-education, higher-income, and higher-rent neighborhoods, where the gap between the payment standard and the listed rent leaves more room to upcharge and the outside pool of tenants presents lower risk.

5 Quasi-Experimental Evidence from Shifts in the Rent Ceiling

The previous section documents how landlords interact with the HCV program through their participation and pricing decisions, and how these patterns vary with the rent ceiling and neighborhood characteristics. This section uses a difference-in-differences design to show that landlords respond along both margins to quasi-exogenous changes in the rent ceiling.

5.1 Background and Empirical Strategy

Complementing the descriptive evidence in Section 4 requires variation in the rent ceiling, the payment standard, that is plausibly orthogonal to local demand and supply conditions. Following prior work (Collinson and Ganong, 2018; Ellen et al., 2025; Eriksen et al., 2025; Park, 2026), we exploit the staggered adoption of Small Area Fair Market Rents (SAFMRs) across PHAs. Conventional payment standards are based on the metro-level Fair Market Rent (FMR) and are therefore largely uniform within a metropolitan area. SAFMRs instead set the market rent benchmark, and hence the payment standard, at the ZIP code level. SAFMR adoption therefore raises the rent ceiling in high-rent ZIP codes and lowers it in low-rent ZIP codes, generating within-metro variation in payment standards. The rollout has been driven largely by HUD, which mandated ZIP-level payment standards for PHAs in 24 metros by 2018 and in an additional 41 metros by 2025, while a few PHAs have been adopting them voluntarily.

We use this rollout in a staggered difference-in-differences design that compares PHAs adopting ZIP-level payment standards earlier in the sample period with PHAs adopting them later. Identification rests on the assumption that, absent adoption, payment standards and landlord behavior in early-adopting PHAs would have evolved in parallel with those in later-adopting PHAs.

We implement this design using 167 PHAs for which we can assign a SAFMR adoption year based either on an observed switch to ZIP-level payment standards or on HUD mandates. For PHAs with an observed switch, we define the adoption year as the first year the switch appears in our hand-

collected data.[15] For non-MTW PHAs designated for mandatory SAFMRs whose switch does not appear in our hand-collected data, we use the year the designation took effect, 2018 for the first round and 2025 for the second. We drop non-designated PHAs without an observed switch and PHAs that already used ZIP-level payment standards when the panel begins. Appendix Figure D.8 shows the number of PHAs in this sample switching from metro- to ZIP-level payment standards from 2015 to 2025.[16]

For observation i (listing or voucher contract) in year t, let g_i denote the year in which the PHA serving i adopts ZIP-level payment standards and $e = t - g_i$ denote event time. We estimate

$$Y_{it} = \sum_{e \neq -1} \beta_e \, \mathbb{1}\{t - g_i = e\} + \mu_{c(i)} + \delta_t + \varepsilon_{it}, \tag{2}$$

normalizing $\beta_{-1} = 0$, where $\mu_{c(i)}$ and δ_t are geography and year fixed effects, with geography measured at the CBSA level for listings and the PHA level for contracts, and standard errors are clustered at the corresponding geography.[17] The coefficients β_e trace the evolution of the outcome Y around adoption relative to the control group. We study four outcomes: the log payment standard, as a first-stage check; an indicator for hard and soft targeting at the listing stage; the listing-to-contract rent change (for linked HCV units); and the within-CBSA poverty rate percentile of the tract in which a voucher holder leases. Because adoption raises the subsidy limit in some ZIP codes and lowers it in others, we estimate Equation (2) separately for ZIP codes where adoption is expected to raise versus lower the payment standard (higher- and lower-PS ZIP codes). We classify each ZIP code by taking the difference between its 40th-percentile rent and the 40th-percentile rent of its CBSA, both measured at baseline in the 2011–2015 American Community Survey (5-year estimates), assigning ZIP codes with a positive difference to the higher-PS group and the rest to the lower-PS group.

We adopt the interaction-weighted estimator of Sun and Abraham (2021), which is robust to heterogeneity in treatment effects across adoption cohorts. We use the 92 PHAs adopting ZIP-level

[15] We require payment standards in both the pre-adoption year and the adoption year to be manually collected. We exclude switches for which either year's payment standard is inferred because the observed switch may lag the true SAFMR adoption. Using such dates would assign treatment too late and could classify already treated PHAs as controls.

[16] For the listing analysis, the unit of observation is the ZIP code rather than the PHA. Because SAFMR adoption occurs at the PHA level and a ZIP code may be served by several PHAs, we assign each ZIP code the treatment timing of its dominant PHA: the PHA with the largest share of the ZIP code's voucher contracts in 2015–2017, provided that share is at least 50%. We restrict the sample to ZIP codes with an average of at least 10 voucher contracts per year. We also drop listing spells first posted before 2019, because the listing-platform field we use to identify hard targeting is sparsely observed in earlier years (see Appendix B.2).

[17] For the listing analyses, we use ZIP-by-bedrooms and state-by-year-month fixed effects to control for varying listings coverage across metropolitan areas and over time.

payment standards in 2023–2025 as the control group.[18] Appendix Table D.3 compares early-adopting, later-adopting, and remaining PHAs in our nationwide payment standard dataset along several dimensions. Although later-adopting PHAs tend to serve more voucher holders, the three groups of PHAs are located in comparable areas in terms of county characteristics, suggesting that later-adopting PHAs provide a plausible comparison group. A robustness check following the empirical strategy in Ellen et al. (2025) yields similar results (see Appendix Figure D.9).[19]

5.2 Results

Figure 8a reports first-stage event studies for the log payment standard separately for higher- and lower-PS ZIP codes. SAFMR adoption shifts the payment standard as expected, in opposite directions across the two groups. Before adoption, both groups evolve in parallel with later adopters, with no significant pre-trends. After adoption, the payment standard rises by roughly 8 percent in higher-PS ZIP codes and falls by about 6 percent in lower-PS ZIP codes, with most effects emerging immediately and persisting over time.

Landlords respond to the change in the rent ceiling already at the listing stage. Figure 8b reports event studies for an indicator that a listing is hard- or soft-targeted. Targeting rises after adoption in both groups of ZIP codes, and by more where the payment standard fell. Averaged over the four years after adoption, the share of targeted listings rises by 1.7 percentage points (standard error 0.6) in lower-PS ZIP codes, from a pre-adoption share of 3.6 percent, and by 0.5 percentage points (standard error 0.2) in higher-PS ZIP codes, from 1.2 percent. Relative to these baselines, both responses amount to roughly 40 to 50 percent. That targeting rises most where the payment standard fell suggests that landlords in these lower-income areas compete more actively for voucher tenants, even though the subsidy itself has decreased, which we characterize in Section 6.

The shift in payment standards triggers additional listing-to-contract rent adjustments (see Figure 8c). After adoption, contract rents in lower-PS ZIP codes fall by roughly 3 percentage points more relative to their listed price, consistent with a tighter ceiling that leaves landlords less headroom to upcharge. In higher-PS ZIP codes, contract rents rise slightly relative to the listed rent, although the estimates are imprecise. This is causal evidence that the rent ceiling shapes pricing decisions, reinforcing the listing-to-contract rent changes documented in Section 4.

[18] Ellen et al. (2025) and Eriksen et al. (2025) instead define the control group by exploiting the cutoff rules HUD used to select the first 24 mandated metropolitan areas, based on the number of voucher holders served, the concentration of voucher holders in low-income Census tracts, and the share of rental units in ZIP codes where the SAFMR exceeds 110 percent of the FMR, using the metropolitan areas just below these thresholds. Our comparison group largely overlaps with theirs, since such PHAs not mandated by 2018 were largely mandated by 2025 and enter our controls.

[19] The listing-to-contract rent adjustment is noisier, because linkage rates are lower early in our sample period, but after adoption its coefficients are more positive in higher-PS areas than in lower-PS areas (panel b). Consistent with this pattern, contract rents rise in higher-PS areas and fall slightly in lower-PS areas (panel c).

Finally, we ask whether these shifts in the payment standard translate into where voucher holders live, an outcome plausibly mediated by the landlord responses just documented. Figure 8d reports effects on the within-CBSA poverty rate percentile of the tract in which a voucher holder leases. After adoption, we observe a downward level shift of about 2 percentiles in the poverty rate of voucher holders' tracts, consistent with SAFMR's intended effect of expanding access to lower-poverty neighborhoods and with prior evidence (Ellen et al., 2025; Eriksen et al., 2025).

Taken together, the event studies show that quasi-exogenous shifts in the rent ceiling move landlord behavior along precisely the margins identified in Section 4: targeting, contract rent setting, and the neighborhoods voucher holders reach. These reduced-form responses motivate and validate the structural model in Section 6, which we use to quantify landlords' valuation of program participation and to evaluate counterfactual payment standard designs.

6 A Model of Landlord Participation and Pricing

Sections 4 and 5 show that landlord participation and pricing respond to the program's design. Downcharging toward the payment standard indicates that some landlords value the payment guarantee enough to accept a lower rent. Upcharging, concentrated among lower-priced units, suggests that others use the kink in the subsidy schedule to raise rents toward the payment standard. In addition, the concentration of voucher targeting in high-poverty neighborhoods points to landlords there finding the program particularly profitable.

This evidence alone, however, cannot answer several questions. How large a rent concession are landlords willing to make in exchange for the payment guarantee, and how does this vary with market conditions? How costly is participation, and how far do upward rent adjustments offset these costs? How would participation, pricing, and voucher holders' access to neighborhoods respond to alternative designs, such as uniform increases in payment standards or stricter enforcement of rent reasonableness? To address these questions, we develop and estimate a three-stage model of landlord participation and rent setting.

6.1 Setup

A market, indexed by t, is a CBSA in a given year. Listing j belongs to market $t(j)$ and has observable characteristics Y_j, including bedroom count and tract-level characteristics (the poverty rate, rental vacancy rate, and log median rent), as well as the applicable payment standard S_j.

The tenant's monthly payment is $p_i(r, S_j) = \tau I_i + (r - S_j)_+$ and the government's housing assistance payment is $\mathrm{HAP}_i(r, S_j) = \min\{r, S_j\} - \tau I_i$, where $\tau = 30\%$. Rent up to the payment

standard can therefore increase without raising the tenant's payment, while each dollar of rent above the payment standard is borne by the tenant. The payment standard creates a kink in the tenant's payment schedule and, in turn, in the landlord's contracting incentives.[20]

The market unfolds in three stages.

1. **Stage 1 (Listing).** Given the market value of the unit R_j^*, the landlord j chooses the listed rent R and the voucher-targeting intensity V.

2. **Stage 2 (Tenant matching).** New vouchers are allocated across listings. The allocation intensity depends on $(R, V; S_j, Y_j)$ and the market-level voucher flow.

3. **Stage 3 (Contracting).** Conditional on a voucher match, the contract rent may differ from the listed rent. Without a voucher match, the landlord receives the expected market payoff associated with its listing choice.

6.2 Stage 1: Listing choice

Each unit has a market value R_j^*, which reflects the unit's quality and is known to the landlord but not observed by the econometrician. Given R_j^*, the landlord chooses the listed rent from the choice set $\mathcal{R}_j(R_j^*)$ and the targeting intensity $V \in \mathcal{V}_j \subseteq \{\text{none}, \text{soft}, \text{hard}\}$. Payoffs are expressed in units of one month of rent revenue.

The landlord's payoff from (R, V) is

$$u_j(R, V; A_t) = \underbrace{\Pi_j^{\text{MKT}}(R)}_{\text{market payoff}} + p_j^{\text{HCV}}(R, V; A_t)\big[\underbrace{\Pi_j^{\text{HCV}}(R, V)}_{\text{voucher payoff}} - \Pi_j^{\text{MKT}}(R)\big] - c_{V,j}^{\text{V}} + \varepsilon_j(R, V), \quad (3)$$

where $p_j^{\text{HCV}}(R, V; A_t)$ is the probability of a voucher match from Stage 2, which depends on the market voucher allocation level A_t, and $c_{V,j}^{\text{V}}$ is the cost of targeting intensity V, with $c_{\text{none},j}^{\text{V}} = 0$. The targeting cost captures the cost of advertising to voucher holders or listing through affordable housing channels and is incurred per listing regardless of whether a voucher match occurs. It may also capture broader differences across landlords, such as familiarity with the program or other persistent differences in willingness to target voucher holders. We allow the targeting cost to vary across CBSAs and with neighborhood characteristics, $c_{V,j}^{\text{V}} = c_{V,c(j)}^{\text{V}} + Y_j'\gamma_V$. The shocks $\varepsilon_j(R, V)$ follow a nested extreme value distribution whose nests are the targeting intensities, with scale σ_R across rents within a targeting intensity and scale $\sigma_V \geq \sigma_R$ across targeting intensities.

[20] At initial lease-up, program rules additionally cap the tenant's total share at $\bar{\tau} = 0.40$ of monthly income, implying the feasibility ceiling $\bar{r}_{ij} = S_j + (\bar{\phi} - \tau)I_i$. For simplicity, we abstract from this cap. Appendix Figure D.10 shows that tenant out-of-pocket payments equal 30 percent of income in the majority of cases and reach 40 percent in very few cases.

With a voucher match, the landlord's payoff is the expected contract rent net of a participation cost,

$$\Pi_j^{\text{HCV}}(R, V) = \bar{r}_j(R, V) - c_j^{\text{HCV}}, \tag{4}$$

where $\bar{r}_j(R, V)$ is the expected contract rent from Stage 3. A match, once it occurs, results in a lease, and the housing authority pays its share of the contract rent directly to the landlord.[21] The participation cost reflects administrative burdens, such as inspections, paperwork, and payment delays, and any aversion to voucher tenants.

Without a voucher match, the landlord leases to a market tenant, and the payoff is

$$\Pi_j^{\text{MKT}}(R) = (1 - \psi_j)R_j^*, \qquad R \in \mathcal{R}_j(R_j^*). \tag{5}$$

A market tenancy yields the landlord the share $1 - \psi_j$ of the market value, which accounts for expected losses from vacancy, nonpayment, and turnover. In the baseline, the choice set $\mathcal{R}_j(R_j^*)$ contains the rents within one percent of the market value, and the market payoff is flat over this range. We examine alternative rent choice sets, including allowing listed rents to vary within five or ten percent of market value and allowing them to take any value while imposing a quadratic penalty on the market payoff for deviations from R_j^*. Key cost parameter estimates are stable across these specifications, as shown in Appendix Table C.3.

The gain from a voucher match is then

$$\Pi_j^{\text{HCV}}(R, V) - \Pi_j^{\text{MKT}}(R) = \underbrace{\bar{r}_j(R, V) - R}_{\text{rent adjustment}} + \underbrace{\psi_j(R)\, R}_{\text{payment guarantee}} - \underbrace{c_j^{\text{HCV}}}_{\text{participation cost}}. \tag{6}$$

The rent adjustment is the expected change from the listed to the contract rent with a voucher tenant, characterized in Stage 3. The payment guarantee is the component of voucher value that scales with the listed rent. For simplicity, we refer to this component as the payment guarantee, though it encompasses several benefits of a voucher tenancy, including expected rent losses under a market tenancy that are avoided under a voucher and the value of longer expected tenure among voucher holders relative to market renters. With $\psi_j(R) \equiv 1 - \Pi_j^{\text{MKT}}(R)/R$, it equals $\psi_j(R)\, R = \psi_j R_j^* + (R - R_j^*)$ within the choice set. We assume that both the guarantee share and the participation cost vary with the neighborhood, $\psi_j = \psi + Y_j'\psi_Y$ and $c_j^{\text{HCV}} = \delta_{c(j)} + Y_j'\delta_Y$. We report robustness to alternative specifications in Appendix Table C.3.

[21] The tenant's own payment, τI_i, remains at risk under a voucher. We abstract from this residual risk, which may be limited because missed tenant payments can jeopardize continued voucher assistance and therefore strengthen tenants' incentives to remain current.

Choice probabilities. The market payoff $(1-\psi_j)R_j^*$ is common to all alternatives and cancels from the choice probabilities. Write

$$\bar{u}_j(R,V;A_t) = p_j^{\mathrm{HCV}}(R,V;A_t)\big[\Pi_j^{\mathrm{HCV}}(R,V) - \Pi_j^{\mathrm{MKT}}(R)\big] - c_{V,j}^{\mathrm{V}} \tag{7}$$

for the remaining payoff without the shock. Conditional on R^*, the choice probabilities are nested logit,

$$P_j(R,V \mid R^*;A_t) = \frac{\exp\{\bar{u}_j(R,V;A_t)/\sigma_R\}}{\exp\{I_j(V \mid R^*;A_t)/\sigma_R\}} \cdot \frac{\exp\{I_j(V \mid R^*;A_t)/\sigma_V\}}{\sum_{V' \in \mathcal{V}_j}\exp\{I_j(V' \mid R^*;A_t)/\sigma_V\}}, \tag{8}$$

where $I_j(V \mid R^*;A_t) = \sigma_R \log \sum_{R' \in \mathcal{R}_j(R^*)} \exp\{\bar{u}_j(R',V;A_t)/\sigma_R\}$ is the value of targeting intensity V before the rent shocks are realized. Integrating over R^* distributed according to F_j,

$$P_j(R,V;A_t) = \int P_j(R,V \mid R^*;A_t)\,\mathrm{d}F_j(R^*). \tag{9}$$

6.3 Stage 2: Tenant matching

Stage 2 allocates the flow of new vouchers across listings. In market t, a fixed voucher flow M_t is allocated across observed listings and an outside pool. For a given listing choice (R,V), define the allocation intensity

$$\lambda_j^{\mathrm{HCV}}(R,V;A_t) = A_t \exp\{x_j(R,V)'\theta\}, \tag{10}$$

$$x_j(R,V)'\theta = \theta_0 \mathbf{1}\{R > S_j\} + \theta_1(S_j - R)_+ + \theta_2(R - S_j)_+ + Y_j'\theta_Y + V'\theta_V. \tag{11}$$

The index $x_j(R,V)'\theta$ measures the listing's attractiveness to voucher holders. The term θ_0 allows a discrete change in attractiveness once the listed rent exceeds the payment standard. The slopes θ_1 and θ_2 allow attractiveness to vary with the distance to the payment standard on either side. A_t scales relative listing attractiveness into expected voucher allocations. It is given by

$$A_t = \frac{M_t}{s_t^0 + \sum_{k:t(k)=t}\exp\{x_k(R_k,V_k)'\theta\}}, \tag{12}$$

where $s_t^0 > 0$ is the attractiveness of units outside the observed listing market. Thus,

$$\lambda_j^{\mathrm{HCV}}(R,V;A_t) = M_t \frac{\exp\{x_j(R,V)'\theta\}}{s_t^0 + \sum_{k:t(k)=t}\exp\{x_k(R_k,V_k)'\theta\}}.$$

The allocation mechanism introduces the main equilibrium interaction across landlords, discussed in Section 6.5.

We map this voucher-allocation intensity $\lambda_j^{\text{HCV}}(R, V; A_t)$ into the probability of at least one voucher match in Stage 1 using

$$p_j^{\text{HCV}}(R, V; A_t) = 1 - \exp\{-\lambda_j^{\text{HCV}}(R, V; A_t)\}. \tag{13}$$

In most cases, allocation intensity is low, so $p_j^{\text{HCV}} \simeq \lambda_j^{\text{HCV}}$.

6.4 Stage 3: Contract rent adjustments

Conditional on a voucher match, the landlord and tenant may agree on a contract rent that differs from the listed rent. The rent adjustment depends on whether the listed rent is below or above the payment standard and how far it is from the standard. Contract rents above the payment standard face an additional friction because the tenant bears the excess rent. Separately, any change from the listed rent incurs an adjustment cost.

Let m_j^* denote the desired adjustment absent contracting frictions:

$$\begin{aligned} m_j^* &\sim N(\mu_j, \sigma_m^2), \\ \mu_j &= \mu_+(S_j - R)_+ - \mu_-(R - S_j)_+ + Y_j'\mu_Y + V_j'\mu_V. \end{aligned} \tag{14}$$

The realized adjustment solves

$$m_j = \arg\min_m \Big\{(m - m_j^*)^2 + \lambda_S(R + m - S_j)_+^2 + \lambda_m|m|\Big\}, \qquad \lambda_S, \lambda_m \geq 0, \tag{15}$$

and the contract rent is $r_j = R + m_j$. The first term keeps the realized adjustment close to the desired adjustment. The second adds a friction to contract rents above the payment standard. The third is a general adjustment cost with a kink at zero, so sufficiently small desired adjustments leave the listed rent unchanged, generating a mass at $r_j = R$. The expected contract rent conditional on a voucher match, which enters Stage 1, is

$$\bar{r}_j(R, V) = R + \mathbb{E}[m_j \mid R, V, Y_j, S_j]. \tag{16}$$

6.5 Equilibrium

Landlords interact through the allocation of voucher tenants. Taking the allocation level A_t as given, each landlord chooses its listed rent and targeting intensity according to Stage 1. The result-

ing choice probability $P_j(R, V; A_t)$ gives the probability that listing j chooses (R, V) conditional on A_t, integrating over the unobserved market value and the listing-payoff shocks.

An equilibrium in market t consists of a scalar A_t^* and the induced listing-choice probabilities $P_j(R, V; A_t^*)$ such that

$$A_t^* = \frac{M_t}{s_t^0 + \underbrace{\sum_{j:t(j)=t} \sum_{R \in \mathcal{R}_j} \sum_{V \in \mathcal{V}_j} P_j(R, V; A_t^*) \exp\{x_j(R, V)'\theta\}}_{T_t(A_t^*)}}. \tag{17}$$

This fixed point captures the feedback between listing choices and the allocation of voucher tenants. A higher A_t raises match probabilities and, where a match is valuable, draws listings toward the payment standard and into targeting. These choices raise the aggregate attractiveness of observed listings T_t, which lowers A_t, so listing choices are strategic substitutes. The equilibrium is unique when $A\,[s_t^0 + T_t(A)]$ is increasing in A, which we check locally at each counterfactual equilibrium, as described in Appendix C.4.

7 Estimation

7.1 Estimation procedure

Model estimation samples. We estimate the model sequentially, working backward from Stage 3 to Stage 1. The Stage 3 sample consists of voucher contracts linked to listings for admissions from 2015 to 2025. We exclude contracts administered by Moving to Work PHAs and matches for which either the listed or contract rent exceeds twice the other. We also require observed targeting status in the linked listing and complete tract covariates. We exclude CBSAs with fewer than 100 linked contracts. The final Stage 3 sample contains 128,489 contracts in 216 CBSAs. The Stage 2 sample consists of listing spells first listed between 2015 and 2025 in these CBSAs. We require an observed targeting status, payment standard, and complete tract covariates, and restrict the sample to zero- to four-bedroom units with listed rents between \$100 and the state-specific 99th percentile. This leaves 29.96 million listings in 210 CBSAs and 2,194 CBSA-years. The Stage 1 sample further restricts the Stage 2 sample to listings from 2019 to 2025, the years in which both soft and hard targeting are measured. This leaves 24.98 million listings. As a robustness check, we estimate Stage 1 on the entire sample period, 2015 to 2025, in which hard targeting is not in the choice set before 2019.

Stage 3: Contract rents. We estimate the Stage 3 parameter vector

$$\vartheta_3 = (\mu_+, \mu_-, \mu_Y, \mu_V, \sigma_m, \lambda_S, \lambda_m)$$

by maximum likelihood using the linked new admission contracts and denote the estimate by $\widehat{\vartheta}_3$. We additionally include CBSA, year, and bedroom fixed effects in the latent mean μ_j. The likelihood combines the probability mass at $r_j = R$ with the continuous density of nonzero adjustments. Appendix C.1 gives the likelihood, expected rent calculation, and specification comparisons.

Stage 2: Tenant matching. The matching procedure may associate several new admission contracts with several candidate listing spells due to fuzzy address matching. Let n_h denote the number of candidate spells associated with contract h. We assign each candidate a share $1/n_h$ of that contract and define the listing-level matching intensity

$$y_j = \sum_h \frac{\mathbf{1}\{h \to j\}}{n_h}.$$

We estimate the Stage 2 parameter vector

$$\vartheta_2 = (\theta_0, \theta_1, \theta_2, \theta_Y, \theta_V)$$

from

$$\mathbb{E}[y_j \mid x_j, t] = \exp\{\alpha_t + x_j'\vartheta_2\}, \qquad t = t(j), \tag{18}$$

where α_t is an unrestricted metro-year fixed effect. We estimate Equation (18) by Poisson Pseudo-Maximum Likelihood (PPML) with CBSA-by-year and bedroom fixed effects, cluster standard errors at the CBSA level, and denote the resulting estimate by $\widehat{\vartheta}_2$. Given $\widehat{\vartheta}_2$, the administrative voucher flow and the linked share of new admissions determine $\widehat{s}_t^0$ and $\widehat{A}_t$ as described in Appendix C.2.

Stage 1: Listing choices. Conditional on $(\widehat{\vartheta}_3, \widehat{\vartheta}_2, \widehat{A})$, the Stage 1 parameters are

$$\vartheta_1 = \big(\sigma_R,\ \sigma_V,\ \psi,\ \psi_Y,\ \delta_Y,\ \gamma_{\text{soft}},\ \gamma_{\text{hard}},\ \{\delta_c, c^{\text{V}}_{\text{soft},c}, c^{\text{V}}_{\text{hard},c}\}_c\big).$$

We estimate ϑ_1 by maximum likelihood,

$$\ell(\vartheta_1) = \sum_j \log P_j(R_j, V_j; \vartheta_1),$$

where P_j is the choice probability in Equation (9) and R_j is the observed rent rounded up to the rent grid, whose step size is one percent of the payment standard, as defined in Appendix C.3.

The distribution F_j is a log-spline density of $\log(R_j^*/R_j^{\text{mkt}})$ whose shape is specific to each CBSA and bedroom count, where the benchmark rent R_j^{mkt} is the median rent of non-targeted listings in the same CBSA, bedroom count, and year, adjusted for whether the listing's tract is typically more or less expensive than its CBSA. Thus, F_j shifts with the tract rent level, while its shape is common within a CBSA and bedroom count. See Appendix C.3 for details. The targeting choice set $\mathcal{V}_j$ always contains no targeting and contains soft or hard targeting only in CBSA-years in which it is observed.

For inference, we account for estimation uncertainty from all three stages. In each of 100 bootstrap replications, we draw ϑ_2 and ϑ_3 from their estimated asymptotic distributions with CBSA-clustered covariance matrices, recompute the allocation levels $\widehat{A}_t$, the match probabilities, and the expected contract rents, and re-estimate ϑ_1. The variance of the Stage 1 estimates across replications measures the uncertainty that Stages 2 and 3 add. We add it to the variance of the Stage 1 maximum likelihood estimator conditional on the Stage 2 and Stage 3 estimates.

Sequential estimation relies on two assumptions. First, conditional on (R, V, Y_j, S_j), the matching and contracting outcomes of Stages 2 and 3 do not depend on the market value, so that voucher holders and PHAs respond to a listing only through its listed rent, targeting intensity, characteristics, and payment standard. Second, landlords know the allocation level of their market, which we measure by $\widehat{A}_t$, and hence the match probability and the expected contract rent of every rent and targeting intensity they can choose.

7.2 Identification

Stages 2 and 3 deliver the match probability and the expected rent adjustment at every rent and targeting intensity. Stage 1 then recovers the cost parameters from how listed rents depart from what the market alone would produce and from the observed targeting shares.

The rent margin identifies the net gain from a voucher match at the payment standard. The match probability drops discretely above the payment standard, while the expected contract rent is continuous there, so the payment standard creates a notch in the payoff. For a unit whose market value is close to the payment standard, the log ratio of the probabilities of listing at the payment

standard, S_j, and just above it, S_j^+, given the targeting intensity V, is approximately

$$\frac{\overbrace{\left[p_j^{\mathrm{HCV}}(S_j, V; A_t) - p_j^{\mathrm{HCV}}(S_j^+, V; A_t)\right]}^{\text{voucher allocation at vs. just above } S_j \text{ (Stage 2)}} \left(\overbrace{\mathbb{E}[m_j \mid S_j, V]}^{\text{expected rent adjustment (Stage 3)}} + \psi_j S_j - c_j^{\mathrm{HCV}}\right)}{\underbrace{\sigma_R}_{\substack{\text{scale of idiosyncratic rent shocks} \\ \text{(Stage 1)}}}}. \tag{19}$$

The drop in the match probability and the expected adjustment are known from Stages 2 and 3 and vary across listings with the allocation level, the neighborhood, the targeting intensity, and the position of the payment standard relative to market values. The excess mass at the payment standard and the missing mass just above it therefore identify the net gain $\psi_j S_j - c_j^{\mathrm{HCV}}$.

Listed rent choices reveal the net gain from a voucher match, not its separate components. A lower administrative burden and a weaker aversion to voucher tenants, for example, shift payoffs in the same way, so we do not attempt to decompose the participation cost. The payment guarantee is distinguished from the participation cost only because it is proportional to the rent. How the gain at the payment standard varies with the level of the payment standard thus identifies ψ, and its average level identifies δ_c. Any part of the guarantee that does not scale with the rent is absorbed by the participation cost, and vice versa. Across neighborhoods, ψ_Y and δ_Y are likewise distinguished only by whether the gradient of the net gain scales with the rent. Because neighborhood characteristics are correlated with rent levels, this decomposition is weakly identified, while the gradient of the net gain itself is not. The poverty gradient of the guarantee share ranges from -0.41 to 0.32 across specifications, for example, while the net gain in high-poverty tracts changes little, as shown in Appendix Table C.3. The net gain is therefore the object the data identify robustly, and it is the one we interpret.

Identifying the targeting parameters requires separating the benefit of targeting from its cost. Observed match rates pin down the benefit directly. A non-targeted listing matches with a voucher holder with probability of about 0.4 percent, and soft and hard targeting raise this roughly fourfold and eightfold, respectively. Because these baselines are low, the gains in expected payoff are small. The estimates we present in the next subsection imply that targeting raises a listing's expected monthly payoff by less than \$10, against targeting costs of about \$200. Thus, conditional on voucher match rates, targeting shares reveal the targeting costs relative to the scale of the idiosyncratic targeting shock σ_V. Differences in the level of targeting shares across metropolitan areas identify $c^{\mathrm{V}}_{\mathrm{soft},c}$ and $c^{\mathrm{V}}_{\mathrm{hard},c}$, and their gradients across neighborhoods identify γ_V.

7.3 Parameter Estimates

Table 2 reports estimates from all three stages. Panel A reports the Stage 3 estimates. Desired rent adjustments are driven primarily by the listing's position relative to the payment standard. Each \$100 of headroom below the standard raises the desired adjustment by \$52 ($\hat{\mu}_+ = 0.52$), while each \$100 above it lowers the desired adjustment by \$112 ($\hat{\mu}_- = 1.12$). Adjustment is also costly, with $\hat{\lambda}_m = \$157$. Consequently, 29 percent of contracts listed at or below the standard retain the listed rent, compared with 30 percent in the model. Among listings above the standard, the corresponding shares are 8 and 14 percent. Conditional on distance to the standard, neighborhood characteristics and targeting have little additional effect, with only tract median rent statistically significant.

Panel B reports the Stage 2 estimates. Listing above the payment standard lowers allocation intensity by 55 percent ($\hat{\theta}_0 = -0.81$), and each additional \$100 above the standard lowers it by another 11 percent. Pricing below the standard generates no additional matches, as $\hat{\theta}_1$ is small and insignificant. Soft targeting raises allocation intensity 4.3-fold and hard targeting 7.7-fold. Voucher allocations also concentrate in poorer and cheaper tracts. A ten percentage point increase in the poverty rate raises allocation intensity by about 20 percent, while a one percent increase in tract median rent lowers it by about one percent. These Stage 3 and Stage 2 estimates are robust to alternative specifications (see Appendix Tables C.1 and C.2).

Panel C reports the Stage 1 estimates. In the average neighborhood, a market tenancy yields the landlord $1-\hat{\psi} = 0.79$ of the market value, so the payment guarantee is worth about 21 percent of the rent. As discussed in Section 6.2, we use the term payment guarantee as a shorthand for voucher-tenancy benefits that scale with rent. These include not only guaranteed government payments but also other benefits, such as the longer expected tenure of voucher holders.[22] The participation cost in the average neighborhood is negative, $-\$133$ per month, with a standard deviation of \$395 across CBSAs. With a ten percentage point higher tract poverty rate, the guarantee share rises by 0.011 and the participation cost by \$80. Targeting costs average \$171 per month for soft targeting and \$221 for hard targeting, and fall by \$9 and \$12 with a ten percentage point higher tract poverty rate.

As discussed in Section 7.2, the payment guarantee and the participation cost are separately identified only through how they scale with rent, so we focus on their combination, reflected in the net gain from a voucher match. Figure 9a shows the net gain, defined as the expected rent adjustment plus the value of the payment guarantee minus the participation cost, across within-CBSA poverty

[22] Voucher households stay at an address 40–50 percent longer than unassisted renters, with a median tenure of 3 versus 2 years and a mean tenure of 5.0 versus 3.6 years. We estimate these tenure differences using the 2015–2023 American Housing Survey panel and a Kaplan-Meier estimator, which recovers the survival function $S(t) = P(\text{tenure} > t)$ non-parametrically from exit rates among tenancies still at risk while accounting for incomplete spells. Because the survey records only the move-in year and brackets exits between interviews, we place move-ins and interviews at mid-year and exits midway between waves. HUD administrative data on voucher tenancies from 2015–2025, observed annually, yield similar estimates for this sample, with a median tenure of 3.2 years and a mean of 4.3 years.

deciles evaluated at the 25th, 50th, and 75th percentiles of the benchmark rent within each CBSA and bedroom count. Within a neighborhood, lower-rent listings gain more, largely because they lie below the payment standard, where expected upward rent adjustments are greater. At a given rent, however, the net gain is larger in low-poverty tracts. At the median rent, it is $516 per month in the lowest poverty quintile compared with $82 in the highest. This difference is driven primarily by higher participation costs in poorer tracts, which may reflect more costly inspections and repairs in older housing stock.

Despite these positive gains, few landlords find it worthwhile to actively target voucher holders. Across listings, the net gain is positive for 71 percent and averages $179 per month, so a voucher tenancy, once matched, would be worth more than a market tenancy for many listings. But the gain enters listing decisions only after being weighted by the low probability of a voucher match. The expected gain is therefore only about $1 per month, and soft and hard targeting raise it by about $3 and $6, respectively, far less than their costs. As a result, fewer than 2 percent of listings find it worthwhile to target voucher holders in the model, as shown in Appendix Table D.5.

Although targeting is rare overall, it is more common in high-poverty tracts, even though the net gain from a voucher match is lower there. Two features explain this pattern. First, targeting is less costly in high-poverty tracts, as shown in Figure 9b. The targeting cost captures not only the cost of advertising to voucher holders but also differences across landlords in familiarity with the program, willingness to participate, and the value of reaching a larger applicant pool. These cost estimates also govern how landlords respond to the changes in program incentives considered in Section 8. As discussed in Section 7.4, the model's response to SAFMR adoption provides an external validation of this margin. Second, conditional on rent and targeting intensity, the match probability is higher in high-poverty tracts, as shown in Figure 9c. This partly reflects voucher holders' search behavior, including attachment to existing neighborhoods, social networks, and other search frictions (Rosenblatt and DeLuca, 2012; Bergman et al., 2024).

7.4 Model Fit

Figure 10 compares model predictions with the data at each stage. Panel (a) shows that the Stage 3 model closely reproduces both the probability and magnitude of rent adjustments on either side of the payment standard, except just above the standard, where the model understates the probability of adjusting because the adjustment cost λ_m leaves small desired adjustments at the listed rent. Panel (b) shows that the Stage 2 model captures how voucher matches vary with the listed rent relative to the payment standard and with tract poverty, vacancy, and median rent, although it overstates the decline in matches just above the standard, predicting a drop of 6.5 rather than 3.4 matches per 1,000 listings. This misfit is especially pronounced among targeted listings, as shown in Appendix

Figure D.11a. Panel (c) shows that the Stage 1 model matches both the distribution of listed rents around the payment standard and the neighborhood gradients in targeting. Across poverty deciles, the targeting share rises from 0.5 to 6.1 percent in the data and from 0.6 to 6.0 percent in the model, while aggregate soft and hard targeting shares are 1.36 and 0.54 percent in the data and 1.35 and 0.53 percent in the model. The model reproduces the excess mass of non-targeted listings exactly at the payment standard but understates the corresponding excess mass among targeted listings, as shown in Appendix Figure D.12a.

We also compare the model with the response of voucher targeting to SAFMR adoption, which is not used in estimation, as additional validation. Re-estimating the difference-in-differences specification from Section 5 after replacing each listing's targeting outcome with its model-predicted probability yields increases in targeting of 0.6 percentage points in lower-PS ZIP codes and 0.3 percentage points in higher-PS ZIP codes, with bootstrap standard errors of 0.06 and 0.04 percentage points. These responses are similar in magnitude to the event study estimates in Section 5.2 and reproduce the larger increase in targeting in lower-PS ZIP codes.[23]

8 Counterfactual Analysis

We use the estimated model to evaluate alternative program designs. Each counterfactual changes one policy instrument and solves the equilibrium of Section 6.5 again, with landlords re-optimizing their targeting and their rents and the flow of new admissions re-sorting across observed listings and the outside pool. Because the voucher flow is fixed, a policy changes where new admissions lease and on what terms, not how many vouchers are used.

In evaluating counterfactual policies, we focus on two goals of the voucher program, where voucher holders live and what the program costs.[24] Our main neighborhood outcome is the mean poverty percentile of the tracts where new admissions lease, where the percentile ranks a tract's 2020 poverty rate among all tracts of its CBSA, defined the same way as in Section 5. We also report the share of new admissions that lease in the lowest-poverty quartile of tracts (Q1), those at

[23] Using the listings and CBSAs in the Stage 1 sample, the corresponding data estimates are 1.5 and 0.5 percentage points, with standard errors of 1.2 and 0.4 percentage points. Neither is statistically different from zero in this smaller sample, but both confidence intervals contain the model predictions.

[24] A central goal of recent voucher policy is to help voucher holders lease in lower-poverty neighborhoods. The Small Area Fair Market Rent rule aims to "help reduce the number of voucher families that reside in areas of high poverty concentration" (U.S. Department of Housing and Urban Development, 2016), and HUD's assessment of housing authorities awards bonus points when voucher families with children live in, or move to, low-poverty tracts, defined relative to the poverty rate of the area (U.S. Department of Housing and Urban Development, 1998). Voucher funding is capped, so a higher cost per voucher reduces the number of families the program can serve. In adopting Small Area Fair Market Rents, HUD noted that if per-voucher costs rose, "fewer households would receive assistance without an overall increase in program funds" (U.S. Department of Housing and Urban Development, 2016).

or below the 25th percentile. Our spending measure is the housing assistance payment plus any sign-up bonus per new admission and month. We summarize cost-effectiveness as the change in spending per percentile point of decline in the mean poverty percentile. Budget-neutral policies hold program-wide spending fixed and therefore show how far the same budget can move voucher holders toward lower-poverty neighborhoods. We also report how each policy affects landlords, through the change in the value of the program per in-market match, and tenants, through changes in payments above the payment standard, as described in Appendix C.4.

8.1 Policy Experiments

We consider four categories of counterfactual policies.

- *Higher payment standards.* Payment standards rise by 10 or 20 percent, either in all tracts, only in the lowest-poverty tracts (Q1), or in tracts below the median poverty percentile (Q1 and Q2). The uniform increase captures greater program generosity, while the targeted increases resemble exception payment standards for low-poverty areas.

- *Rent-indexed payment standards.* Each tract's standard is set in proportion to its benchmark rent R_j^{mkt}, as defined in Appendix C.3. The multiplier is common to the CBSA, bedroom count, and year and is chosen so that the mean payment standard across listings in each CBSA, bedroom count, and year equals its current value. This policy is a stylized version of universal Small Area Fair Market Rents and raises standards in expensive tracts while lowering them in cheaper ones.[25] A budget-neutral variant sets every indexed standard by a common percentage so that program-wide spending is unchanged, and a hold-harmless variant allows standards to rise but leaves them unchanged where the rent-indexed standard would be lower.

- *Sign-up bonus.* The PHA pays the landlord a one-time bonus for each new voucher lease, equal to one or two months of the contract rent, either for all leases or only for leases in Q1 tracts or in Q1 and Q2 tracts.[26] We spread the one-time bonus over a 24-month tenancy when computing the monthly payoff. Because the bonus is proportional to contract rent, it provides a larger incentive where rents are higher.

- *Rent-reasonableness enforcement.* The PHA reviews each proposed rent increase above the listed rent and rejects it with probability $1 - \rho$. If rejected, the contract rent remains at the

[25] Small Area Fair Market Rents are set by ZIP code, while the counterfactual sets standards by tract, the geography at which the benchmark rent varies in our setting.

[26] This is motivated by a recent HUD policy. Under the Landlord Incentives cohort of the Moving to Work (MTW) expansion, 28 PHAs selected in 2022 received waivers to offer landlord incentives outside standard program rules, with the aim of expanding the pool of participating landlords and raising voucher holders' lease-up rates. Signing bonuses of up to one month of contract rent were the incentive that participating PHAs most often adopted (Kumar et al., 2024).

listed rent. At $\rho = 0$, upward adjustments are entirely eliminated. A budget-neutral variant uses the savings from enforcement at $\rho \in \{0.5, 0\}$ to finance higher payment standards, either in all tracts, in Q1 and Q2 tracts, or, combined with rent indexing, as a common increase in the indexed standards financed by both savings.

In the model estimation sample, we link 16.8 percent of new voucher contracts to observed rental listings. We therefore need an assumption about how the outside market responds to counterfactual program designs when recomputing the equilibrium in Section 6.5. The response depends on how similar the outside pool is to units advertised on online rental platforms. This pool includes leases in place, units found through friends, family, or the housing authority, and units advertised online but missing from our listings. Survey evidence on how voucher holders find their units implies that between 17 and 75 percent of the outside pool may respond to program design like observed listings in the same market and neighborhood.[27] We use the midpoint, 46 percent, as the baseline and report robustness at both bounds. Appendix C.4 describes the calibration and other computational details.

8.2 Results

Table 3 reports outcomes for all new voucher admissions under the baseline outside-pool responsiveness. Appendix Table D.4 restricts to admissions matched to observed listings, Appendix Table D.5 reports targeting shares and tenant payments, and Appendix Table D.6 reports outcomes at both bounds of the outside-pool responsiveness. In the status quo, the mean poverty percentile of new admissions is 70.6, and only 7.2 percent lease in Q1 tracts. All dollar figures below are per month.

A uniform increase is an expensive way to move voucher holders toward lower-poverty tracts. A 10 percent increase in all tracts lowers the mean poverty percentile by only 0.19 points and raises the Q1 share from 7.22 to 7.37 percent, while spending rises by $94 per admission (7 percent), mostly on admissions that would have leased anyway under the old standard. Each percentile point costs $499 per admission. The same payment standard increase exclusive to Q1 tracts lowers the mean percentile by 0.43 points at $10 per admission, or $24 per percentile point, about 20 times less. Extending the increase to Q1 and Q2 tracts lowers the mean percentile by 0.84 points at $27 per admission, or $32 per percentile point. Its Q1 share is slightly lower than under the Q1 standard increase, 7.83 against 7.94 percent, because most of the voucher holders it moves lease in

[27] In the national study of voucher success rates by Finkel and Buron (2001), 21 percent of successful voucher holders leased in place. In a 2024 survey of voucher holders in the Twin Cities (Metropolitan Council, 2026), between 25 and 54 percent reported hearing about their current home through an online website. The lower bound assumes that only units found on the open market respond like observed listings, while the upper bound assumes that all units except leases in place do so.

Q2 tracts.[28] Doubling any of these increases roughly doubles both the effects and the costs. The value of the program to landlords of observed listings rises by $102 per voucher match under the uniform 10 percent increase and by $15 and $37 under the increases in Q1 tracts and in Q1 and Q2 tracts.

Rent-indexed payment standards, as under Small Area Fair Market Rents, raise standards in expensive tracts and lower them in cheap ones without designating particular tracts. The mean poverty percentile falls by 0.89 points and the Q1 share rises from 7.22 to 7.87 percent. Among in-market admissions, the mean percentile falls by 2.06 points, as shown in Appendix Table D.4. This is close to the roughly 2 percentile decline following SAFMR adoption in Section 5.[29] The policy also saves $102 per admission (7.7 percent). This is because although the mean standard across listings is unchanged, new admissions are concentrated in high-poverty tracts, where benchmark rents are low and the standard falls. Contract rents there are often set at the standard and fall with it, so the mean contract rent of new admissions falls by $95, and landlords lose $76 per voucher match. Spending the saving on a 9.8 percent increase in every indexed standard makes the policy budget neutral and lowers the mean percentile by 1 point, while landlords gain $35 per voucher match. The hold-harmless variant lowers the mean percentile by 0.36 points at $39 per admission, or $108 per percentile point, between the uniform and the targeted increases.

A sign-up bonus makes voucher tenants more attractive to landlords but moves voucher holders toward higher-poverty tracts. A bonus of one month of the contract rent raises the mean poverty percentile by 0.11 points and a two-month bonus by 0.24 points.[30] The bonus raises the payoff from a voucher match in every tract, whereas a higher standard raises it most where rents are near or above the standard, which is mostly in low-poverty tracts, so the voucher matches the bonus adds are mostly in high-poverty tracts. A one-month bonus costs $56 per admission, most of it paid on admissions that would have leased anyway. Paying the bonus only for leases in Q1 tracts, or in Q1 and Q2 tracts, barely moves voucher holders. It lowers the mean percentile by 0.01 and 0.03 points, at about $500 per percentile point, 16 to 21 times the cost of a higher payment standard in the same tracts. A targeted payment standard is far more effective than a targeted bonus because it works through two channels. Both policies raise the landlord's gain from a voucher match, but this channel is weak because any given listing has a low probability of matching with a voucher holder, so landlords' listing decisions respond little. A higher payment standard additionally expands the set of listings priced at or below the standard, raising their match rates even when landlords do not change their listings.

[28] Raising standards by 10 percent in Q2 tracts alone lowers the mean percentile by 0.43 points, as in Q1 tracts, but at $39 per percentile point. The same increase in Q3 tracts lowers it by only 0.15 points, and in Q4 tracts it raises it by 0.78 points.

[29] The two estimates are not directly comparable because they are based on different sets of CBSAs.

[30] The bonus enters the monthly payoff spread over a tenancy of 24 months.

Rent-reasonableness enforcement saves spending at little cost to the neighborhood outcome. Each 25 percentage point reduction in the approval probability saves about $11 per admission, and ruling out upward adjustments saves $45 (3.4 percent). The mean poverty percentile falls slightly, by 0.14 points. Upward adjustments are largest for listings below the payment standard, which are concentrated in high-poverty tracts, so enforcement lowers the gain from a match most there. Landlords lose $57 per voucher match, because they can recover only part of the forgone adjustment by listing higher within the window.

We next examine how the savings from tighter rent-reasonableness enforcement can be used to finance higher payment standards and shift voucher holders across neighborhoods. Spent on higher standards in all tracts, the savings at $\rho = 0.5$ and $\rho = 0$ fund increases of 2.9 and 7.6 percent, which lower the mean poverty percentile by only 0.14 and 0.33 points. Landlords lose only $3 and $4 per voucher match, against $28 and $57 under enforcement alone, because the higher standard offsets most of the forgone adjustments. Spent on standards in Q1 and Q2 tracts, the same savings fund increases of 10.3 and 28.2 percent and lower the mean percentile by 0.90 and 1.97 points. The latter is the largest effect of any budget-neutral policy we consider and is close to the effect of SAFMR adoption. Concentrating the saving in Q1 tracts alone would require increases of 27 percent at $\rho = 0.5$ and more than 49 percent at $\rho = 0$, because few voucher holders lease there. Combined with rent indexing, both savings fund a common increase of 11.9 and 14.8 percent in the indexed standards, which lowers the mean percentile by 1.06 and 1.13 points while landlords gain $32 and $29 per voucher match.

The calibration of the outside-pool responsiveness matters for where new admissions lease but hardly for spending. Across its bounds, as shown in Appendix Table D.6, the 10 percent increase in Q1 tracts lowers the mean poverty percentile by 0.25 to 0.60 points, at $38 to $18 per percentile point. The budget-neutral rent index lowers it by 0.57 to 1.43 points, and budget-neutral enforcement at $\rho = 0$ spent in Q1 and Q2 tracts by 1.26 to 2.55 points. A more responsive outside pool moves more of the outside admissions with the observed listings in the same neighborhood, which amplifies every shift across neighborhoods. The changes in spending differ by at most $7 per admission across the bounds, because the payments on outside admissions follow from the policy's standards, bonuses, and enforcement whatever the response of the outside pool.

Overall, the counterfactual analysis suggests that the most effective way to move voucher holders toward lower-poverty neighborhoods is to target payment standards to those areas, either directly through higher standards in low-poverty tracts or through standards indexed to local rents. A higher standard both expands the set of listings priced at or below it and raises landlords' returns to voucher participation where rents are near or above the standard, conditions more common in low-poverty tracts. By contrast, a sign-up bonus raises landlords' returns in every tract, draws additional matches

mainly in high-poverty tracts, and is paid on admissions that would have leased anyway. Even when targeted to low-poverty tracts, it shifts few voucher holders. When additional funding is unavailable, stricter review of rent adjustments can finance targeted increases in payment standards. Using the savings to raise standards in tracts below the median poverty percentile, or to fund rent-indexed standards, shifts voucher holders toward lower-poverty neighborhoods without increasing total spending.

These results have several limitations. We hold the flow of new admissions fixed, so the counterfactuals change where voucher holders lease and on what terms but not the number who successfully lease. We also hold market rents fixed and therefore do not capture spillovers to the unassisted rental market. The budget-neutral policies balance housing assistance payments and bonuses only, and they do not account for the administrative cost of stricter rent-reasonableness review, so they may overstate the payment-standard increase that enforcement can finance. Finally, the response of the outside pool, which accounts for most new admissions, is calibrated from survey evidence rather than estimated, and the effects on neighborhood allocation are sensitive to this assumption.

9 Conclusion

We assemble the first national linkage of online rental listings to administrative voucher contracts, together with a newly constructed panel of the program's price ceilings, to study landlord participation and pricing in the Housing Choice Voucher program. The data reveal that landlords respond to the program's payment rule at every stage. Targeting of voucher holders concentrates in high-poverty neighborhoods and predicts participation by an order of magnitude. Rents bunch at the program's ceiling, modestly in listings, especially targeted ones, and much more strongly in contracts, indicating substantial adjustment between listing and contracting. The adjustment runs in both directions. Among linked voucher units, rents rise between listing and contract in 37.5 percent of cases and fall in 34.3 percent, by about 8 and 9 percent on average, respectively, with no comparable adjustments in market transactions over the same range of rents. Quasi-experimental variation from ZIP-level payment standards further shows that both participation and pricing respond when the rent ceiling changes.

To interpret these patterns, we develop and estimate a three-stage model of landlord voucher targeting, matching, and pricing. The counterfactuals yield several policy implications. Raising payment standards only in low-poverty neighborhoods shifts voucher holders toward them at less than one-twentieth of the cost of a uniform increase, while rent-indexed payment standards shift them in the same direction and reduce spending. Sign-up bonuses, by contrast, raise what landlords receive but lead to more voucher contracts in high-poverty neighborhoods, and even when

paid only in low-poverty neighborhoods they do little to move voucher holders. Tighter review of rent reasonableness saves spending with little effect on where voucher holders lease, and these savings can fund higher payment standards in lower-poverty neighborhoods or rent-indexed payment standards at no net cost. Because we hold the flow of new admissions fixed, these results speak to where voucher holders lease and on what terms rather than how many succeed in leasing. Future work could incorporate this dimension by modeling voucher-holder search, landlords' screening and refusal of voucher applicants, and rent negotiation between landlords and voucher holders explicitly.

These lessons extend beyond housing. In in-kind programs that rely on voluntary private providers and limit what the program pays, program design affects not only how much providers receive but also which providers participate and where, which may also impact the quality of the good being provided. Broad increases in payments spend heavily on inframarginal providers, while more targeted changes can alter participation at lower cost. The distribution of willing providers, not only of eligible recipients, can therefore be central to what the program delivers.

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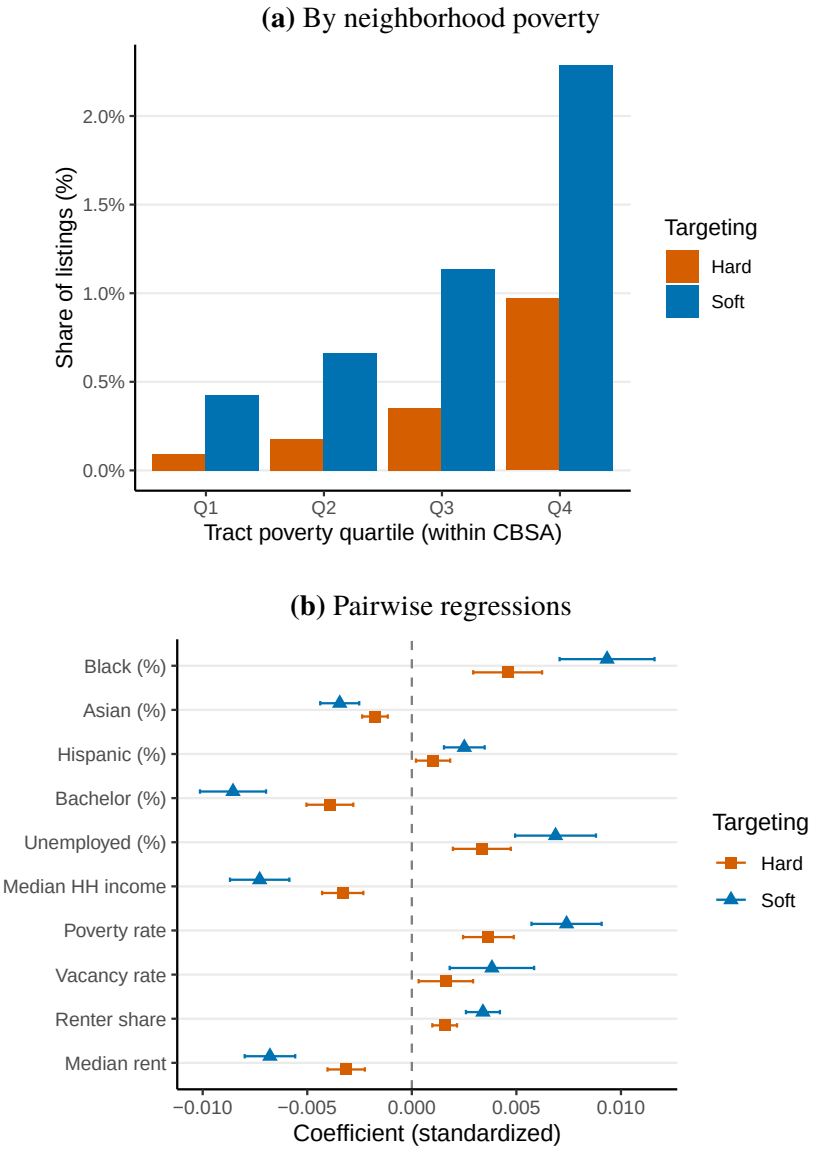
Figures and Tables

Table 1: Summary Statistics

	HCV New Admissions		RentHub Listings	
	All	Linked	Scrape level	Spell level
	(1)	(2)	(3)	(4)
# Observations	1,482,120	201,863	421,477,315	38,948,845
# CBSAs	924	860	928	928
Rent (USD)				
25th Percentile	644	781	1,350	1,381
Median	850	1,100	1,800	1,900
75th Percentile	1,217	1,650	2,340	2,597
Mean	1,019.2	1,306.9	1,980.9	2,146.8
Standard Deviation	565.3	716.1	1,030.4	1,195.5
Payment Standard (USD)				
25th Percentile	755	882	1,139	1,345
Median	990	1,233	1,538	1,776
75th Percentile	1,378	1,810	2,027	2,383
Mean	1,162.4	1,449.2	1,661.5	1,930.1
Standard Deviation	601.8	765.5	696.9	807.3
Number of Bedrooms				
25th Percentile	1	1	1	2
Median	2	2	2	2
75th Percentile	3	3	3	3
Mean	1.88	1.86	2.13	2.27
Standard Deviation	0.96	0.98	1.07	1.06
Square Footage				
25th Percentile	—	672	800	850
Median	—	861	1,094	1,174
75th Percentile	—	1,100	1,477	1,602
Mean	—	945.7	1,208.7	1,299.9
Standard Deviation	—	493.3	580.8	683.0
Targeting composition (%)				
Hard-targeted	—	6.54	0.15	0.39
Soft-targeted	—	6.92	0.71	1.12

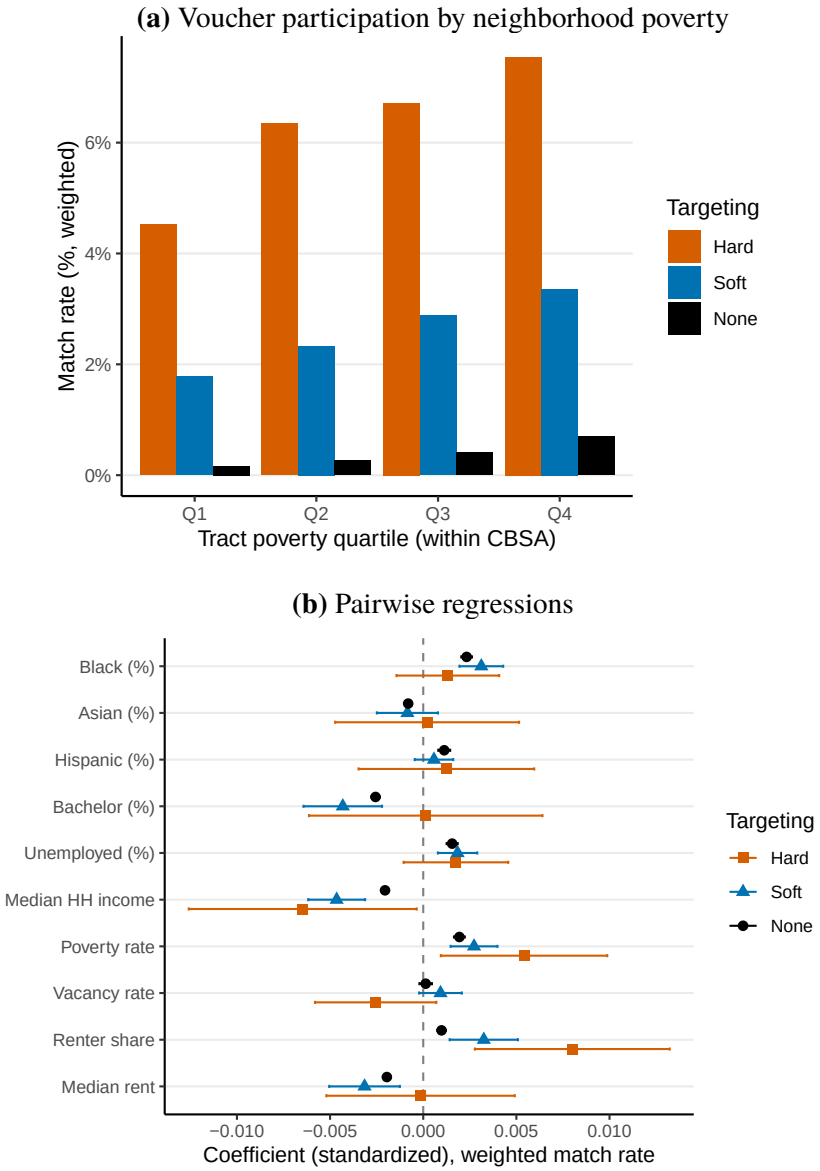
Note. The table reports summary statistics of rental unit characteristics, including rents (contract for HCV, listed for RentHub), number of bedrooms, payment standards, and square footage. Columns (1) and (2) use HUD voucher contract data restricted to new admissions, with Column (2) further restricted to admissions linked to a RentHub listing, with square footage information obtained from RentHub. Columns (3) and (4) use the RentHub sample restricted to listing spells first posted in 2015–2025. Column (4) reports statistics at the spell level, and Column (3) at the level of the underlying raw scrapes that compose those spells. Payment standard statistics in Columns (3) and (4) are computed on the subsample with non-missing payment standards. Summary statistics for that subsample are reported in Appendix Table D.1.

Figure 1: Landlord targeting of voucher holders by neighborhood characteristics



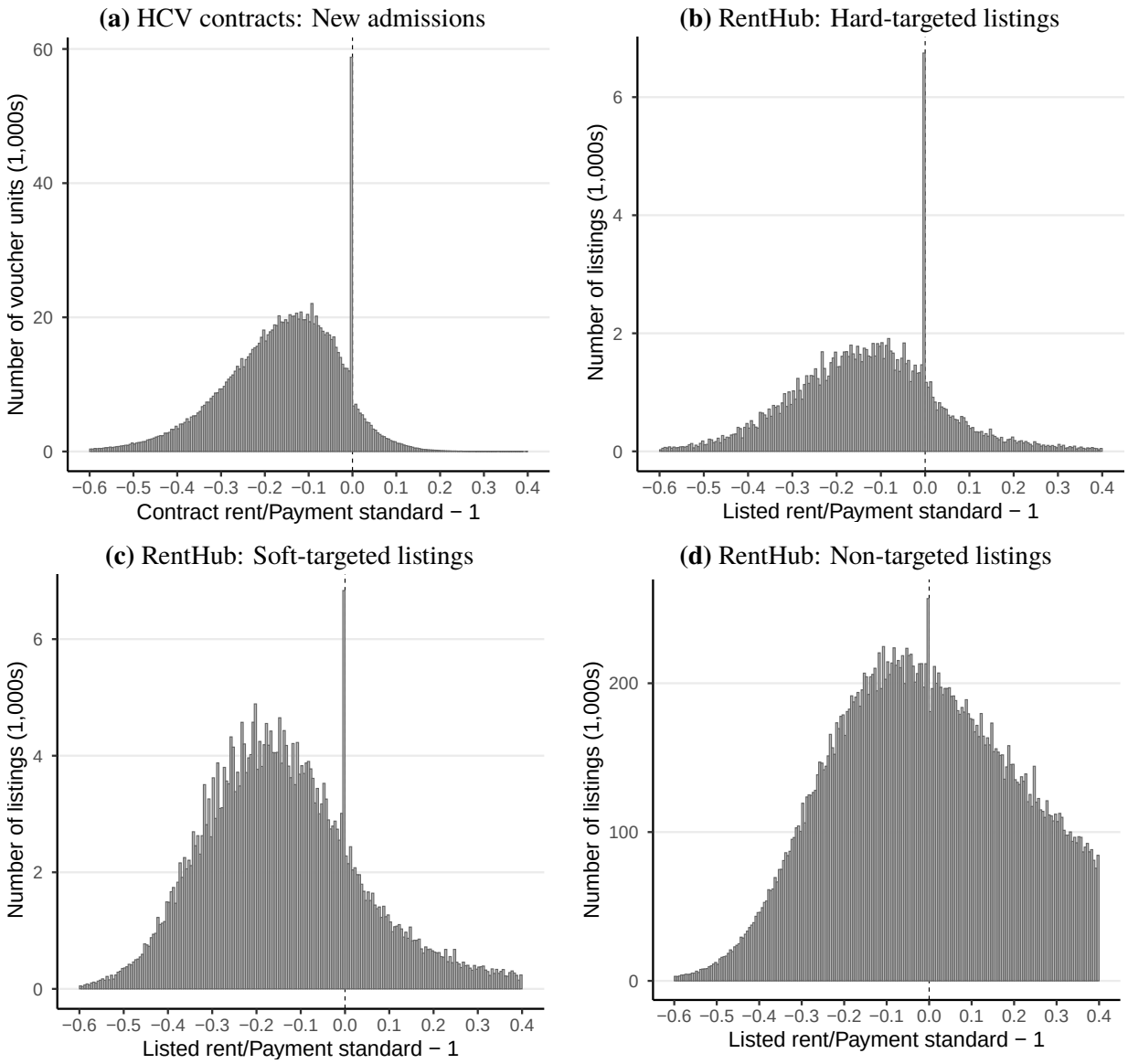
Note. Panel (a) shows the share of RentHub listings classified as hard-targeted (orange) or soft-targeted (blue) within each poverty quartile of the listing's Census tract, where quartiles are defined within the CBSA. Panel (b) plots coefficients from separate listing-level regressions in which the outcome is an indicator for hard-targeted (orange) or soft-targeted (blue) status and the regressor is a tract-level characteristic standardized within the CBSA. All regressions include CBSA-by-year and CBSA-by-bedrooms fixed effects, and standard errors are clustered at the tract level. Hard-targeted listings appear on affordable housing platforms, and soft-targeted listings appear on general platforms and contain voucher-seeking language, as defined in Section 3.2. Census characteristics are obtained from the 2016–2020 American Community Survey (5-year estimates).

Figure 2: Voucher participation across listing spells by targeting and neighborhood characteristics



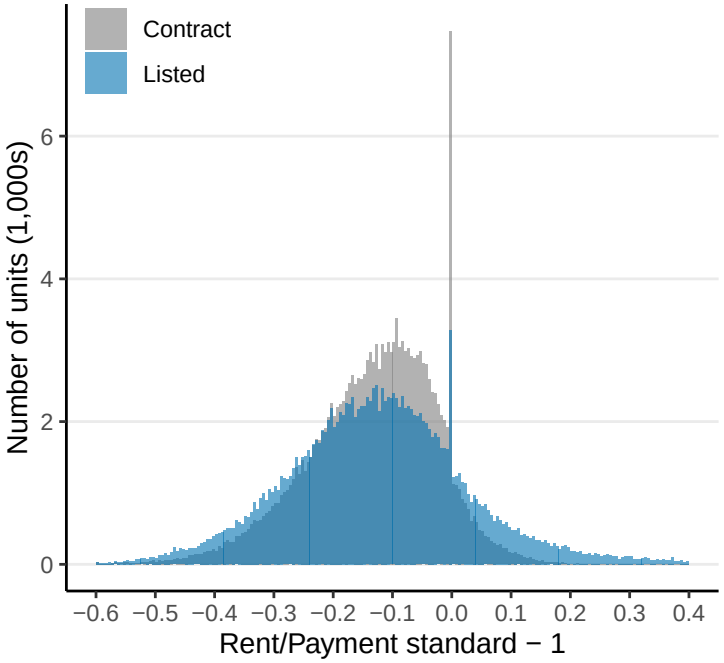
Note. Panel (a) shows the share of RentHub listing spells linked to an HCV unit within each poverty quartile of the listing's Census tract, where quartiles are defined within the CBSA, separately for hard-targeted (orange), soft-targeted (blue), and non-targeted (black) listings. Panel (b) plots coefficients from separate listing-level regressions in which the outcome is an indicator for whether the listing spell is linked to an HCV unit and the regressor is a tract-level characteristic standardized within the CBSA, estimated separately by targeting status. All regressions include CBSA-by-year and CBSA-by-bedrooms fixed effects, and standard errors are clustered at the tract level. Hard-targeted listings appear on affordable housing platforms while soft-targeted listings appear on general platforms and contain voucher-soliciting language, as defined in Section 3.2. Census characteristics are obtained from the 2016–2020 American Community Survey (5-year estimates). The linkage weight assigns a listing spell, for each HCV unit linked to it, a fraction equal to one over the number of spells that unit is linked to, so every unit contributes a total weight of one and unlinked spells contribute zero, as defined in Section 3.4. The estimated intercepts, which reflect the level differences in linkage rates across targeting types, are 0.0706 (s.e. 0.0048) for hard-targeted listings, 0.0294 (s.e. 0.0014) for soft-targeted, and 0.0038 (s.e. 0.0002) for non-targeted.

Figure 3: Histogram of contract and listed rents relative to the payment standard



Note. The panels show the histogram of contract (panel a) and listed (panels b–d) rents relative to the payment standard, measured as (rent/payment standard) − 1. Panel (a) uses new admissions from HCV administrative data, 2015–2025. Panels (b)–(d) use RentHub listings over the same period, restricted to the geographic areas and years for which payment standards are available, as described in Section 3.3. Hard-targeted listings appear on affordable housing platforms, and soft-targeted listings appear on general platforms and contain voucher-seeking language, as defined in Section 3.2. Bins are 0.5 percentage points wide. The vertical dashed line marks zero deviation from the payment standard.

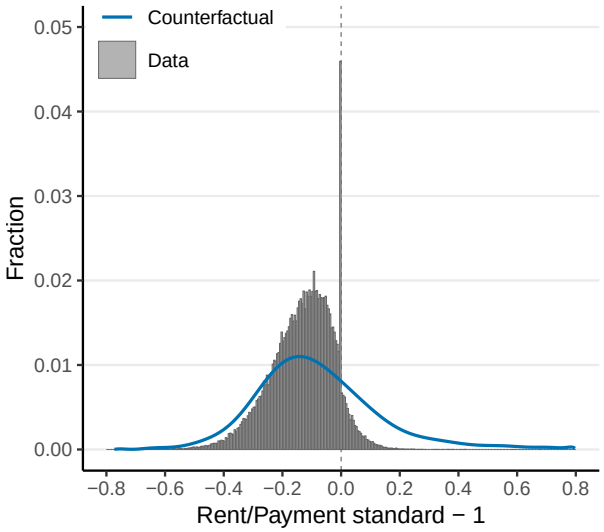
Figure 4: Histogram of contract and listed rents for HCV units linked to listings



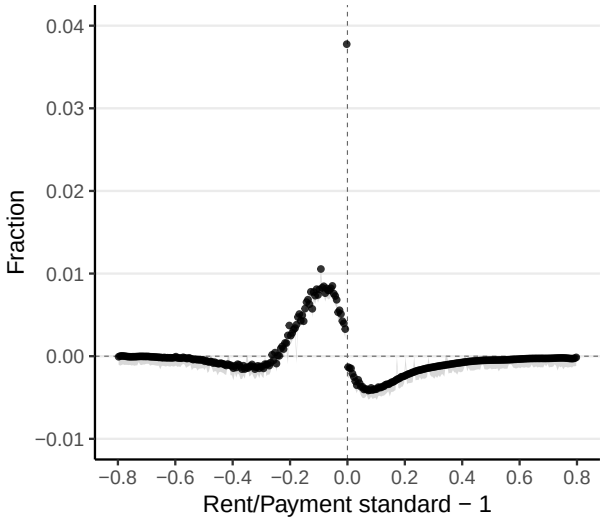
Note. The plot shows the histogram of contract (gray) and listed (blue) rents relative to the payment standard, measured as (rent/payment standard) − 1. The sample includes new admissions in HCV administrative data from 2015 to 2025 that are linked to a RentHub listing spell, as described in Section 3.4. Bins are 0.5 percentage points wide.

Figure 5: Differences-in-bunching: Contract vs counterfactual rents for linked units

(a) Contract vs. counterfactual listed rent

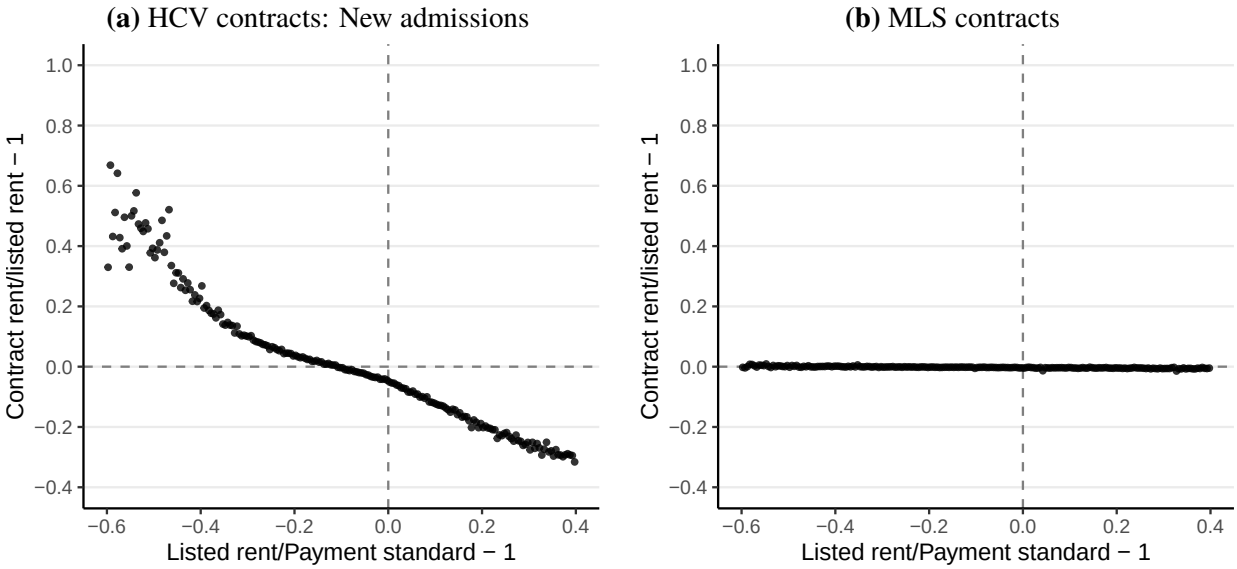


(b) Contract rent - counterfactual listed rent



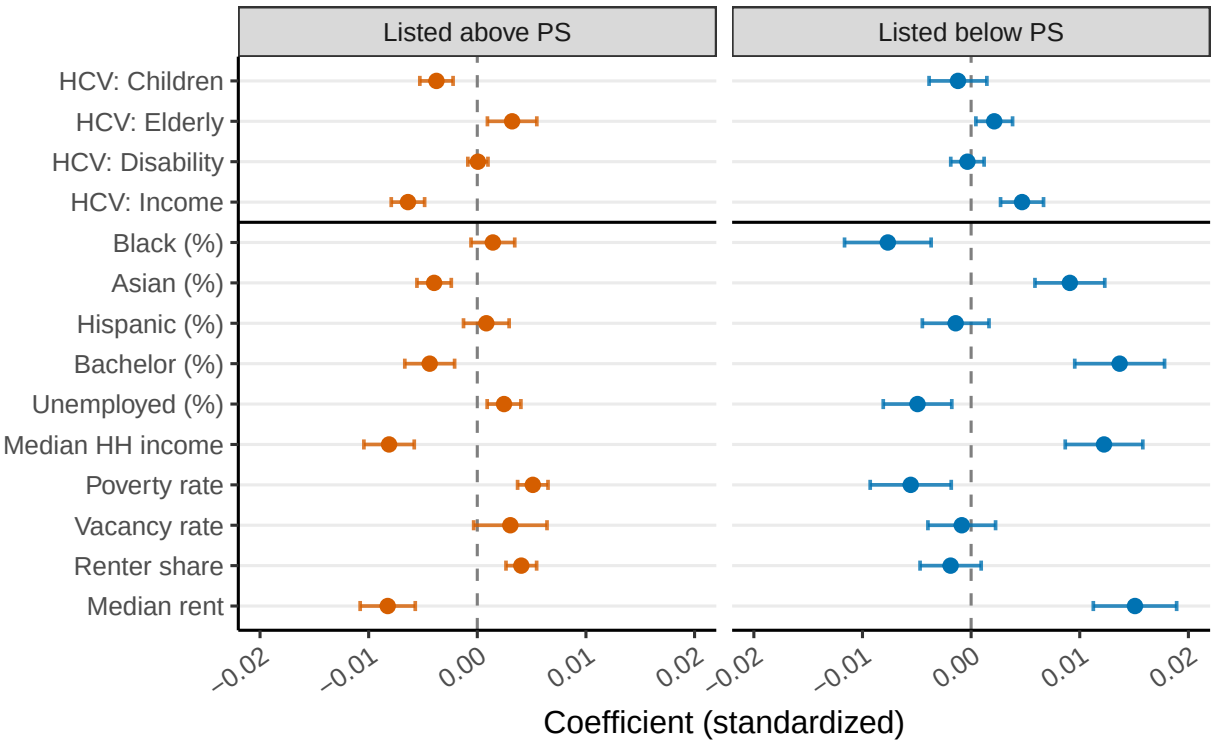
Note. Panel (a) plots the histogram of contract rents for new admissions in HCV administrative data, 2015–2025, that are linked to a RentHub listing spell (gray), together with the counterfactual distribution of listed rents (black), computed from linked listed rents using the differences-in-bunching estimator described in Section 4. Panel (b) plots the difference between the two, with 95% confidence bands (gray) from the iterative bootstrap procedure described there. Bins are 0.5 percentage points wide.

Figure 6: Listing-to-contract rent changes around payment standards, by market



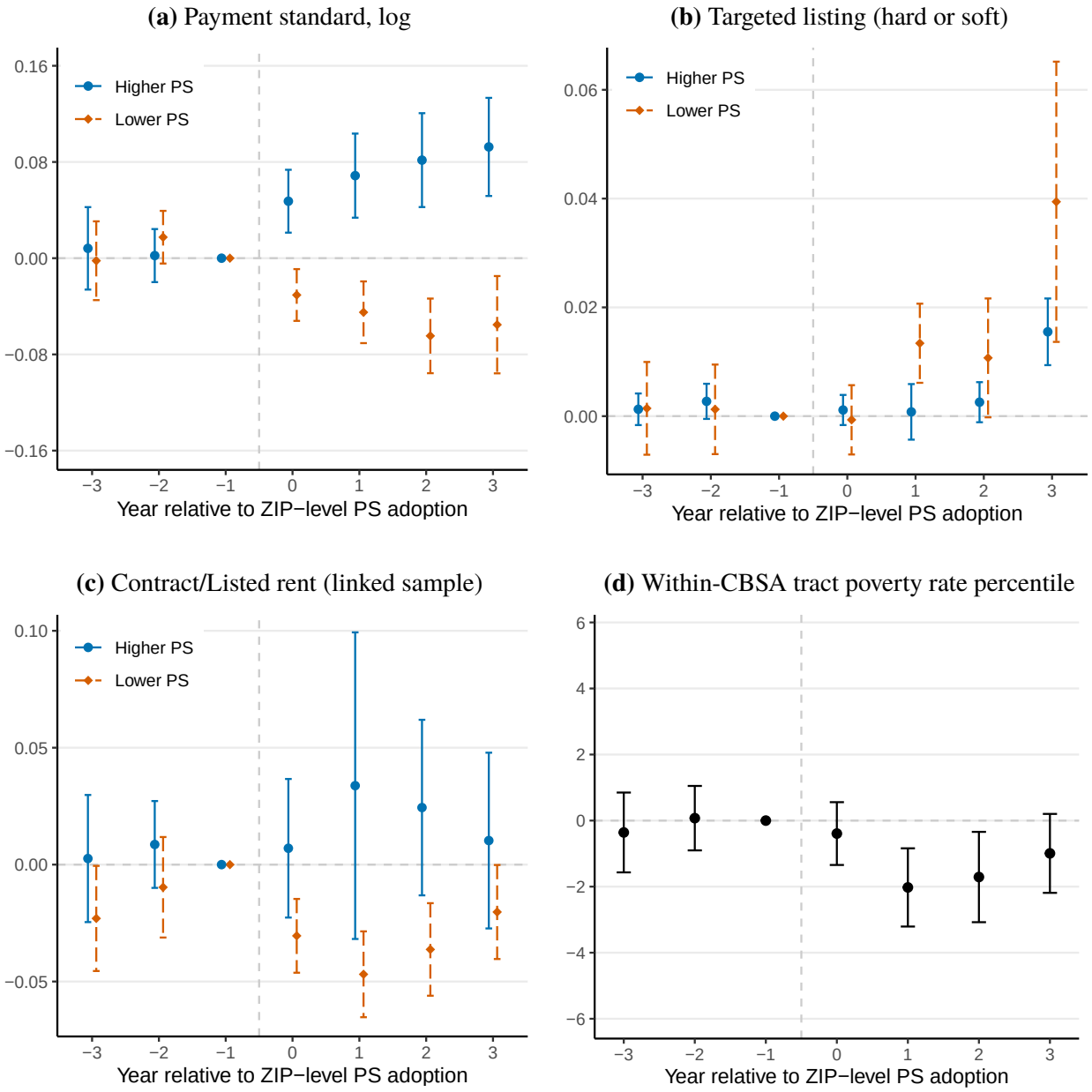
Note. The figure shows binscatter plots of the mean rent change from the listing to the contract stage, as a function of the listed rent relative to the payment standard, measured as (listed rent/payment standard) − 1. Panel (a) uses 2015–2025 HCV new admissions linked to RentHub listing spells, and panel (b) uses contracts for units listed in Multiple Listing Service (MLS) as a comparison market. Bins are 0.5 percentage points wide.

Figure 7: Pairwise regressions of listing-to-contract rent changes on voucher-holder and tract characteristics for linked units, by listed rent position relative to payment standards



Note. Each panel reports the coefficient from a separate regression of the relative difference between contract and listed rent, defined as (contract rent/listed rent) − 1, on a single standardized characteristic. All regressions control for the listed rent and include PHA-by-year and PHA-by-bedroom fixed effects, and standard errors are clustered by PHA. Regressions are estimated separately by whether the listed rent lies above the payment standard. The left panel uses the negative of the relative difference, and the right panel uses the relative difference itself. Census characteristics are drawn from the 2016–2020 American Community Survey (5-year estimates).

Figure 8: Event studies: The impacts of payment standard adoption at the ZIP-level, by sign of the change of payment standard



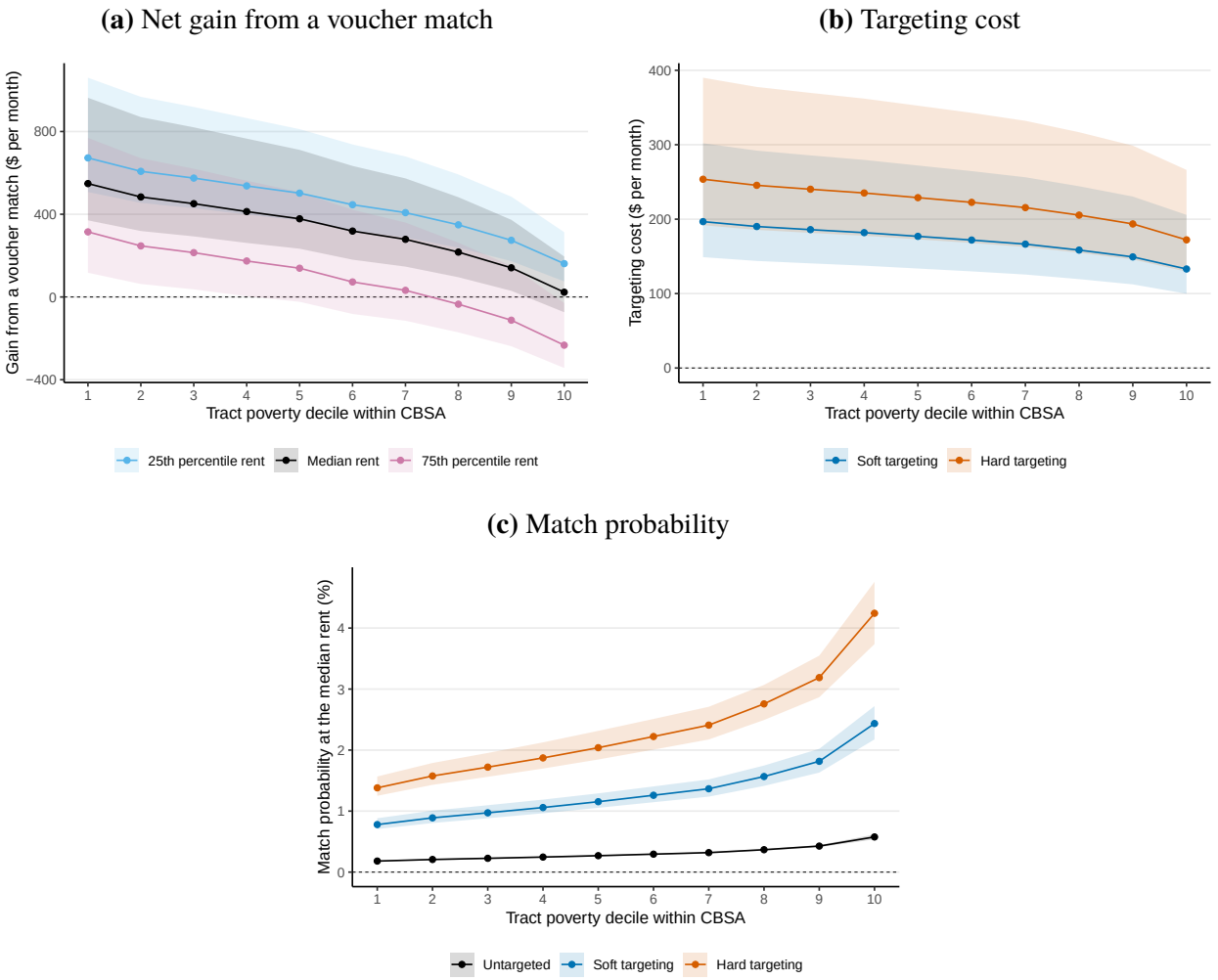
Note. The figure plots the event study coefficients β_e from Equation (2), estimated with the interaction-weighted estimator of Sun and Abraham (2021), together with 95% confidence intervals. Each panel uses a different outcome: panel (a), the log payment standard reported by HUD; panel (b), an indicator for a listing being hard- or soft-targeted; panel (c), the listing-to-contract rent change; and panel (d), the within-CBSA tract poverty rate percentile. Panels (a) and (d) use HCV new admissions over 2015–2025; panel (b) uses RentHub listing spells over the same period; and panel (c) uses the HCV-listings linked sample. In panels (a)–(c), each panel reports two separate regressions: one for ZIP codes where ZIP-level PS adoption is predicted to raise the payment standard (blue) and one where it is predicted to lower it (orange). For this purpose, we classify ZIP codes by the difference between a ZCTA's 40th-percentile rent and its CBSA's 40th-percentile rent, both measured in the 2011–2015 American Community Survey (5-year estimates).

Table 2: Model parameter estimates

	Estimate	Std. error
Panel A. Stage 3: contract rent adjustment		
Headroom below S, μ_+ (\$ per \$)	0.519	(0.026)
Distance above S, μ_- (\$ per \$)	1.121	(0.014)
Tract poverty rate	−1.71	(11.11)
Tract vacancy rate	−18.74	(16.77)
Log tract median rent	−32.50	(14.78)
Soft targeting	1.99	(5.10)
Hard targeting	6.28	(8.44)
Dispersion of desired adjustment, σ_m (\$)	178.0	(12.5)
Payment standard penalty, λ_S	0.440	(0.093)
Adjustment cost, λ_m (\$)	157.4	(8.0)
Panel B. Stage 2: allocation of new vouchers		
Above S, θ_0	−0.806	(0.106)
Headroom below S, θ_1 (per \$100)	−0.022	(0.012)
Distance above S, θ_2 (per \$100)	−0.112	(0.038)
Tract poverty rate	1.785	(0.178)
Tract vacancy rate	−0.417	(0.296)
Log tract median rent	−0.954	(0.059)
Soft targeting	1.467	(0.055)
Hard targeting	2.047	(0.063)
Panel C. Stage 1: listed rent and targeting		
Listing-choice shocks		
Scale across rents, σ_R (\$)	6.10	(0.89)
Nesting, σ_R/σ_V	0.178	(0.021)
Scale across targeting intensities, σ_V (\$)	34.3	(6.5)
Guarantee share, ψ_j		
Average neighborhood, ψ	0.207	(0.038)
Tract poverty rate	0.113	(0.030)
Tract vacancy rate	−0.203	(0.050)
Log tract median rent	0.102	(0.026)
Participation cost, c_j^{HCV} (\$)		
Mean δ_c across CBSAs	−132.9	(33.4)
Tract poverty rate	803.2	(126.3)
Tract vacancy rate	−720.9	(105.2)
Log tract median rent	−247.8	(34.1)
Soft targeting cost (\$)		
Mean $c^{\text{V}}_{\text{soft},c}c$ across CBSAs	171.0	(32.6)
Tract poverty rate	−87.2	(16.1)
Tract vacancy rate	−23.7	(5.3)
Log tract median rent	63.8	(11.7)
Hard targeting cost (\$)		
Mean $c^{\text{V}}_{\text{hard},c}$ across CBSAs	221.3	(42.0)
Tract poverty rate	−123.2	(23.5)
Tract vacancy rate	−14.6	(4.3)
Log tract median rent	75.4	(14.0)

Notes. This table reports estimates from the three stages of the model. Panel A reports maximum likelihood estimates of the Stage 3 contract rent model. Index coefficients are in dollars of desired adjustment. σ_m, λ_S, and λ_m are estimated in logs, with standard errors calculated using the delta method. Panel B reports PPML estimates of the Stage 2 allocation model. θ_1 and θ_2 are scaled per \$100. Standard errors in Panels A and B are clustered by CBSA. Panel C reports maximum likelihood estimates of the Stage 1 listing choice model, conditional on the Stage 2 and Stage 3 estimates. Stage 1 costs and values are in dollars per month. Neighborhood characteristics are centered at their listing-weighted means. Standard errors combine the variance conditional on the Stage 2 and Stage 3 estimates with the bootstrap variance from their estimation described in Section 7.1.

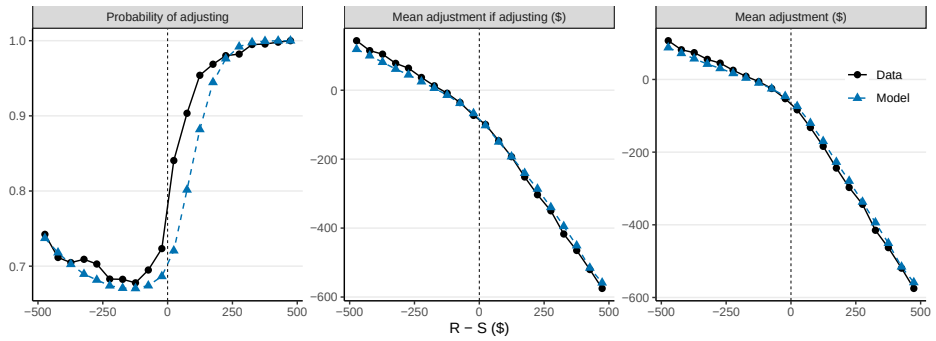
Figure 9: Voucher gain, targeting costs, and match probabilities by poverty



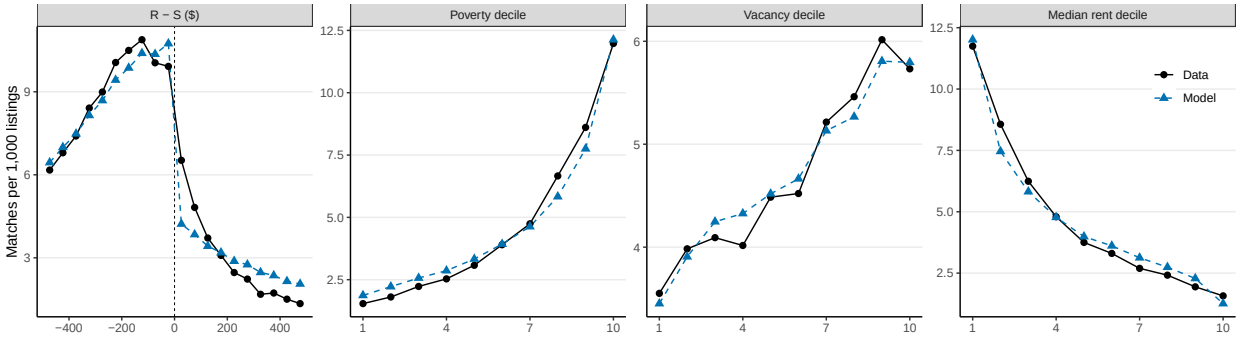
Notes. This figure depicts model-implied values by within-CBSA decile of the tract poverty rate, using listing-weighted cutoffs. Panel (a) shows the net gain from a voucher match for a non-targeted unit listed at market value, defined as the expected rent adjustment plus the value of the payment guarantee minus the participation cost, evaluated at the 25th, 50th, and 75th percentiles of benchmark rent within CBSA and bedroom count. Panel (b) shows soft and hard targeting costs. Panel (c) shows the voucher match probability at the median benchmark rent by targeting intensity. Shaded areas show 95 percent bootstrap confidence intervals.

Figure 10: Model fit

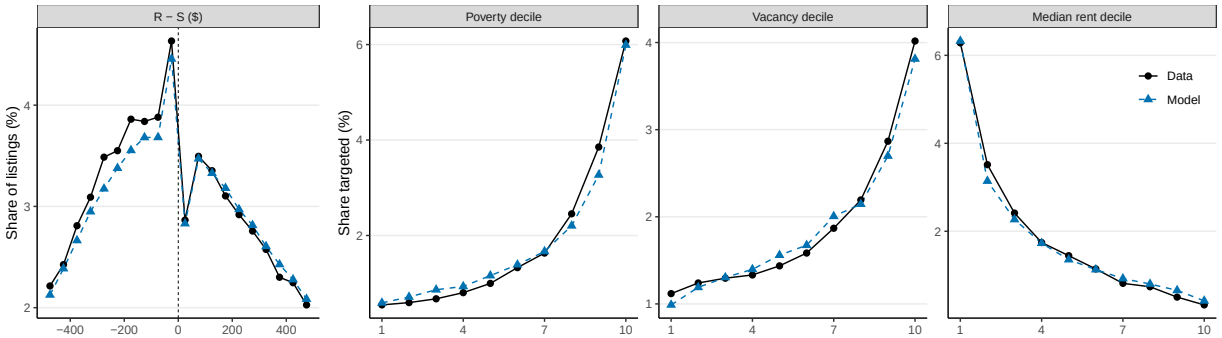
(a) Stage 3: listing-to-contract rent adjustments



(b) Stage 2: voucher matches



(c) Stage 1: listed rents and targeting



Notes. Panel (a) shows the probability of adjusting the rent between listing and contract, the mean adjustment given an adjustment, and the unconditional mean adjustment for linked contracts, by $R - S$. Panel (b) shows voucher matches per 1,000 listings by $R - S$ and tract covariate deciles. Panel (c) shows the share of listings by $R-S$ and the share of listings that target voucher holders, soft or hard, by tract covariate deciles. $R-S$ is grouped in \$50 bins over $(-500, 500]$, and tract poverty, vacancy, and median rent in deciles with listing-weighted cutoffs. Model values are expectations at each observation's listing and neighborhood characteristics under the estimates in Table 2.

Table 3: Counterfactual outcomes

	Poverty Q1 (%)	Δ Poverty percentile	Spending ($)	Δ Spending ($)	Cost per percentile ($)	Δ Landlord per match ($)
Status quo	7.22		1,332			
Increased payment standards						
+10%, all tracts	7.37	−0.19	1,426	93.7	499	102
+20%, all tracts	7.50	−0.33	1,502	169.9	520	185
+10%, Q1 tracts	7.94	−0.43	1,342	10.2	24	15
+20%, Q1 tracts	8.48	−0.75	1,351	19.4	26	28
+10%, Q1–Q2 tracts	7.83	−0.84	1,359	26.7	32	37
+20%, Q1–Q2 tracts	8.27	−1.45	1,382	50.2	35	69
Rent-indexed standards						
Indexed	7.87	−0.89	1,230	−102.1	–	−76
Hold harmless	7.55	−0.36	1,371	38.8	108	49
Budget neutral (+9.8%)	7.97	−1.00	1,332	0.0	–	35
Sign-up bonus						
1 month, all tracts	7.18	0.11	1,388	55.8	–	60
2 months, all tracts	7.12	0.24	1,444	111.6	–	118
1 month, Q1 tracts	7.24	−0.01	1,337	4.6	499	6
1 month, Q1–Q2 tracts	7.24	−0.03	1,345	12.9	516	16
Rent-reasonableness enforcement						
$\rho = 0.5$	7.26	−0.08	1,309	−22.7	–	−28
$\rho = 0$	7.28	−0.14	1,287	−45.4	–	−57
Budget-neutral enforcement						
$\rho = 0.5$, all tracts (+2.9%)	7.30	−0.14	1,332	0.0	–	−3
$\rho = 0.5$, Q1–Q2 tracts (+10.3%)	7.84	−0.90	1,332	0.0	–	4
$\rho = 0.5$, indexed (+11.9%)	8.01	−1.06	1,332	0.0	–	32
$\rho = 0$, all tracts (+7.6%)	7.41	−0.33	1,332	0.0	–	−4
$\rho = 0$, Q1–Q2 tracts (+28.2%)	8.64	−1.97	1,332	0.0	–	11
$\rho = 0$, indexed (+14.8%)	8.06	−1.13	1,332	0.0	–	29

Notes. This table reports counterfactual outcomes pooled across the 2019–2025 new voucher admissions, including those matched to observed rental listings and those allocated to the outside pool, under the baseline outside-pool responsiveness, $\zeta = 0.46$, defined in Appendix C.4. Appendix Table D.4 reports the outcomes among in-market admissions, those matched to observed listings. The poverty percentile ranks a tract's 2020 poverty rate among all tracts of its metropolitan area from 0 to 100, as in Section 5, and Q1 denotes the tracts at or below the 25th percentile. Poverty Q1 is the share of new admissions in Q1 tracts, and Δ Poverty percentile is the change in the mean poverty percentile of the tracts of new admissions, which is 70.6 in the status quo. Spending includes the housing assistance payment and any sign-up bonus per admission and month, and Δ Spending is its change. Cost per percentile is Δ Spending divided by the decrease in the mean poverty percentile and is reported when spending rises and the mean percentile falls. Δ Landlord per match is the change in the total value of the program to the landlords of observed listings, as defined in Appendix Equation C.15, divided by the number of in-market voucher matches in the status quo, per month. Sign-up bonuses are proportional to the contract rent. Budget-neutral policies hold program-wide spending fixed, with the corresponding increase in payment standards reported in parentheses; for the indexed policies, the increase applies to the rent-indexed standards.

Appendix

A Payment Standards: Dataset Construction

A.1 Overview

Payment standards are the maximum gross rent a Public Housing Authority (PHA) will subsidize under the Housing Choice Voucher (HCV) program. Each PHA sets its own payment standards, expressed as a dollar amount per bedroom size, within an area defined by HUD's Fair Market Rents (FMR) or, for designated metropolitan areas, Small Area Fair Market Rents (SAFMR). Because payment standards are not directly reported in any centralized HUD administrative file, we construct them from two complementary sources: (i) systematic inference from HUD's Form 50058, and (ii) manual collection from PHA websites and Freedom of Information Act requests. We reconcile the two sources into a single dataset. The HUD-inferred data cover February 2015 through September 2025 (the span of available Form 50058 microdata), and the manually collected data extend through the end of 2025. The remainder of this section describes the sample of PHAs covered by the dataset and the decisions we made to assign a payment standard to each PHA-period combination.

A.2 Sample of PHAs

We start from the sample of PHAs present in the HUD Public and Indian Housing (PIH) 50058 administrative records. We impose the following sequential restrictions:

1. *Size*: We retain PHAs that administered at least 1,000 voucher units in 2015, yielding 481 PHAs (after dropping the Virgin Islands authority, which is out of our scope). This restriction is made to guarantee a minimum of voucher lease-up activity to reduce noise in our inference methodology.

2. *Moving to Work (MTW) exclusion*: When inferring payment standards from HUD data, we exclude MTW PHAs because these PHAs have missing information in the payment standard field. This exclusion leaves a sample of 472 PHAs for inferred payment standards.

We classify each PHA as *regional* or *local*. Regional and state PHAs are identified by the third character of the PHA code being "9," or by the keywords "state" or "region" appearing in the formal PHA name or contact email. Regional PHAs receive additional restrictions described below.

A.3 HUD-Inferred Payment Standards

We infer payment standards from HUD administrative data for each PHA-period combination, where a period is the interval during which a given set of FMR or SAFMR values is in effect according to HUD. These periods usually begin in October of each year, when HUD issues updated FMRs/SAFMRs, and end either when mid-year updates take effect or at the following October.

A.3.1 Modal Method

For each PHA, period, and bedroom size, we identify the *most frequent* (modal) payment standard from the observed distribution of contracts among leased-up units, dropping any payment standards outside the \$100–\$8,000 range. Formally, let $\mathcal{S}_{j,t,b}$ denote the multiset of payment standards reported in PHA j's lease-up records during FMR period t for bedroom size b, and let $\hat{s}_{j,t,b}$ denote the modal value. A PHA-period cell is declared "valid" if, for either 1-bedroom *or* 2-bedroom units, the modal share clears a minimum threshold τ, which we set at 55 percent, and the cell has at least 10 observations. Formally, the restriction can be expressed as

$$\frac{\#\{r \in \mathcal{S}_{j,t,b} : r = \hat{s}_{j,t,b}\}}{|\mathcal{S}_{j,t,b}|} \geq \tau, \qquad \tau \geq 0.55, \tag{A.1}$$

together with $|\mathcal{S}_{j,t,b}| \geq 10$. When a cell qualifies, we assign the modal value to all bedroom sizes 0–4 in that PHA-period. When inference fails for isolated periods, we fill gaps as follows: a single missing period takes the prior period's value; a two-period gap is filled only if the values on both sides agree. We use newly admitted voucher holders (action type code = 1 with admission year $\geq$ 2015) as the primary sample, and keep results using all voucher admissions as a fallback.

Overall, this method yields valid payment standards for approximately 454 PHAs across at least some portion of the study period.

A.3.2 FMR– and SAFMR–Multiples Method

As a complementary approach, and given that most PHAs set payment standards as a fixed percentage of FMR, we infer the "FMR multiple" that PHAs use to set payment standards. For each PHA, FMR period, and bedroom size, we test whether a given share of observed payment standards matches any of the canonical multiples:

$$m \in \{0.90,\ 0.95,\ 1.00,\ 1.05,\ 1.10,\ 1.15,\ 1.20,\ 1.25\}$$

applied to the county-level FMR. We assign a multiple m if:

$$\frac{\#\{s \in \mathcal{S}_{j,t,b} : s = m \times \text{FMR}_{c,t,b}\}}{|\mathcal{S}_{j,t,b}|} \geq \tfrac{1}{3}, \tag{A.2}$$

subject to $|\mathcal{S}_{j,t,b}| \geq 10$, retaining the highest-share multiple when several qualify. FMR values come from HUD's publicly available FMR schedule. We fill 1- and 2-period gaps using the same logic as in the modal method, and likewise run the inference separately on new admissions (primary) and all admissions (fallback). For regional PHAs operating across multiple counties, we restrict inference to PHA-by-county cells with at least 100 active vouchers in 2015 to ensure sufficient data density for reliable inference.

For PHAs in SAFMR-designated metropolitan areas, we follow an analogous procedure that infers the SAFMR multiple using HUD's ZIP-level Small Area Fair Market Rents. SAFMR-multiple inference is run as a single pass (not split into new vs. all admissions). SAFMR inference begins in October 2018, the month in which the SAFMR mandate took effect for most designated PHAs.[1] Inferred SAFMR multiples are then mapped from each PHA to the ZIP codes within the PHA's HUD metro area using HUD's published ZIP-to-metro crosswalk.

Overall, this method yields valid payment standards for approximately 404 and 73 PHAs for FMR multiples and SAFMR multiples, respectively, across at least some portion of the study period. Figure A.1 shows the frequency of several multiples of FMR and SAFMR in our final dataset using 2-bedroom payment standards for which this variable is non-missing.

A.4 Manual Collection of Payment Standards

For a subset of large or SAFMR-covered PHAs, we supplement inference with payment standards collected directly from PHA documents, including administrative plans, FOIA responses, and archived web pages. We target the largest 110 PHAs considered for the SAFMR implementation in 2018, of which we obtain data for 76. The collected data record the number of bedrooms, the payment standard, the geographic unit at which it applies, and the effective date.

Most of these PHAs set payment standards at the metropolitan area or ZIP code level, though several publish them at other geographies: county, submarket/tier, or Census tract. We account for this variation by constructing the manual panel separately for two sets of geographies.

Metro, county, and submarkets. We merge the metro, county, and submarket datasets into a single PHA-geography-bedroom panel. This is feasible because submarket definitions generally

[1] Eight PHAs with documented early adoption are assigned SAFMR-inferred standards starting earlier, based on visual inspection of their lease-up records: IL101, TN004, TX009, TX392, TX434, TX436, TX559, and TX901.

map to county boundaries.[2]

ZIP codes. We separately process PHAs that publish ZIP code-specific standards, the majority of which are PHAs in SAFMR-mandated metropolitan areas. For agencies that list payment standards for explicitly named ZIP codes alongside an *"All Other ZIP codes"* category (`CA006`, `CA072`, `NY409`), we expand the *"All Other ZIP codes"* category to all remaining ZIP codes in the PHA's county jurisdiction using HUD's USPS 2015 ZIP-to-county crosswalk.

A.5 Final Payment Standard Dataset

We combine the manual and inferred datasets into a unified monthly panel in three steps. First, we construct a PHA-level timeline by pooling all effective dates from the manual schedules and the inferred FMR/SAFMR period boundaries. Within each PHA, we assign the geographic level (metro or ZIP) in each time segment to the most granular source available, giving precedence to manual over inferred data. A PHA may therefore transition from metro-level to ZIP-level coverage mid-panel as manual ZIP-specific schedules become available. We also apply a small number of PHA-specific corrections at this stage, dropping inferred metro-level values when we can verify from the schedule that the PHA was using ZIP-level standards, and dropping legacy metro rows from the manual data once ZIP-level information takes over.

Second, we expand the timeline to monthly resolution and populate each row with every available source-specific payment standard value. After this step, each row carries all inferred and manual estimates alongside one another, regardless of which source we ultimately treat as canonical.

Third, we apply the priority hierarchy in Table A.1 to select a canonical payment standard from all three methods, assigning each observation the highest-priority (lowest-numbered) non-missing value. Figure A.2 reports the number of PHAs covered by each source. Figure A.3 assesses agreement across sources by plotting the distribution of percent deviations between modal-inferred and manually collected payment standards, restricted to PHA-month cells where both are available. The two sources match exactly in roughly 70 percent of these observations.

The resulting dataset is our final payment standards dataset, which contains payment standard information across a total of 474 PHAs.

[2] A small number of PHAs whose submarket definitions are based on custom maps without correspondence to any administrative boundary, such as Atlanta, cannot be included in this step.

A.6 ZIP-Level Version of the Payment Standard Dataset

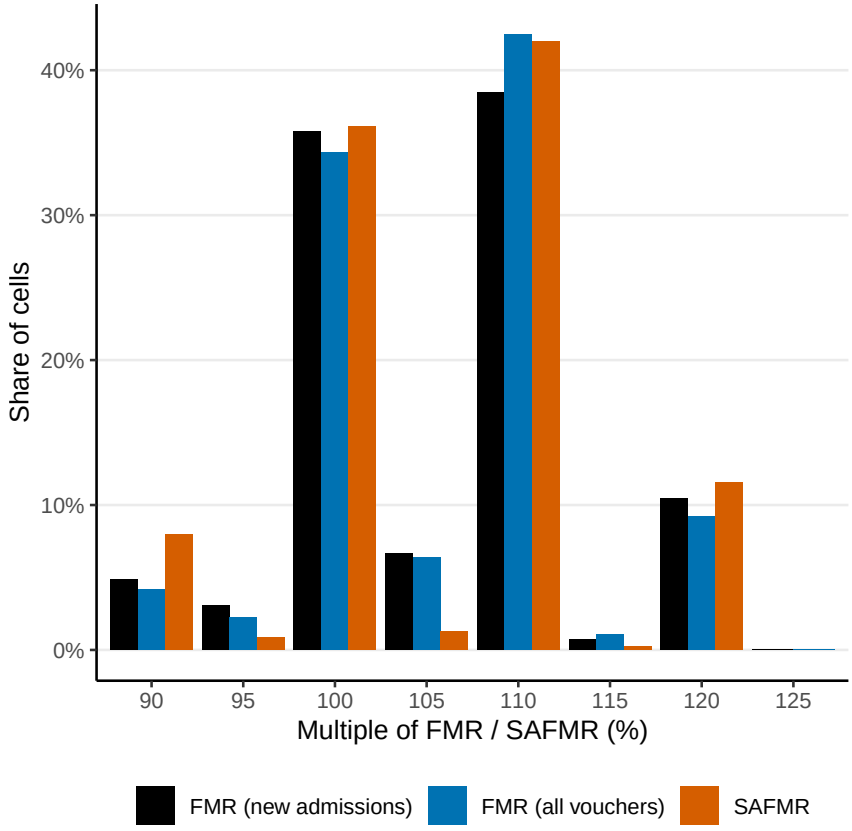
For the analyses with rental listings, where it is not clear which PHA serves the listing's unit, we build a curated dataset of payment standards at the ZIP code level from the dataset in the previous subsection.

We first expand metro, county, and submarket rows to the ZIP code level using a ZCTA-to-county crosswalk (or, for New England states where FMR values vary at the county-subdivision level, a ZCTA-to-county-subdivision crosswalk). For metro-level payment standards, we only capture ZIP codes of the county where the PHA is located.[3]

For ZIP codes covered by more than one PHA, we retain the PHA with the most voucher holders in that ZIP code. If no PHA has voucher holders there, we retain the largest PHA by total Section 8 unit count. Each (ZIP code, bedroom size, month) cell appears exactly once in this curated panel.

[3]For multi-county metro PHAs such as the New York City Housing Authority, which spans five boroughs, we replicate metro-level rows across all counties served.

Figure A.1: Frequency of FMR/SAFMR multiples in inferred payment standards



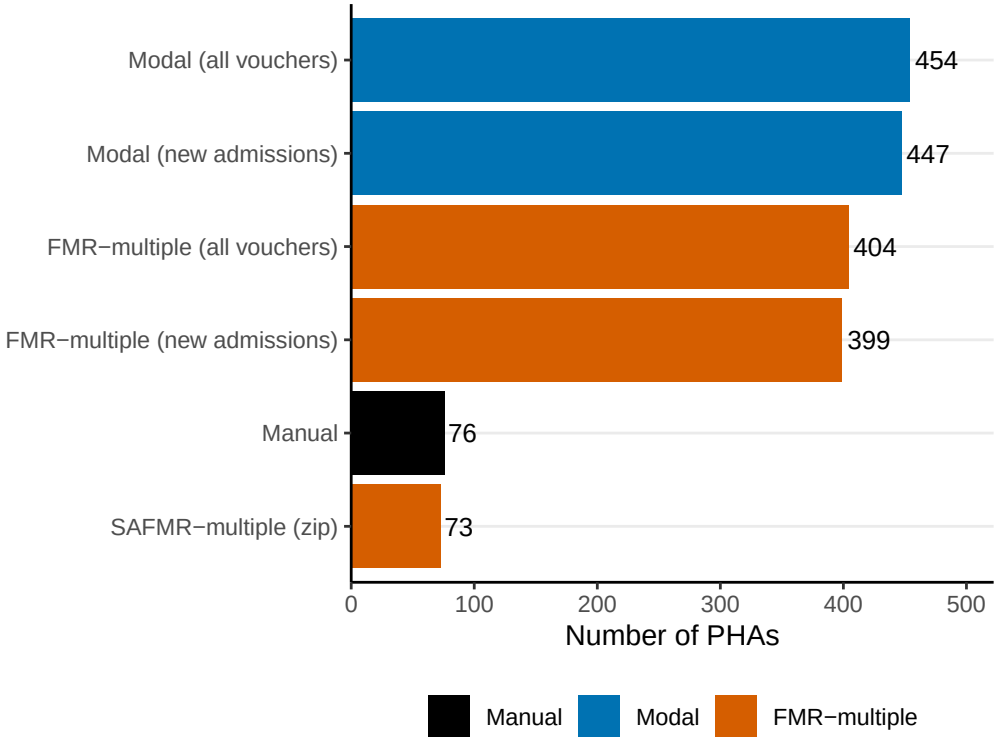
Note. The figure shows the share of PHA-month combinations assigned to each canonical multiple of FMR or SAFMR, restricted to the sample of 2-bedrooms and PHA-months for which the multiples method yields a non-missing payment standard.

Table A.1: Payment standard priority hierarchy

Priority	Source	Column
1	Manual (directly collected)	`ps_manual`
2	HUD-inferred, modal PS, new admissions	`ps_infer_new`
3	HUD-inferred, FMR multiple, new admissions	`ps_infer_fmr_new`
4	HUD-inferred, modal PS, all vouchers	`ps_infer_all`
5	HUD-inferred, FMR multiple, all vouchers	`ps_infer_fmr_all`
6	HUD-inferred, SAFMR multiple (ZIP-level only)	`ps_infer_safmr`

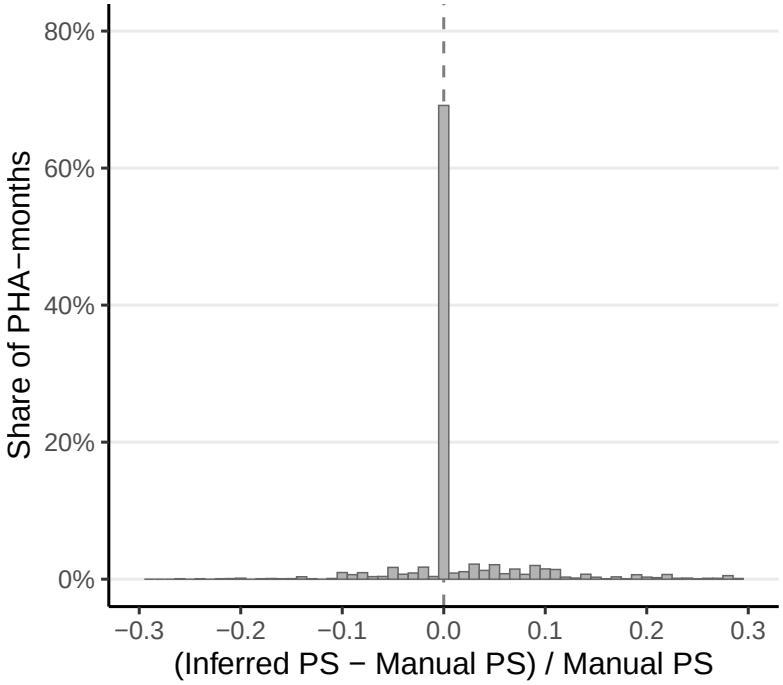
Note. The table reports the order in which sources are used to assign a payment standard (PS) to each PHA, period, and bedroom cell when more than one source covers it; the highest-priority available source is used.

Figure A.2: Number of PHAs by source of payment standard data



Note. The figure shows the number of unique PHAs with non-missing payment standard information from each source used to construct the final dataset.

Figure A.3: Agreement between modal-inferred and manually collected payment standards



Note. The figure plots the histogram of the relative deviation between modal-inferred and manually collected payment standards, using 1 percentage point bins. The sample includes all PHA-month observations for 2-bedroom units with non-missing values for both sources. Each PHA-month receives equal weight.

B RentHub Listing Data Details

This appendix documents the RentHub listing panel used in Section 3.2. We describe how we chain scrapes into listing spells (Appendix B.1) and how we classify voucher targeting (Appendix B.2).

B.1 Constructing listing spells

As described in Section 3.2, we chain a unit's consecutive scrapes into a listing spell and start a new spell whenever the gap between scrapes exceeds 60 days. Figure B.1 plots the distribution of these within-unit gaps, measured across covered periods so that statewide coverage holes do not inflate the tail. The mass sits at short gaps, with a median of 6 days and about 88 percent of gaps at 21 days or fewer, which reflects the weekly to biweekly scrape cadence of an active listing. Beyond this cadence the distribution thins out quickly. Gaps of 31 to 45 days account for about 3 percent of the total, gaps of 46 to 90 days for about 1.5 percent, and the distribution then flattens into a long thin tail of genuine re-listings beyond 90 days. The 60-day break sits at the transition between the cadence-driven mass and this flat tail and leaves roughly 97 percent of consecutive gaps within a spell. The exact threshold is not consequential. Moving it from 45 to 90 days changes the number of spells by under 10 percent. A 60-day cutoff partitions a unit's scrapes into spells that average about seven scrapes spanning roughly seven weeks on the market.

We date each spell by its first scrape rather than the listing's posting date. The posting date is almost always present, but only about 42 percent of units carry a single posting date across their scrapes because the field is refreshed over the life of a listing. We therefore use the first scrape date as the measure of when a spell first appears in the data.

B.2 Voucher targeting classification

Voucher Hard-Targeting

RentHub records a management or listing company for many scrapes. We map each distinct company string to one of six groups using a hand-built crosswalk. The groups are affordable housing platforms (e.g., AffordableHousing.com), affordable housing management organizations, mainstream rental platforms, institutional owners or managers, brokerages, and a residual other category. We aggregate to the spell by counting the distinct companies in each group. We classify a spell as hard-targeted if it lists on an affordable housing platform or with an affordable housing management organization, since these channels exist specifically to serve voucher holders.

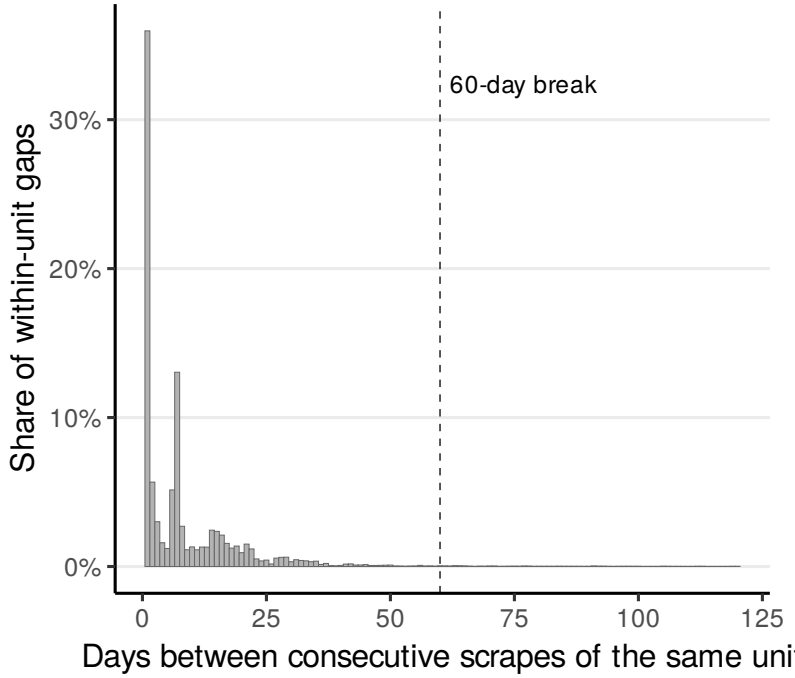
The company and platform metadata are sparse in the early years and fill in sharply from 2019 onward. Company coverage is between 3 and 6 percent in 2015–2017 and 12 percent in 2018, then jumps to 69 percent in 2019. Because a hard-targeted listing can be identified only through this data field, hard targeting is effectively unmeasured before 2019, at below 0.02 percent of spells, and those early-year shares should be read as uninformative rather than as lower bounds. From 2019 on, the hard-targeted share runs from about 0.2 to 1.0 percent of spells.

Figure B.2 shows the composition of company groups among the spells that carry company information. Affordable housing platforms and organizations together account for about 0.6 percent of these spells. The bulk are mainstream platforms at about 46 percent, the residual other category at 22 percent, brokerages at 20 percent, and institutional managers at 12 percent. Across all spells, including the roughly 40 percent with no company information, affordable platforms and organizations make up 0.35 percent.

Voucher Soft-Targeting

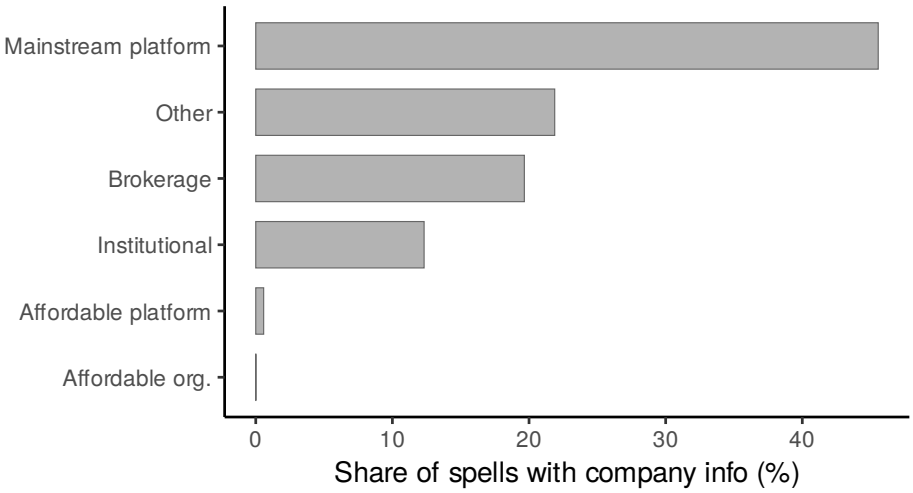
We detect voucher language in the listing description with a dictionary of regular expressions developed through manual review of a random sample of listings. Each listing spell inherits a voucher flag if any of its scrapes match. The dictionary is organized as a general voucher-mention flag, `i_hcv`, which fires on any voucher or subsidy reference such as "section 8," "housing choice voucher," or "HAP contract," and excludes non-housing senses of the word such as gift, promo, parking, meal, and travel vouchers. Nested under it are four subcategories. `i_hcv_yes` captures acceptance or welcome, as in "section 8 welcome," "vouchers accepted," or "voucher holders welcome." `i_hcv_no` captures explicit refusal, as in "no section 8," "private pay only," or "section 8 need not apply." `i_hcv_sec8_only` captures project-based units that require a voucher, as in "section 8 only" or "must have a voucher." `i_hcv_boilerplate` captures fair-housing and compliance template text, which carries no acceptance or refusal signal. Our soft-targeted category is the set of listings that explicitly welcome vouchers (`i_hcv_yes`), with acceptance keywords screened against a refusal list so that phrases such as "not authorized to accept section 8" are not read as acceptance. The full pattern set is in the replication files. The description field itself is present for about 73 percent of spells in 2015–2016, 51 percent in 2017, and 37 percent in 2018, and for 95 to 100 percent of spells from 2019 on, so soft targeting, like hard targeting, is measured on a thinner base before 2019.

Figure B.1: Distribution of within-unit gaps between consecutive scrapes



Note. The figure depicts the share of within-unit gaps by number of days between consecutive scrapes of the same unit, pooled across units and truncated at 120 days. Gaps spanning a statewide coverage hole are excluded. The dashed line marks the 60-day spell break.

Figure B.2: Company-group composition among spells with company information



Note. The figure depicts the shares among spells with any non-missing company information, for spells first listed in 2015–2025. The 40 percent of spells with no company information are excluded from the figure and are concentrated in the pre-2019 years.

C Model Computational Details

This appendix presents the computational details underlying Section 7 and Section 8.

C.1 Stage 3 estimation

Write $d_j = R - S_j$ and let

$$\mu_j = \mu_+(-d_j)_+ - \mu_-(d_j)_+ + Y_j'\mu_Y + V'\mu_V + \nu_j, \tag{C.1}$$

where ν_j is the sum of the CBSA, year, and bedroom fixed effects of contract j. The baseline includes an effect for every retained CBSA and therefore no global intercept, and the year and bedroom effects are relative to omitted categories. Conditional on μ_j, the latent desired adjustment satisfies

$$m_j^* \sim N(\mu_j, \sigma_m^2).$$

The realized adjustment minimizes

$$(m - m_j^*)^2 + \lambda_S(d_j + m)_+^2 + \lambda_m|m|, \qquad \lambda_S, \lambda_m \geq 0. \tag{C.2}$$

Let $a_m = \lambda_m/2$ and $c_j = \lambda_S(d_j)_+$. The first-order conditions imply that the listed rent remains unchanged whenever

$$c_j - a_m \leq m_j^* \leq c_j + a_m. \tag{C.3}$$

The implied probability mass at the listed rent is therefore

$$\pi_j^0 = \Phi\left(\frac{c_j + a_m - \mu_j}{\sigma_m}\right) - \Phi\left(\frac{c_j - a_m - \mu_j}{\sigma_m}\right). \tag{C.4}$$

When $R < S_j$, the payment-standard penalty is inactive at $m = 0$, so $c_j = 0$, but the inaction probability can still vary with the distance to the standard through the latent mean μ_j. When $R > S_j$, both the latent mean and the location of the inaction interval vary with $R - S_j$.

For a nonzero realized adjustment m, the corresponding latent desired adjustment is

$$m_j^*(m) = m + \lambda_S(d_j + m)_+ + a_m \operatorname{sgn}(m), \tag{C.5}$$

with Jacobian

$$\frac{\partial m_j^*(m)}{\partial m} = 1 + \lambda_S \mathbf{1}\{R + m > S_j\}. \tag{C.6}$$

Thus the continuous density of a nonzero adjustment ($m \neq 0$) is

$$f_m(m \mid d_j, \mu_j) = \frac{1 + \lambda_S \mathbf{1}\{d_j + m > 0\}}{\sigma_m} \phi\left(\frac{m + \lambda_S (d_j + m)_+ + a_m \operatorname{sgn}(m) - \mu_j}{\sigma_m}\right). \quad \text{(C.7)}$$

The log likelihood combines $\log \pi_j^0$ for contracts that remain at the listed rent with $\log f_m(m_j \mid d_j, \mu_j)$ for nonzero adjustments. We estimate all Stage 3 parameters jointly.

Stage 1 and the counterfactuals use the expected contract rent

$$\bar{r}_j(R, V; S_j) = R + \int m(x; R, S_j, \lambda_S, \lambda_m) \frac{1}{\sigma_m} \phi\left(\frac{x - \mu_j(R, V, S_j)}{\sigma_m}\right) \mathrm{d}x, \quad \text{(C.8)}$$

where $m(x; R, S_j, \lambda_S, \lambda_m)$ is the solution to Equation (C.2) when $m_j^* = x$. The solution is piecewise linear in x, so the expectation can be evaluated numerically or using truncated normal moments. A counterfactual change in the payment standard affects expected contract rent through two channels: it changes the desired adjustment through $\mu_j(R, V, S_j)$ and changes the penalty on contract rents above the standard through $(R + m - S_j)_+^2$. Changes in rent adjustment frictions are implemented through λ_m.

Specification comparisons. We evaluate several alternatives to the baseline specification. First, we re-estimate the model using only 2023–2025, when the linked listing data have substantially better coverage. Second, we replace CBSA fixed effects with PHA fixed effects, while retaining year and bedroom fixed effects. Third, we set $\lambda_S = 0$ while retaining μ_+ and μ_-, which tests whether the above-payment standard penalty improves fit beyond the direct dependence of desired adjustments on the distance to the payment standard. Fourth, we replace the quadratic penalty with a linear one, $\lambda_S(R+m-S_j)_+$, which adds a kink at $r_j = S_j$ and therefore a second mass point at the payment standard. Fifth, we remove μ_+ and μ_-, corresponding to the more restrictive specification in which the latent desired adjustment does not depend directly on $R - S_j$. Sixth, we replace the Normal latent shock with a standardized logistic distribution. Seventh, we express the adjustment relative to the listed rent rather than in dollars. Eighth, we estimate the model separately for listings weakly below and above the payment standard.

We do not consider separate adjustment frictions for upward and downward changes: with a full set of CBSA fixed effects in μ_j, a shift of λ_m^+ against λ_m^- moves the inaction interval and every interior desired adjustment by the same amount and is absorbed by the fixed effects, so only their average is identified. We compare likelihood-based fit criteria only across specifications defined on the same outcome scale and estimated on the same sample. Note that the 2023–2025 specification uses a different sample, and under the linear penalty contracts at the payment standard enter the likelihood as probabilities rather than densities. Across all specifications, we additionally compare

fit to the probability of adjustment, the mean adjustment conditional on adjusting, the unconditional mean adjustment, and the share of contracts at the payment standard across bins of $R - S_j$.

C.2 Stage 2 estimation

Let L_t be the fraction of new admissions linked to the final listing panel, and let

$$T_t^{\text{obs}} = \sum_{j:t(j)=t} \exp\{x_j(R_j, V_j)'\widehat{\vartheta}_2\}.$$

For $0 < L_t < 1$, the outside-mass calibration is

$$\widehat{s}_t^0 = \frac{1 - L_t}{L_t} T_t^{\text{obs}}, \qquad \widehat{A}_t^{\text{obs}} = \frac{M_t}{\widehat{s}_t^0 + T_t^{\text{obs}}} = \frac{M_t L_t}{T_t^{\text{obs}}}. \tag{C.9}$$

Thus the model exactly reproduces the linked share in every interior market-year. For market-years with no links and the small number of market-years with $L_t \geq 1$, where the exact formula is unavailable, we fill s_t^0 using an M_t-weighted additive metro and year smoother estimated on the interior market-years. For a metropolitan area with no interior year, the metro component is replaced by the M_t-weighted mean metro component.

The first-step PPML includes unrestricted metro-year fixed effects to identify ϑ_2 from within-market variation in listing attractiveness. We treat these fixed effects as nuisance parameters rather than as estimates of the structural allocation level A_t. Given $\widehat{\vartheta}_2$, the administrative flow M_t and linkage share L_t determine the structural level through Equation (C.9).

Specification comparisons. We compare the baseline allocation index with four alternatives estimated by PPML on the same listings and outcome, each recalibrated with the same outside-mass procedure. First, we drop the discrete term, setting $\theta_0 = 0$, so attractiveness is continuous at the payment standard. Second, we cap both distance terms at \$500, replacing $(S_j - R)_+$ and $(R - S_j)_+$ with $\min\{(S_j - R)_+, 500\}$ and $\min\{(R - S_j)_+, 500\}$, so the slopes are identified from listings within \$500 of the payment standard and the far tails do not determine them. Third, we replace the CBSA-by-year fixed effects with additive CBSA and year fixed effects, which use variation across years within a metropolitan area and across areas within a year. Fourth, we let the discrete term differ by targeting status, replacing $\theta_0 \mathbf{1}\{R > S_j\}$ with $\mathbf{1}\{R > S_j\}(\theta_0 + V'\theta_{0V})$, since targeted listings may reach voucher holders for whom a rent above the payment standard is less of a barrier. We compare the specifications on the PPML pseudo-log-likelihood, which is comparable because the sample and outcome are identical, and on fit to the linked-match rate across bins of $R - S_j$, targeting status, and tract-covariate deciles.

Table C.2 reports the results. The targeting effects and the neighborhood gradients change little across specifications. Without the discrete term, the slope below the payment standard turns positive and the slope above it nearly doubles, but the model misses the drop at the standard, predicting -1.3 matches per 1,000 listings against -3.4 in the data, and the pseudo log likelihood falls by 4,317. Capping the distance terms at \$500 halves the discrete term and triples the slope above the standard. It fits matches around the standard best, with a drop of -4.9 and an RMSE of 0.44 against 0.78 for the baseline, and it raises the pseudo log likelihood by 2,242. We nevertheless keep the uncapped specification as the baseline, because Stage 1 evaluates the allocation intensity at rents up to twice the benchmark rent, whereas the capped estimates imply no further decline beyond \$500 above the standard, a range they do not identify. Additive CBSA and year fixed effects leave the payment-standard terms within one standard error of the baseline, and the pseudo log likelihood is lower because the specification has about 1,800 fewer fixed effects. Letting the discrete term differ by targeting status concentrates it among non-targeted listings. The term is -0.90 for non-targeted, -0.08 for soft-targeted, and 0.14 for hard-targeted listings, and the pseudo log likelihood rises by 857 with two more parameters. Because 98 percent of listings are non-targeted, the predicted drop at the standard barely changes. The baseline keeps a common discrete term, so Stage 1 applies the drop in the match probability above the standard to targeted listings as well.

C.3 Stage 1 estimation

Market rent benchmark. Each listing has a benchmark rent R_j^{mkt} that locates comparable units in its metropolitan area and tract. It combines a metropolitan level with a time-invariant tract position, both computed from the non-targeted listings of the Stage 2 sample,

$$R_j^{\text{mkt}} = \rho_j \bar{R}_j, \qquad \rho_j = \frac{N_j \tilde{\rho}_j + 10}{N_j + 10}, \tag{C.10}$$

where $\bar{R}_j$ is the median rent of non-targeted listings with the metropolitan area, bedroom count, and year of listing j, and $\tilde{\rho}_j$ is the median ratio $R/\bar{R}$ among the N_j non-targeted listings with the tract and bedroom count of listing j, pooled over 2015 to 2025. The tract's relative rent ρ_j shrinks the raw ratio $\tilde{\rho}_j$ toward one with the weight of ten listings, which limits the noise in tracts with few listings, where a single listing would otherwise set its own benchmark.

The benchmark does not enter payoffs directly. It sets the range of rents in the choice problem and normalizes the distribution of market values F_j, which is defined on R^*/R_j^{mkt}. This normalization matters for identification. Payment standards are mostly common to a metropolitan area, bedroom count, and year, so a distribution of market values defined in dollars at that level could place a step at the payment standard and absorb the voucher incentives. The tract position shifts the

location of the standard within F_j from tract to tract. Because it does not vary over time, a policy shock to rents in particular ZIP codes does not move the benchmark of the listings it affects.

Grid and window. Rents and market values lie on a grid anchored at the payment standard, $R_{jk} = S_j e^{kh}$ for the integers k with $0.40\, R_j^{\mathrm{mkt}} \le R_{jk} \le 2.00\, R_j^{\mathrm{mkt}}$, with step $h = 0.01$. The grid is uniform in log rent, so every grid point stands for a rent interval of the same relative width, and the payment standard is a grid point, $R_{j0} = S_j$. An observed rent in $(R_{j,k-1}, R_{jk}]$ is assigned to grid point k, so listings at the payment standard and up to one step below it are assigned to S_j, and listings just above it to the next grid point, on the side of the standard where they belong. The rent choice set $\mathcal{R}_j(R_j^*)$ of a market value at grid point l is the set of grid points k with $|k - l| \le \bar{k}$, truncated at the ends of the grid. The baseline sets $\bar{k} = 1$, so that the listed rent lies within one percent of the market value. Listings with R_j/R_j^{mkt} outside $[0.40, 2.00]$ are excluded, which drops 0.63 percent of the listings from 2019 to 2025.

Distribution of market values. Market values R^* are not observed. The distribution F_j depends on the listing only through its benchmark rent and its group $\mathcal{G}(j)$, the metropolitan area and bedroom count: $x = \log(R^*/R_j^{\mathrm{mkt}})$ has the log-spline density

$$f_{\mathcal{G}}(x) \propto \exp\Big\{ \sum_{b=1}^{8} \xi_{\mathcal{G}b}\, \chi_b(x) \Big\}, \tag{C.11}$$

where $\chi_1, \ldots, \chi_8$ are cubic B-splines with interior knots at the sextiles of $\log(R_j/R_j^{\mathrm{mkt}})$ among non-targeted listings. The density is smooth and flexible enough to fit the skewness and heavy upper tail of rents, with separate coefficients $\xi_{\mathcal{G}}$ for each of the 1,018 groups. On the grid of listing j, the probability of market value l is $q_j(l) \propto f_{\mathcal{G}(j)}(\log R_{jl} - \log R_j^{\mathrm{mkt}})$.

We estimate F_j in a first step and hold it fixed when estimating ϑ_1. Without the voucher program, a listing chooses each rent in the window around its market value with equal probability. Non-targeted listings match with a probability of 0.4 percent on average, so their rents depart little from this benchmark. We therefore fit $\xi_{\mathcal{G}}$ by maximum likelihood to the grid rents of non-targeted listings under the model without the program, with a ridge penalty $\|\xi_{\mathcal{G}}\|^2/2$, using expectation maximization: starting from the direct fit of $\log(R_j/R_j^{\mathrm{mkt}})$, we alternate between updating the posterior over market values and Newton steps on the coefficients until the log likelihood rises by less than one. As a robustness check, we estimate $\xi_{\mathcal{G}}$ jointly with ϑ_1 using the posteriors of all listings.

Likelihood. With $p_j(k, V)$ the Stage 2 match probability at grid rent k under $\widehat{A}_t$ and $\bar{r}_j(k, V)$ the Stage 3 expected contract rent, the payoff in Equation (7) of listing at grid rent k with market value

l is

$$\bar{u}_j(k,l,V) = p_j(k,V)\big[\bar{r}_j(k,V) - (1-\psi_j)R_{jl} - c_j^{\mathrm{HCV}}\big] - c_{V,j}^{\mathrm{V}}.$$

The expected adjustment $\bar{r}_j(k,V) - R_{jk}$ is interpolated bilinearly from a table over the Stage 3 index, in steps of \$5, and over $R - S_j$, in steps of \$10, each entry integrating Equation (C.8) over 201 quantiles of the desired adjustment. With $E_j(k,l,V) = \exp\{\bar{u}_j(k,l,V)/\sigma_R\}$ and $Z_j(l,V) = \sum_{k:|k-l|\le\bar{k}} E_j(k,l,V)$, the probability of choosing grid rent k and targeting intensity V is

$$P_j(k,V) = \sum_l q_j(l)\,\frac{E_j(k,l,V)}{Z_j(l,V)} \cdot \frac{Z_j(l,V)^{\sigma_R/\sigma_V}}{\sum_{V'\in\mathcal{V}_j} Z_j(l,V')^{\sigma_R/\sigma_V}}, \tag{C.12}$$

where the sum over l runs over the market values whose window contains k. Listings with the same tract, bedroom count, year, and payment standard share the grid, the payoffs, and the distribution of market values, so the likelihood is computed on the number of such listings that choose each grid rent and targeting intensity.

We maximize the log likelihood with Newton steps based on the outer product of the listing-level scores and a backtracking line search, until every step is below 0.01 standard errors or the log likelihood rises by less than 0.01 over three iterations. The metropolitan participation cost δ_c enters payoffs only through the gain from a match, which is multiplied by the match probability, so the likelihood is nearly flat in δ_c where match probabilities are close to zero. We therefore subtract $\sum_c \delta_c^2/(2\cdot 1000^2)$ from the log likelihood. The penalty keeps these levels finite and has a negligible effect where the data are informative, since the estimated levels have a standard deviation of \$395 across metropolitan areas. Standard errors from the outer product of the scores are conditional on the Stage 2 and Stage 3 estimates and on F_j, and the bootstrap below adds the uncertainty from the first two.

Bootstrap. Each of 100 replications draws the Stage 2 slopes from a normal distribution with the CBSA-clustered covariance matrix of the PPML estimates and the Stage 3 parameters from a normal distribution with the CBSA-clustered covariance matrix of the maximum likelihood estimates. Given the Stage 2 draw, the outside mass s_t^0 and the allocation level A_t are recalibrated as in Equation (C.9). With both Stage 3 and Stage 2 draws, the match probabilities and expected contract rents are recomputed at every grid rent and targeting intensity, and Stage 1 is re-estimated from the baseline estimates, holding F_j and $\bar{k}$ fixed. The total variance of the Stage 1 parameters is the conditional variance from the outer product of the scores plus the variance of the re-estimates across replications. It raises the standard errors of the common parameters by a factor of 2 to 17 and those of the metropolitan levels by a median factor of 1.8 for the participation cost and 14 and 10 for the soft and hard targeting costs. For model-implied outcomes, each replication evaluates

the outcome at the re-estimated parameters plus a draw from the conditional sampling distribution, and the standard error is the standard deviation across replications.

Specification comparisons. Table C.3 compares the baseline with eleven alternatives, each changing one element. Panel A changes the rent margin. Two specifications widen the window to $\bar{k} = 5$ and $\bar{k} = 10$. One replaces the window with a quadratic market cost of departing from the market value, $\Pi_j^{\mathrm{MKT}}(R) = (1 - \psi_j)R_j^* - (\eta/2)(\log R - \log R_j^*)^2$ with $\eta = \sigma_R/\omega^2$ and $\omega = 0.03$, so that without the program listed rents are normally distributed around the market value with a standard deviation of three percent. One halves the grid step to $h = 0.005$ with $\bar{k} = 2$, which keeps the window at one percent. Panel B changes where neighborhood characteristics enter (only the guarantee share, only the participation cost, or not the targeting costs), the estimator (F_j estimated jointly with ϑ_1, or a single logit with $\sigma_V = \sigma_R$), and the sample. The large-metropolitan-area sample keeps the 195 metropolitan areas with at least 1,000 listings and makes a targeting intensity available in a metropolitan-area-year only if at least ten listings use it. The 2015–2025 sample excludes hard targeting from the choice set before 2019, when affordable housing platforms and organizations are not recorded reliably, and drops the few listings recorded as hard targeted in those years. Log likelihoods are comparable across specifications estimated on the same listings and grid.

The likelihood prefers the baseline rent margin. The five- and ten-percent windows lower the log likelihood by 2,334 and 18,294, and the quadratic market cost by 1,499. All three raise $\hat{\sigma}_R$ to \$21–\$30, so the rent shocks rather than the voucher payoffs absorb the spread of rents near the standard, and the excess mass of soft-targeted listings at the standard falls to at most 2 percent, against 18 percent in the baseline and 62 percent in the data. Halving the grid step leaves the estimates nearly unchanged. Where neighborhood characteristics enter shifts the split between the guarantee and the participation cost but changes the net gain little. With characteristics only in the guarantee share, $\hat{\psi}$ rises to 0.34 and its poverty gradient turns negative. With characteristics only in the participation cost, $\hat{\psi}$ falls to 0.18. Both lower the log likelihood, by 2,062 and 301 with three fewer parameters, and leave the net gain in the highest poverty quintile between \$340 and \$358. Targeting costs that vary only across metropolitan areas lower the log likelihood by 111,983. The model then cannot reproduce the rise in targeting with poverty, with 2.3 percent of listings soft targeted in the highest poverty decile against 4.2 percent in the data, and the net gain becomes flat across poverty quintiles. Estimating F_j jointly raises the log likelihood by 1,422 with 8,144 more parameters and leaves the estimates unchanged. A single logit lowers the log likelihood by 2,462 and produces no excess mass at the standard. The large-metropolitan-area and 2015–2025 samples yield similar estimates. In all specifications except metropolitan targeting costs, the net gain falls from \$732–\$1,168 in the lowest poverty quintile to \$302–\$469 in the highest.

C.4 Counterfactual computation

Status quo and the outside pool. Estimation uses the observed choices and the estimated allocation level $\widehat{A}_t$, whereas a counterfactual uses the choices that the model predicts. To compare like with like, we recalibrate the outside mass so that the predicted choices reproduce the estimated allocation level,

$$\tilde{s}_t^0 = \frac{M_t}{\widehat{A}_t} - T_t(\widehat{A}_t), \tag{C.13}$$

with T_t from Equation (17). The status quo is then an equilibrium at $A_t^* = \widehat{A}_t$, and every counterfactual is compared with it. The recalibration also absorbs the attractiveness of the listings outside the grid of rents. Across markets, the median of $|\tilde{s}_t^0/\widehat{s}_t^0 - 1|$ is 0.5 percent.

Let the poverty percentile of a tract rank its 2020 poverty rate among all tracts of its metropolitan area from 0 to 100, as in Section 5, and let $q = 1, \ldots, 4$ index its quartiles, holding the assignment fixed, with Q1 the tracts at or below the 25th percentile. We split the outside mass across quartiles, $\tilde{s}_t^0 = \sum_q \tilde{s}_{tq}^0$, in proportion to the voucher households in each quartile that the model does not assign to observed listings, so that the status quo reproduces the distribution of voucher households across neighborhoods.[4] Let $T_{tq}(A)$ be the attractiveness of the observed listings in quartile q under the policy at allocation level A, so that $T_t = \sum_q T_{tq}$, and let T_{tq}^{sq} be its status-quo value. In a counterfactual, the outside mass of quartile q is

$$s_{tq}^0(A) = \tilde{s}_{tq}^0 \left[(1-\zeta) + \zeta \, \frac{T_{tq}(A)}{T_{tq}^{\text{sq}}} \right], \tag{C.14}$$

where ζ is the share of the outside mass whose attractiveness moves with that of the observed listings in the same market and quartile, and the equilibrium solves $A_t^* = M_t / [\sum_q s_{tq}^0(A_t^*) + T_t(A_t^*)]$. With $\zeta = 0$, the outside pool is fixed, and a policy that makes observed listings more attractive draws admissions from it. With $\zeta = 1$, the outside pool responds as observed listings do, so a policy moves admissions across neighborhoods but not between observed listings and the outside pool within a neighborhood. Splitting the outside pool by quartile keeps a policy confined to Q1 tracts from moving the outside pool in other neighborhoods.

We calibrate ζ as follows. Let a be the share of new admissions whose units respond to a policy as observed listings do, including the observed listings themselves. Because 16.8 percent of new admissions in the Stage 1 markets are linked to observed listings, $\zeta = (a - 0.168)/(1 - 0.168)$. Voucher holders who lease in place, 21 percent of successful voucher holders in Finkel and Buron (2001), stay in the unit they already rent, so their location cannot respond. The others found their

[4] Voucher households in a tract are the vouchers in use in HUD administrative data, averaged over 2023 to 2025, the years recorded on 2020 tracts, and their distribution across quartiles is used in every year. We scale them to the flow M_t, subtract the status-quo matches to observed listings in the quartile, and set negative differences to zero.

unit on the open market or through friends, family, or the housing authority. Units found on the open market respond as observed listings do. Units found through personal or agency referrals reach voucher holders through other channels, but their landlords still decide whether to accept a voucher at the same payment standard, bonus, and review of rents, so they may respond as well. For the lower bound, only the units found on the open market respond. In a 2024 survey of voucher holders in the Twin Cities, between 25 percent, through HousingLink, the region's affordable housing listing site, and 54 percent, through any website, heard of their current home online (Metropolitan Council, 2026). At the midpoint of this range, $a = 0.79 \times 0.395 = 0.31$ and $\zeta = 0.17$. For the upper bound, every unit except leases in place responds, so $a = 0.79$ and $\zeta = 0.75$. We use the midpoint, $\zeta = 0.46$, as the baseline, so that about half of the units found through referrals respond, and report results at both bounds.[5]

Counterfactual policies. New payment standards are rounded to the nearest point of the status-quo grid, $S_j e^{nh}$ for an integer n, so the grid of rents and market values and the distribution f_j over it do not change. The rounding moves a standard by at most half a percent, so the 10 and 20 percent increases are increases of $e^{0.10} - 1 = 10.5$ and $e^{0.18} - 1 = 19.7$ percent. The match probabilities and expected contract rents are recomputed at every grid rent with the new standard. The rent-indexed standard of a listing is its benchmark rent R_j^{mkt} times a factor common to its metropolitan area, bedroom count, and year, chosen so that the listing-weighted mean standard is unchanged. The hold-harmless variant sets the standard to the larger of the status-quo and the indexed standard.

A bonus of ι months of the contract rent enters the voucher payoff as $\iota\, \bar{r}_j(R, V)/D$ added to the expected contract rent in $\bar{u}_j(k, l, V)$, with a tenancy of $D = 24$ months.[6] We set $\iota \in \{1, 2\}$ in all tracts and $\iota = 1$ in Q1 tracts or in Q1 and Q2 tracts only. The bonus is accounted for separately from the contract rent and is paid on every new admission in the tracts where it applies, including the outside admissions.

The stricter rent-reasonableness counterfactual approves a proposed increase over the listed rent

[5]Respondents could name more than one source, so 25 percent understates and 54 percent overstates the share that named any website, which sums the shares naming HousingLink (25 percent), other affordable housing websites (12 percent), and general websites (17 percent). More than a third heard of their home through friends or family and 12 percent through the housing authority. The shares come from the online survey of current voucher holders of two housing authorities and refer to their most recent search. The share leasing in place comes from the 2000 sample of 48 large metropolitan housing authorities.

[6]A tenancy in a voucher unit ends when the household moves within the program or leaves it. Among families entering the program from 1995 to 2002, 14.5 percent of those still in the program had moved within two years of leasing up and 45 percent within five years (Feins and Patterson, 2005), and the average voucher household that left the program between 2000 and 2015 had participated for 4.9 years (McClure, 2018). Twenty-four months is therefore at the short end of the length of a first voucher tenancy, which also accounts for the landlord's discounting of future payments. Because the bonus enters only through its value per month of tenancy, a bonus of ι months with $D = 24$ is equivalent to a bonus of 2ι months with $D = 48$, so the two bonus sizes we consider also bracket longer tenancies.

with probability ρ and signs the contract at the listed rent when the increase is denied. The expected adjustment is then $\rho\,\mathbb{E}[m_j\mathbf{1}\{m_j > 0\}] + \mathbb{E}[m_j\mathbf{1}\{m_j < 0\}]$, so increases are scaled down while concessions below the listed rent are unaffected. The change enters Stage 1 through the expected contract rent $\bar{r}_j(R, V)$, computed with the same quadrature, holding the desired adjustments and the match probabilities fixed.

The budget-neutral policies spend the savings from the rent index, from enforcement, or from both on a common increase in the payment standards, applied in all tracts or only in the two lowest poverty quintiles, so that program-wide spending on new admissions is unchanged. Because standards move in steps of h, we find the two adjacent step counts whose spending brackets the status quo and interpolate the outcomes linearly between them. Across the values of ζ, the budget-neutral increases range from 2.9 (enforcement alone, in all tracts) to 33 percent (index and enforcement, in the two lowest poverty quintiles).

Equilibrium and outcomes. The allocation level enters only through the allocation intensity, which is proportional to A_t. At level A, the match probability at a grid rent and targeting intensity is $1 - (1 - \hat{p})^{A/\widehat{A}_t}$, where $\hat{p}$ is the probability at $\widehat{A}_t$ used in estimation, and the attractiveness $\exp\{x_j(R, V)'\theta\}$ is $-\log(1 - \hat{p})/\widehat{A}_t$. Given A, the choice probabilities follow from Equation (C.12), and $T_{tq}(A)$ sums the attractiveness over grid rents and targeting intensities, weighted by the choice probabilities and the number of listings. Given the allocation levels, markets are independent, so we solve $\log A = \log M_t - \log[\sum_q s^0_{tq}(A) + T_t(A)]$ market by market, starting from $\widehat{A}_t$ and taking secant steps until the largest change in $\log A$ across markets is below 10^{-6}. We verify market by market that $A\,[\sum_q s^0_{tq}(A) + T_t(A)]$ is increasing at the equilibrium, by checking that it exceeds M_t at $1.01A^*_t$, so that no other equilibrium lies just above it.

New admissions in quartile q are $A^*_t[s^0_{tq}(A^*_t) + T_{tq}(A^*_t)]$, of which $A^*_t T_{tq}(A^*_t)$ lease through observed listings, and they sum to M_t across quartiles. Because τI_i does not change, a policy changes the housing assistance payment per admission by the change in $\mathbb{E}[\min\{r, S_j\}]$, plus the bonus. For matches to observed listings, these follow from Stage 3 at the counterfactual choices. The contract rents of outside admissions are not observed, so we assign the outside admissions of a market and quartile the status-quo distribution of the model's matches to observed listings there, relative to the payment standard, and apply the policy's standard, bonus, and Stage 3 objects without re-pricing. The mean poverty percentile weights the percentile of each tract by its new admissions. Cost-effectiveness is the change in spending per admission divided by the decrease in the mean poverty percentile, both computed among the same admissions. Because budget neutrality refers to program-wide spending, a budget-neutral policy can change spending per admission through observed listings.

Landlord value. We measure landlord incidence by the program's value to the landlord of listing j, per month and relative to a market without voucher matches,

$$W_j = \sigma_V \int \Big\{ \log \sum_{V \in \mathcal{V}_j} \exp\{I_j(V \mid R^*; A_t)/\sigma_V\} - \log \sum_{V \in \mathcal{V}_j} \exp\{I_j^0(V \mid R^*)/\sigma_V\} \Big\} \, \mathrm{d}F_j(R^*), \tag{C.15}$$

where $I_j(V \mid R^*; A_t)$ is the inclusive value of targeting intensity V in Equation (9) and $I_j^0(V \mid R^*)$ is the same inclusive value with no voucher matches, $p_j^{\mathrm{HCV}} = 0$, so that $\bar{u}_j(R, V) = -c_{V,j}^{\mathrm{V}}$. The two terms are the landlord's expected payoff from choosing a rent and targeting intensity, with and without voucher matches. Their difference combines the gain from a match, the match probabilities, and the option values of re-pricing within the window and of targeting, averaged over market values, including listings that rarely match.

The change in W_j under a policy accounts for landlords' re-optimization and, through the allocation level A_t^*, for the lower match probabilities of listings that become relatively less attractive. Because most listings rarely match, W_j is small per listing, \$0.98 per month at the status quo, so we report the change in the sum of W_j over observed listings per status-quo in-market voucher match (\$238 at the status quo). The measure covers only the landlords of observed listings and holds market rents fixed.

Table C.1: Stage 3 specification comparisons

	Baseline	2023–2025	PHA FE	$\lambda_S = 0$	Linear penalty	No $\mu_\pm$	Logistic	Proportional	Split by $R \lessgtr S$
μ_+	0.519	0.530	0.543	0.520	0.514	–	0.522	0.507	0.501
	(0.026)	(0.027)	(0.025)	(0.027)	(0.026)		(0.025)	(0.013)	(0.026)
μ_-	1.121	1.132	1.112	1.131	1.121	–	1.106	1.156	1.040
	(0.014)	(0.015)	(0.013)	(0.015)	(0.015)		(0.014)	(0.016)	(0.007)
σ_m	178.0	212.2	175.2	173.7	180.5	296.9	158.6	0.134	183.0
									[153.5]
λ_S	0.440	0.378	0.427	0	110.1	2.555	0.498	0.587	0.716
									[0.118]
λ_m	157.4	188.9	157.0	153.2	159.4	220.0	126.8	0.121	168.9
									[117.2]
Log likelihood ($\times 10^3$)	-680.6	-337.1	-678.7	-681.2	-669.4	-728.2	-672.8	-671.8	-679.1
RMSE: $\mathbb{P}$(adjust)	4.3	4.0	4.3	4.9	3.8	4.9	4.2	4.1	3.4
RMSE: $\mathbb{E}[m \mid \text{adjust}]$	12.8	13.4	12.2	13.3	12.1	121.0	12.0	12.9	13.1
RMSE: $\mathbb{E}[m]$	9.7	10.8	9.1	11.3	9.2	96.2	9.0	7.4	8.5

Notes. This table reports maximum likelihood estimates of Stage 3 under alternative specifications. CBSA-clustered standard errors are in parentheses. σ_m and λ_m are in dollars and $\mu_\pm$ in dollars per dollar, except in the proportional specification, where they are shares of the listed rent. The linear penalty's λ_S is in dollars per dollar above S. The log likelihoods of the linear penalty and of 2023–2025 are not comparable with the others as it runs on a subsample. The split specification estimates μ_+ on listings at or below S and μ_- on listings above it, reports σ_m, λ_S, and λ_m for listings at or below S and, in brackets, above S, and sums the two log likelihoods. RMSEs are over the \$50 bins of $R - S$ in Figure 10, in percentage points for the probability of adjusting and in dollars for the mean adjustments.

Table C.2: Stage 2 specification comparisons

	Baseline	No jump	Capped at \$500	Additive FE	Jump by V
Above S, θ_0	-0.806	–	-0.408	-0.774	-0.899
	(0.106)		(0.050)	(0.115)	(0.106)
$\times$ soft targeting	–	–	–	–	0.824
					(0.079)
$\times$ hard targeting	–	–	–	–	1.036
					(0.068)
Headroom below S, θ_1 (per \$100)	-0.022	0.020	-0.036	-0.016	-0.021
	(0.012)	(0.008)	(0.018)	(0.015)	(0.012)
Distance above S, θ_2 (per \$100)	-0.112	-0.199	-0.339	-0.114	-0.106
	(0.038)	(0.048)	(0.027)	(0.036)	(0.037)
Tract poverty rate	1.785	1.841	1.799	1.792	1.784
Log tract median rent	-0.954	-1.002	-0.935	-0.970	-0.947
Soft targeting	1.467	1.512	1.460	1.460	1.361
Hard targeting	2.047	2.097	2.035	2.025	1.914
Pseudo log likelihood	$-761{,}197$	$-765{,}513$	$-758{,}955$	$-775{,}263$	$-760{,}340$
Drop at S (data: -3.4)	-6.5	-1.3	-4.9	-6.3	-6.4
RMSE by $R - S$	0.78	1.43	0.44	0.77	0.74

Notes. This table reports PPML estimates of Stage 2 under alternative specifications on the same 29,908,026 listings, each recalibrated with the same outside-mass procedure. CBSA-clustered standard errors are in parentheses. Capped at \$500 caps the distance terms at \$500. Additive FE uses CBSA and year fixed effects instead of CBSA-by-year fixed effects. Jump by V replaces the jump at S with $\mathbf{1}\{R > S_j\}(\theta_0 + V'\theta_{0V})$. Drop at S is the number of matches per 1,000 listings in the $(0, 50]$ bin of $R - S$ minus that in the $(-50, 0]$ bin. RMSE is over the \$50 bins of $R - S$ in $(-500, 500]$, weighted by listings, in matches per 1,000 listings.

Table C.3: Stage 1 specification comparisons

Panel A. Rent margin

	Baseline	Five-percent window	Ten-percent window	Quadratic market cost	Half grid step
σ_R (\$)	6.10	21.58	30.29	21.33	5.61
σ_V (\$)	34.3	35.8	46.4	35.4	34.6
ψ	0.207	0.222	0.249	0.223	0.211
ψ_Y, tract poverty rate	0.113	0.041	-0.028	0.031	0.120
Soft targeting cost (\$)	171.0	178.5	231.8	176.4	172.8
Net value, Q1 (\$)	786	881	1,168	866	789
Net value, Q5 (\$)	351	360	469	344	363
Δ Log likelihood	0	$-2{,}334$	$-18{,}294$	$-1{,}499$	–
Parameters	645	645	645	645	645
Excess mass, soft (%)	18	2	1	2	25[†]
Excess mass, hard (%)	34	6	5	7	47[†]
Targeting RMSE (pp)	0.12	0.12	0.12	0.12	0.12

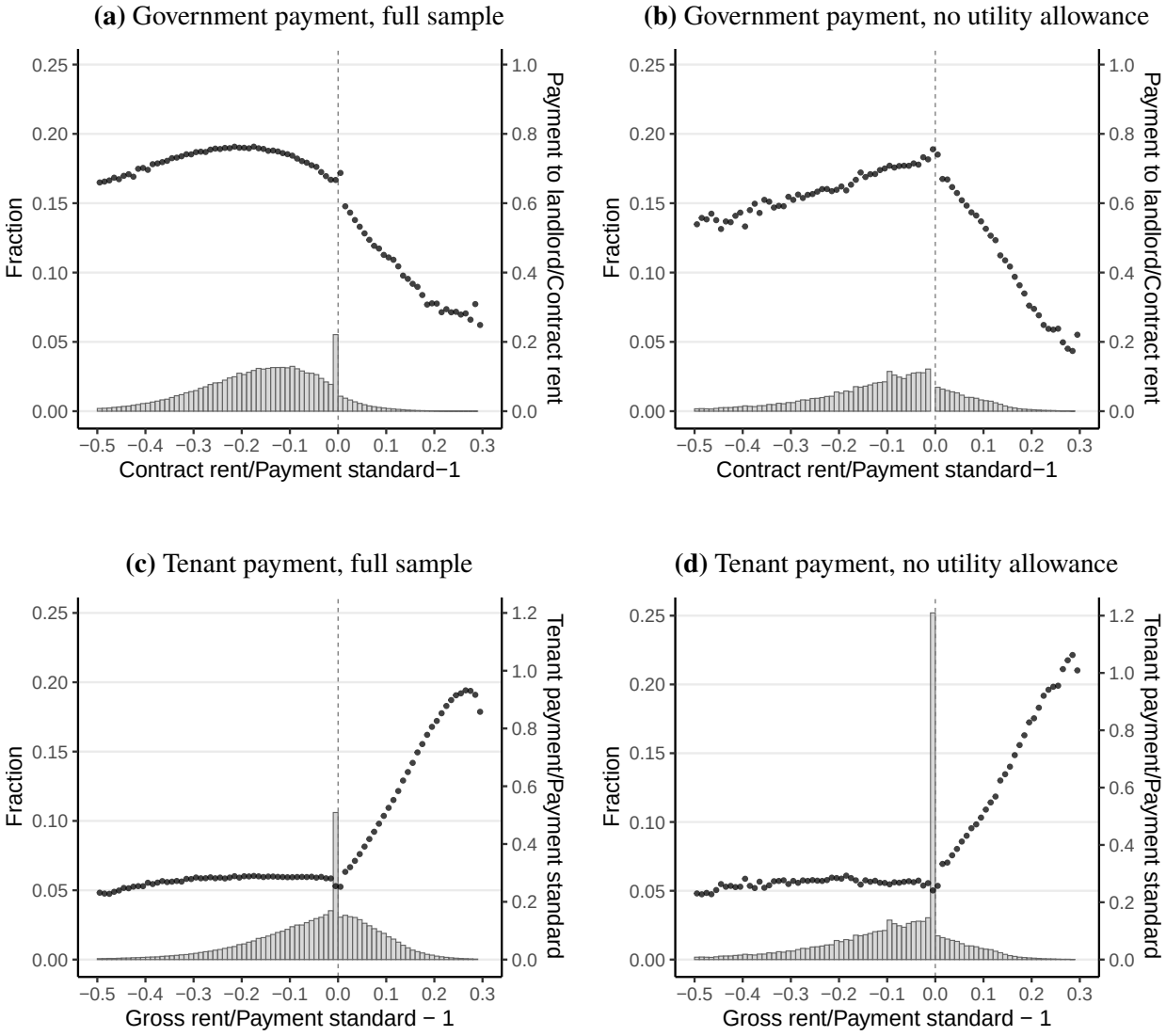
Panel B. Neighborhood terms, estimator, and sample

	Y in ψ_j only	Y in c_j^{HCV} only	Metropolitan targeting costs	F_j estimated jointly	Single logit	Large metros	2015–2025
σ_R (\$)	7.34	5.79	6.09	6.08	33.40	6.19	6.63
σ_V (\$)	36.1	34.0	28.6	34.3	33.4	34.6	30.9
ψ	0.336	0.181	0.201	0.206	0.230	0.214	0.275
ψ_Y, tract poverty rate	-0.409	0	0.319	0.113	0.041	0.104	-0.041
Soft targeting cost (\$)	179.8	169.6	136.5	170.8	166.7	171.4	157.5
Net value, Q1 (\$)	1,008	732	593	784	841	811	780
Net value, Q5 (\$)	358	340	590	350	334	360	302
Δ Log likelihood	$-2{,}062$	-301	$-111{,}983$	$+1{,}422$	$-2{,}462$	–	–
Parameters	642	642	639	8,789	644	600	645
Excess mass, soft (%)	15	19	27	18	-1	18	13[‡]
Excess mass, hard (%)	33	35	47	34	-1	34	29[‡]
Targeting RMSE (pp)	0.12	0.12	0.63	0.12	0.12	0.12	0.10

Notes. This table reports Stage 1 estimates under alternative specifications, holding the Stage 2 and Stage 3 estimates fixed. Δ Log likelihood is relative to the baseline and is not reported when the grid or the sample differs. Soft targeting cost is the average across metropolitan areas. Net value is the payment guarantee net of the participation cost, $\psi_j R_j^{\mathrm{mkt}} - c_j^{\mathrm{HCV}}$, in dollars per month, averaged over listings in the lowest (Q1) and highest (Q5) quintiles of the tract poverty rate. Excess mass is the share of soft- or hard-targeted listings priced at the payment standard relative to the five grid points below it, minus one. Targeting RMSE is the error in the soft-targeting share across deciles of tract poverty, vacancy, and median rent, in percentage points.

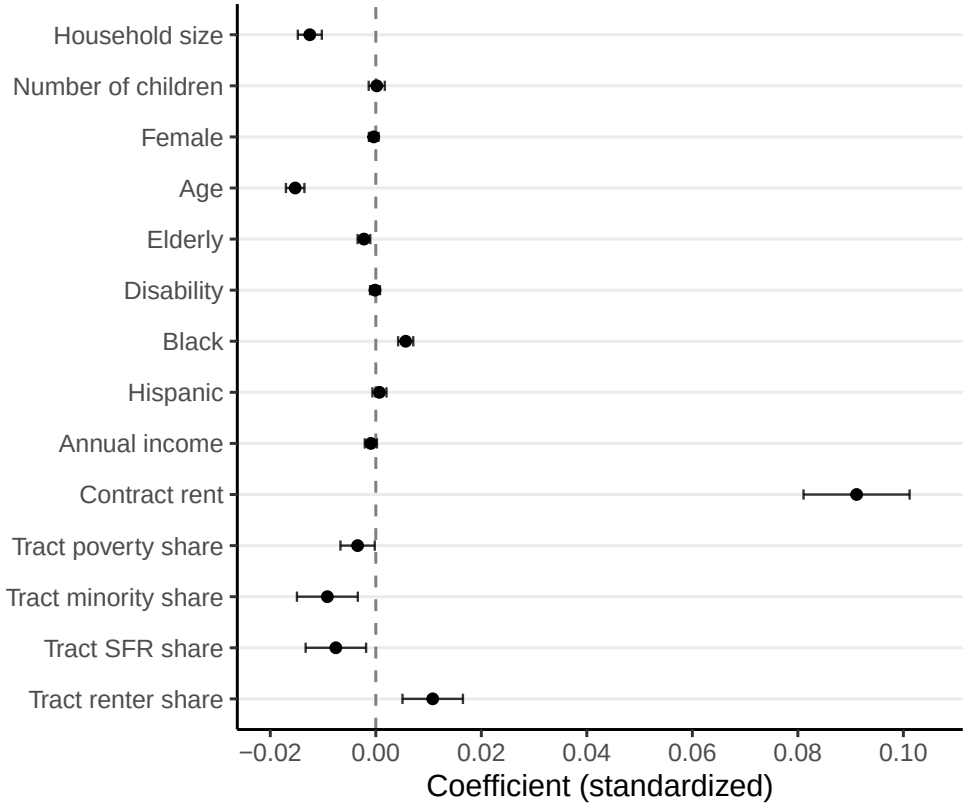
D Supplementary Figures and Tables

Figure D.1: Government and tenant payments under the HCV Program



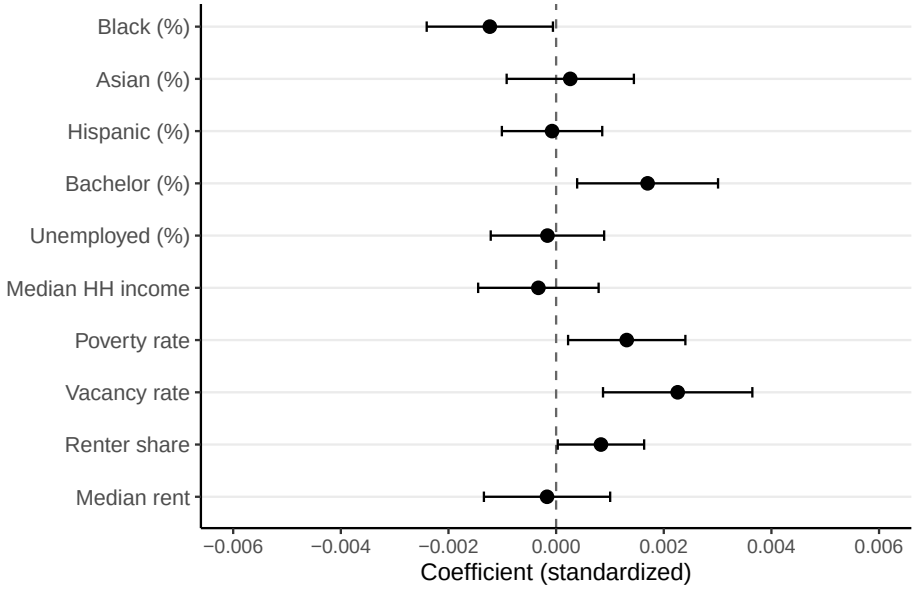
Note. The gray bars show the distribution of contract rents (panels a and b) or gross rents (panels c and d) for newly admitted voucher holders in HUD administrative data, expressed as the percentage deviation from the corresponding payment standard. The sample includes voucher holders admitted between 2015 and 2025 in the year when they were admitted. In panels (b) and (d), we restrict the sample to contracts with no utility allowance (i.e., where contract rent equals gross rent). The histogram uses 1 percentage-point-wide bins. The gray dashed line indicates a zero deviation from the payment standard. The black dots in panels (a) and (b) show the average share of the contract rent paid by the government in each bin, and, in panels (c) and (d), the average share of the gross rent paid by voucher holders.

Figure D.2: Regression of linkage indicator on voucher holder and neighborhood characteristics



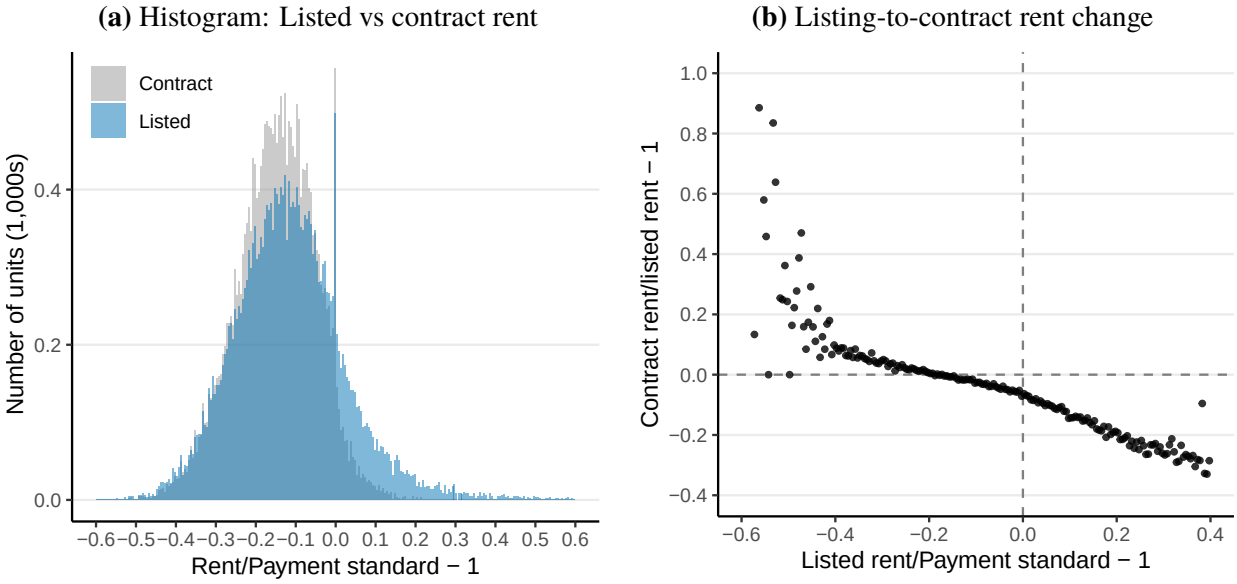
Note. This figure plots coefficients and 95% confidence intervals from pairwise regressions of an indicator for whether an HCV contract is linked to a RentHub listing spell on voucher-holder and neighborhood characteristics, the latter being standardized within PHA. The sample includes HCV new admissions from 2015 to 2025. All regressions include PHA-by-year and PHA-by-bedroom fixed effects, with standard errors clustered at the PHA level. All independent variables are obtained from HUD administrative data.

Figure D.3: Bunching at payment standards in HCV contracts by neighborhood characteristics



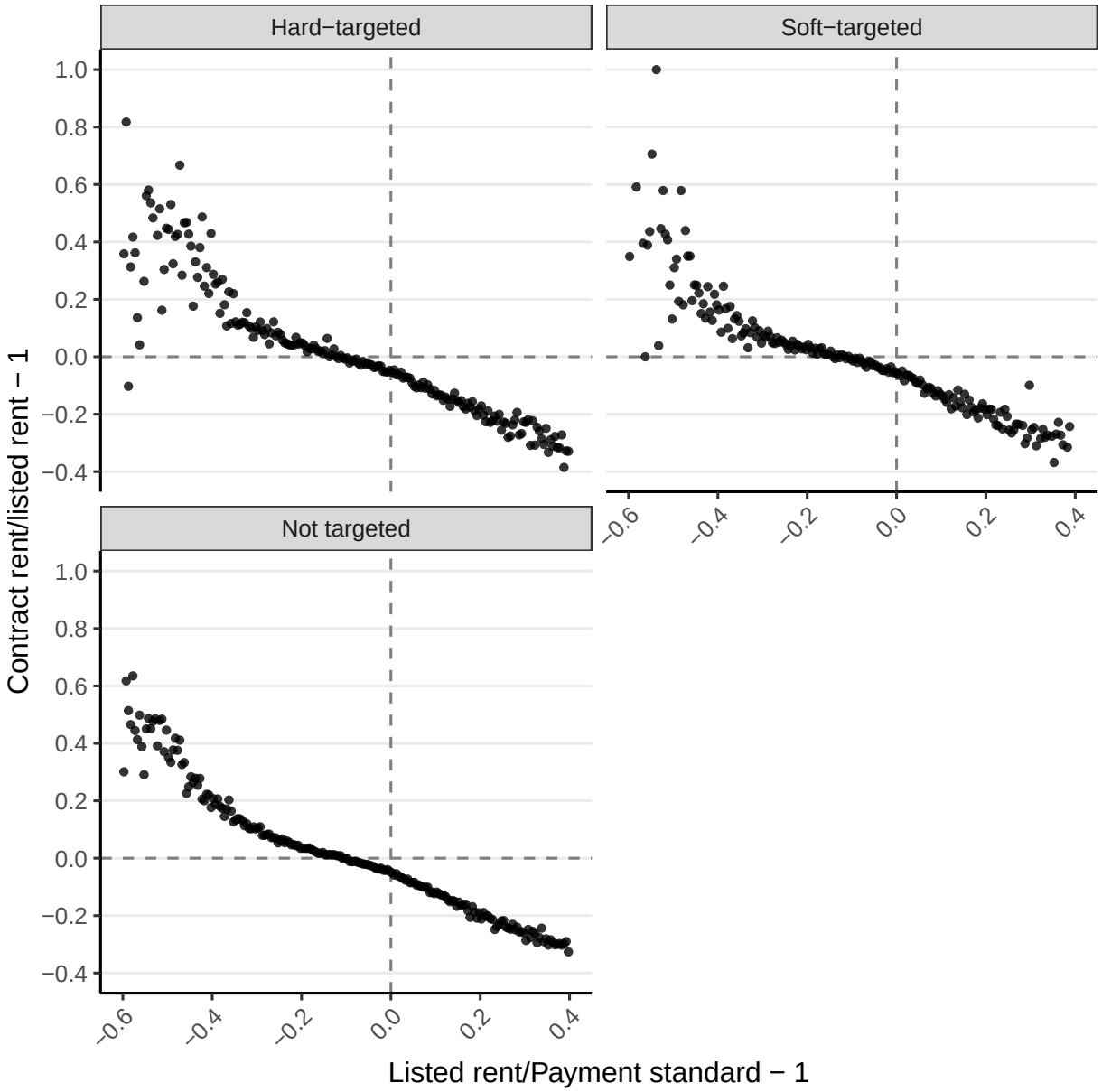
Note. The figure plots coefficients from separate HCV contract-level regressions in which the outcome is an indicator for whether the contract rent equals the payment standard and the regressor is a tract-level characteristic standardized within the PHA. All regressions include PHA-by-year and PHA-by-bedrooms fixed effects, and standard errors are clustered at the PHA level. The sample includes all new admissions in HCV administrative data, 2015–2025. Census characteristics are obtained from the 2016–2020 American Community Survey (5-year estimates).

Figure D.4: Distribution of contract rents, listed rents, and listing-to-contract rent changes for single-unit addresses



Note. Panel (a) shows the histogram of contract (gray) and listed (blue) rents relative to the payment standard, measured as (rent/payment standard) − 1. The sample includes new admissions in HCV administrative data from 2015 to 2025 that are linked to a RentHub listing spell, as described in Section 3.4, where the address is classified as being single-unit according to both the HCV administrative and RentHub datasets. Panel (b) shows a binscatterplot of the mean rent change from the listing to the contract stage, as a function of the listed rent relative to the payment standard, measured as (listed rent/payment standard). Bins are 0.5 percentage points wide.

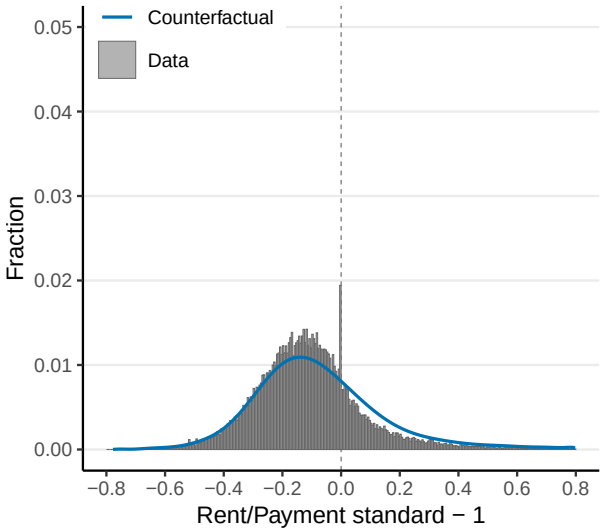
Figure D.5: Listing-to-contract rent changes around payment standards by targeting



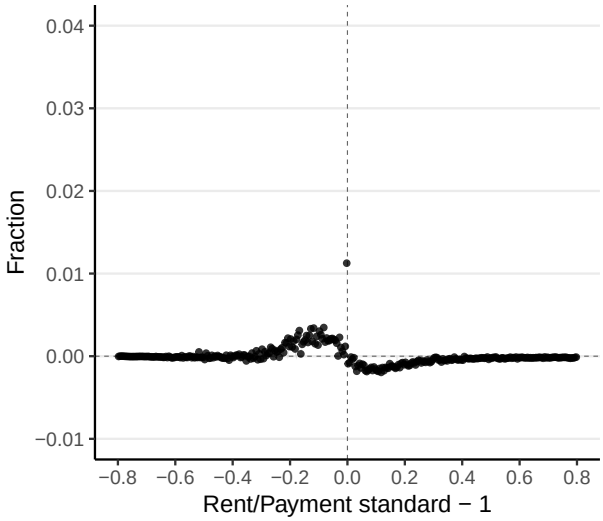
Note. The figure shows binscatter plots of the mean rent change from the listing to the contract stage, as a function of the listed rent relative to the payment standard, measured as (listed rent/payment standard) − 1. The first panel is restricted to hard-targeted listings; the second, soft-targeted listings; the bottom panel, to non-targeted listings as defined in Section 3.2. The samples use 2015–2025 HCV new admissions linked to RentHub listing spells. Bins are 0.5 percentage points wide.

Figure D.6: Differences-in-bunching: Listed vs counterfactual listed rents for linked units

(a) Listed vs. counterfactual listed rent



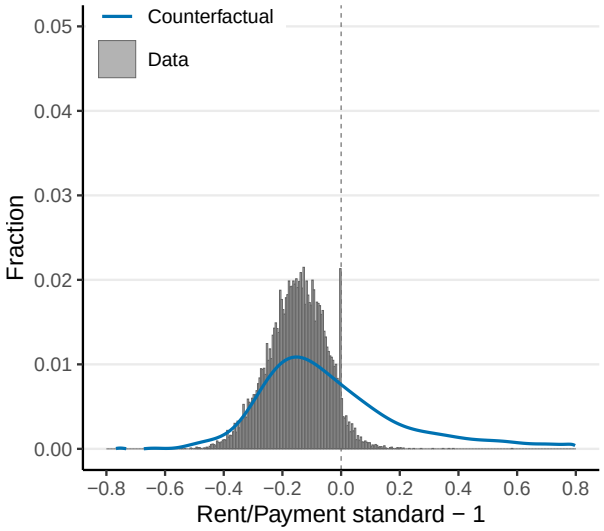
(b) Listed rent - counterfactual listed rent



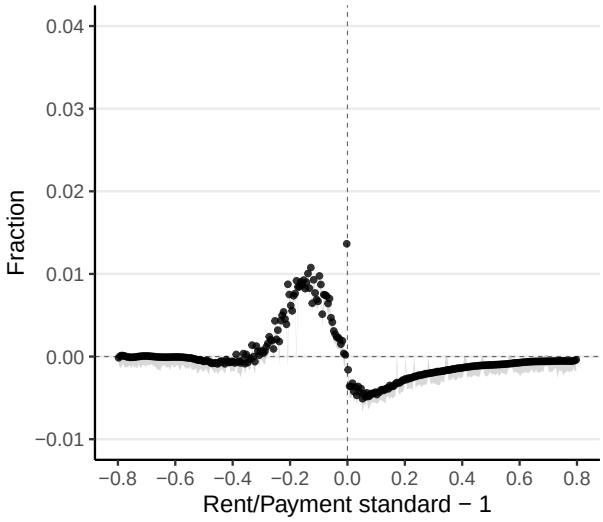
Note. Panel (a) plots the histogram of listed rents for new admissions in HCV administrative data, 2015–2025, that are linked to a RentHub listing spell (gray), together with the counterfactual distribution of listed rents (black), computed from linked listed rents using the differences-in-bunching estimator described in Section 4. Panel (b) plots the difference between the two, with 95% confidence bands (gray) from the iterative bootstrap procedure described there. Bins are 0.5 percentage points wide.

Figure D.7: Differences-in-bunching (single-unit addresses): Contract vs counterfactual rents

(a) Contract vs. counterfactual listed rent

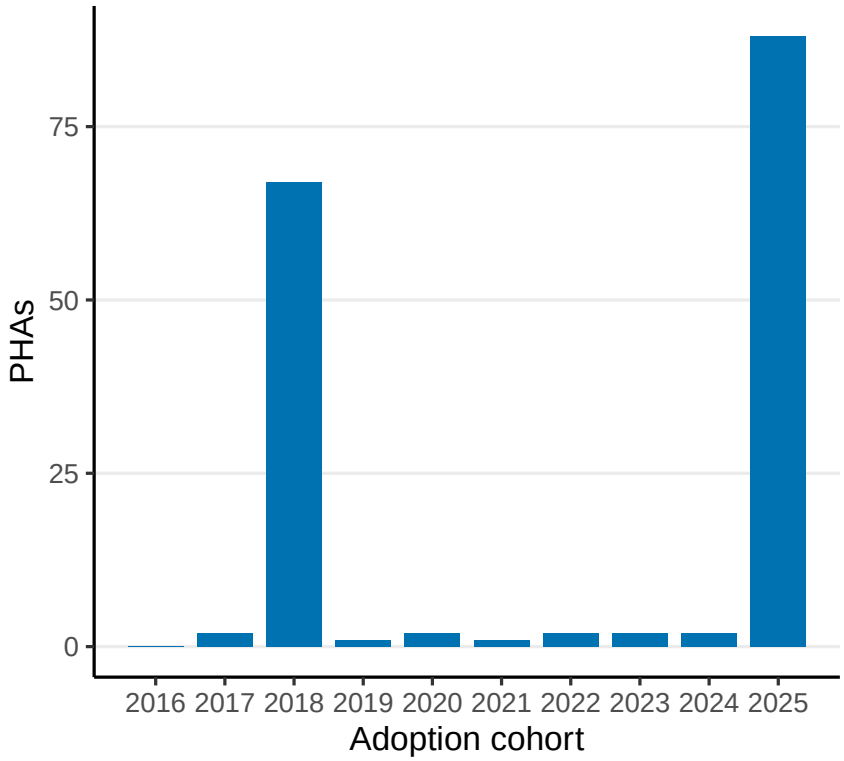


(b) Contract rent - counterfactual listed rent



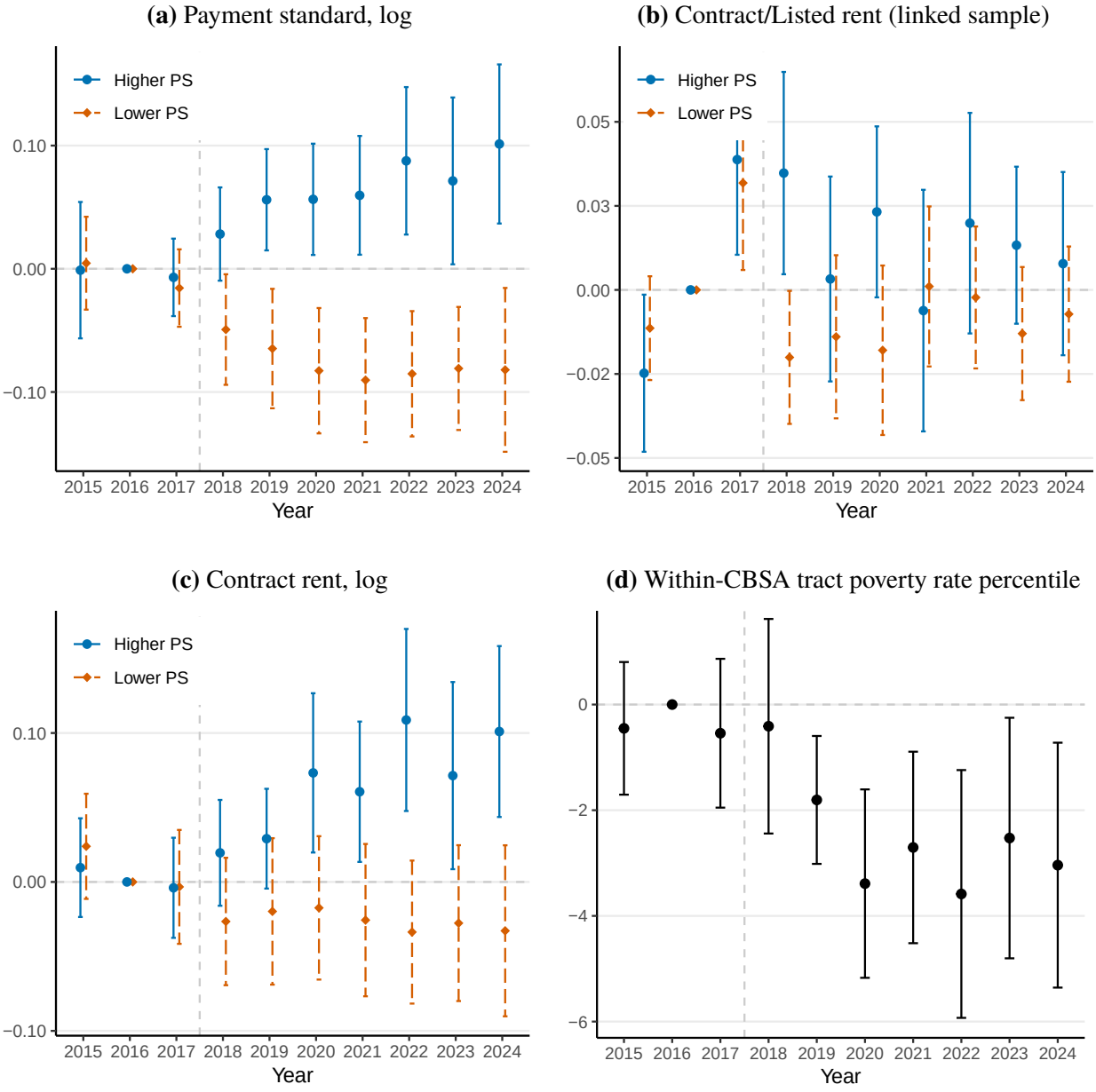
Note. Panel (a) plots the histogram of contract rents for new admissions in HCV administrative data, 2015–2025, that are linked to a RentHub listing spell (gray), together with the counterfactual distribution of listed rents (black), computed from linked listed rents using the differences-in-bunching estimator described in Section 4. The sample is restricted to HCV units that we identify as single-unit addresses using voucher administrative data and RentHub rental listing information. Panel (b) plots the difference between the two, with 95% confidence bands (gray) from the iterative bootstrap procedure described there. Bins are 0.5 percentage points wide.

Figure D.8: Histogram of cumulative number of PHAs adopting ZIP-level payment standards



Note. This figure shows a histogram of the cumulative number of PHAs that adopt ZIP-level payment standards between 2015 and 2025 in the sample used in Section 5.1.

Figure D.9: Robustness: The impacts of SAFMR mandates by sign of the change of payment standard, following Ellen et al. (2025)



Note. The figure plots estimates of β_τ and 95% confidence intervals from the event study specification: $Y_{it} = \sum_{\tau \neq 2016} \beta_\tau \mathbb{1}\{t = \tau\} \times \text{Treated}_{c(i)} + \mu_{c(i)} + \delta_t + \varepsilon_{it}$, where $c(i)$ denotes the PHA of unit i and 2016 is the omitted year. $\text{Treated}_{c(i)}$ equals 1 if the PHA falls under the SAFMR mandate and 0 if it belongs to the comparison group constructed following Ellen et al. (2025). The sample includes 182 treated and 187 comparison PHAs. Standard errors are clustered by PHA. Each panel uses a different outcome: panel (a), the log payment standard reported by HUD; panel (b), the listing-to-contract rent change; panel (c), the log contract rent reported by HUD; and panel (d), the within-CBSA tract poverty rate percentile. Panels (a), (c), and (d) use HCV new admissions over 2015–2025; and panel (b) uses the HCV-listings linked sample. Each panel reports two separate regressions: one for ZIP codes where ZIP-level PS adoption is predicted to raise the payment standard (blue) and one where it is predicted to lower it (orange). For this purpose, we classify ZIP codes by the difference between a ZCTA's 40th-percentile rent and its CBSA's 40th-percentile rent, both measured in the 2011–2015 American Community Survey (5-year estimates).

Table D.1: Summary Statistics: Payment Standard Subsamples

	HCV New Admissions		RentHub Listings	
	All	Linked	Scrape level	Spell level
	(1)	(2)	(3)	(4)
# Observations	938,522	150,354	352,400,345	32,027,791
# CBSAs	512	437	653	653
Rent (USD)				
25th Percentile	699	800	1,350	1,400
Median	914	1,130	1,815	1,953
75th Percentile	1,304	1,700	2,390	2,655
Mean	1,095.2	1,345.7	2,011.7	2,208.1
Standard Deviation	589.4	732.1	1,059.1	1,224.6
Payment Standard (USD)				
25th Percentile	813	912	1,139	1,345
Median	1,071	1,272	1,538	1,776
75th Percentile	1,498	1,878	2,027	2,383
Mean	1,253.2	1,495.9	1,661.5	1,930.1
Standard Deviation	638.7	787.8	696.9	807.3
Number of Bedrooms				
25th Percentile	1	1	1	1
Median	2	2	2	2
75th Percentile	3	2	3	3
Mean	1.86	1.84	2.08	2.24
Standard Deviation	0.95	0.96	1.06	1.05
Square Footage				
25th Percentile	—	666	787	840
Median	—	857	1,072	1,150
75th Percentile	—	1,100	1,430	1,584
Mean	—	938.6	1,185.3	1,283.8
Standard Deviation	—	482.2	574.8	689.0
Targeting composition (%)				
Hard-targeted	—	7.48	0.16	0.44
Soft-targeted	—	7.16	0.72	1.16

Note. The table reports summary statistics for the subsamples with non-missing payment standards. The sample and variable definitions follow Table 1: Columns (1) and (2) use HUD voucher contract data restricted to new admissions, with Column (2) further restricted to admissions linked to a RentHub listing. Columns (3) and (4) use RentHub listings between 2015 and 2025 at the level of the underlying raw scrapes and the spell, respectively. Each column is further restricted to observations with a non-missing payment standard.

Table D.2: Robustness: Differences-in-bunching estimates in linked HUD-rental listings analyses

	Main	Polynomials			1 pp	Pre-2020	Absolute	Single
		$k = 12$	$k = 16$	$k = 24$	bins		deviation	unit
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel A: Total (Contract - Counterfactual listed)								
Excess mass (EM)	0.038	0.038	0.038	0.038	0.041	0.034	0.035	0.014
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.001)
Missing mass above (MMA)	-0.215	-0.214	-0.214	-0.216	-0.221	-0.114	-0.229	-0.298
	(0.005)	(0.005)	(0.005)	(0.005)	(0.003)	(0.004)	(0.001)	(0.010)
Missing mass below (MMB)	0.194	0.193	0.193	0.194	0.193	0.095	0.193	0.290
	(0.005)	(0.005)	(0.005)	(0.005)	(0.003)	(0.004)	(0.001)	(0.010)
Panel B: Stage 1 (Listed - Counterfactual listed)								
Excess mass (EM)	0.011	0.011	0.011	0.011	0.012	0.006	0.009	0.011
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Missing mass above (MMA)	-0.089	-0.089	-0.089	-0.090	-0.090	-0.026	-0.100	-0.157
	(0.003)	(0.003)	(0.003)	(0.003)	(0.001)	(0.002)	(0.001)	(0.005)
Missing mass below (MMB)	0.082	0.081	0.081	0.082	0.082	0.022	0.091	0.154
	(0.003)	(0.003)	(0.003)	(0.003)	(0.001)	(0.002)	(0.001)	(0.005)
Panel C: Stage 2 (Contract - Listed)								
Excess mass (EM)	0.027	0.026	0.027	0.027	0.029	0.029	0.026	0.002
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.001)
Missing mass above (MMA)	-0.126	-0.125	-0.125	-0.126	-0.131	-0.088	-0.128	-0.141
	(0.005)	(0.005)	(0.005)	(0.005)	(0.003)	(0.003)	(0.000)	(0.011)
Missing mass below (MMB)	0.112	0.111	0.112	0.112	0.111	0.073	0.102	0.137
	(0.005)	(0.006)	(0.005)	(0.005)	(0.003)	(0.003)	(0.000)	(0.011)

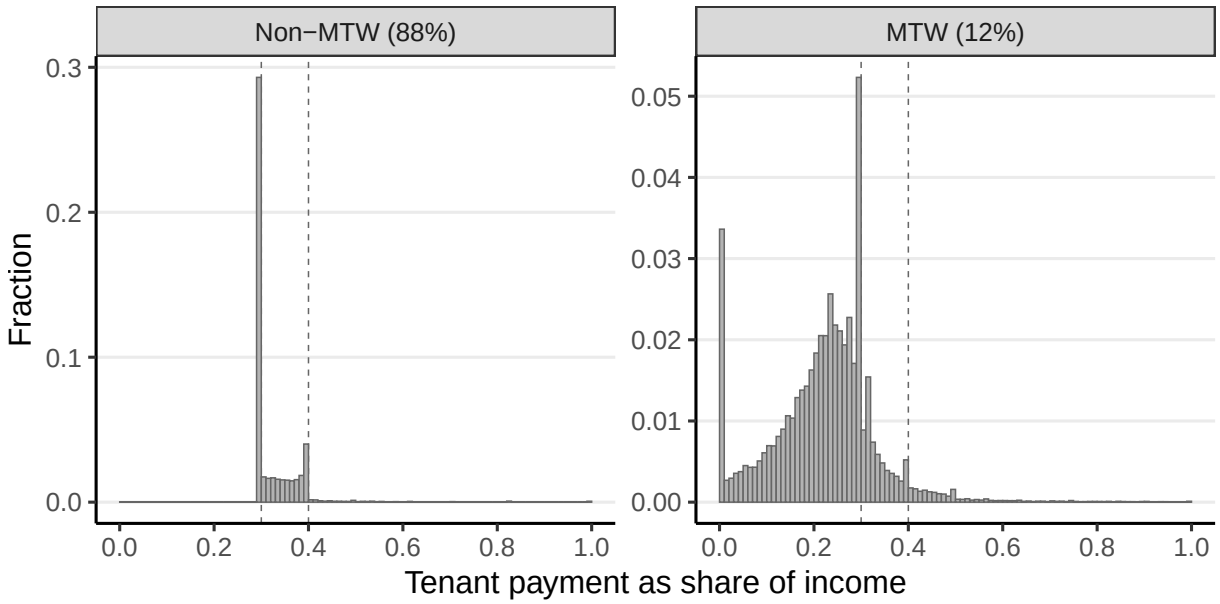
Note. Each column reports estimates of Equation (1) for an alternative sample or specification: (1) the main sample; (2–4) alternative polynomial orders for the counterfactual; (5) 1-percentage-point rent bins; (6) the pre-COVID period only; (7) rent deviations from the payment standard measured in absolute rather than relative terms; and (8) voucher units in single-unit addresses. Panel A reports bunching estimates from the difference between the contract rent and counterfactual listed rent distributions; Panel B, between the listed rent and counterfactual listed rent distributions; Panel C, between the contract rent and listed rent distributions. Bootstrap standard errors (1,000 iterations), following the procedure detailed in Section 4, are reported in parentheses.

Table D.3: Summary statistics: Early vs. later ZIP-level payment standard adopters

	(1) Early adopters (2016–22)	(2) Later adopters (2023–25)	(3) Never ZIP-level
Panel A: PHA characteristics			
Voucher units	4,494	5,230	3,789
Payment standard (mean, 2020)	1,326	1,149	1,146
Number of bedrooms (mean, 2020)	1.90	1.81	1.80
MTW share	0.03	0.04	0.05
Panel B: County characteristics 2020			
Bachelor degree share	0.36	0.33	0.32
Black share	0.17	0.20	0.11
Hispanic share	0.18	0.18	0.16
Median HH income	78,615	69,083	68,201
Poverty rate	0.11	0.14	0.14
Vacancy rate	0.10	0.10	0.10
Renter share	0.35	0.40	0.38
Median bdg age	47	47	46
Median rent	1,209	1,030	979
Renter turnover	0.24	0.24	0.25
Observations (PHAs)	75	92	260

Note. The table reports sample means for three groups of PHAs in the national payment standards dataset described in Section 3.3: PHAs that adopt ZIP-level payment standards in 2016–2022, those adopting in 2023–2025, and never-adopters. Panel A reports the number of voucher units served by each PHA as of 2024, and the average payment standard and number of bedrooms across all voucher units in 2020. Panel B reports Census characteristics of the county in which the PHA is located, obtained from the 2016–2020 American Community Survey (5-year estimates).

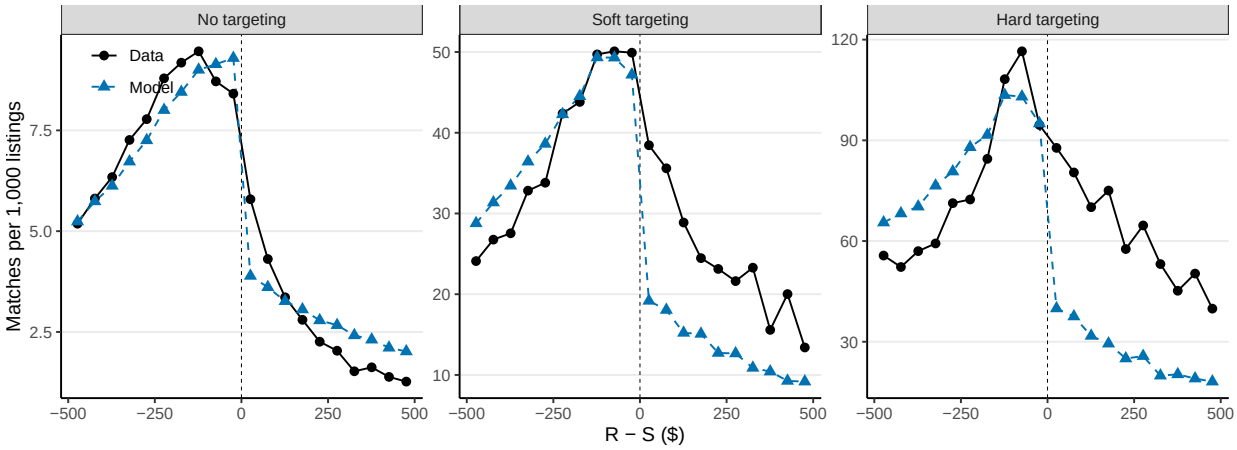
Figure D.10: Tenant Payment as a Share of Income



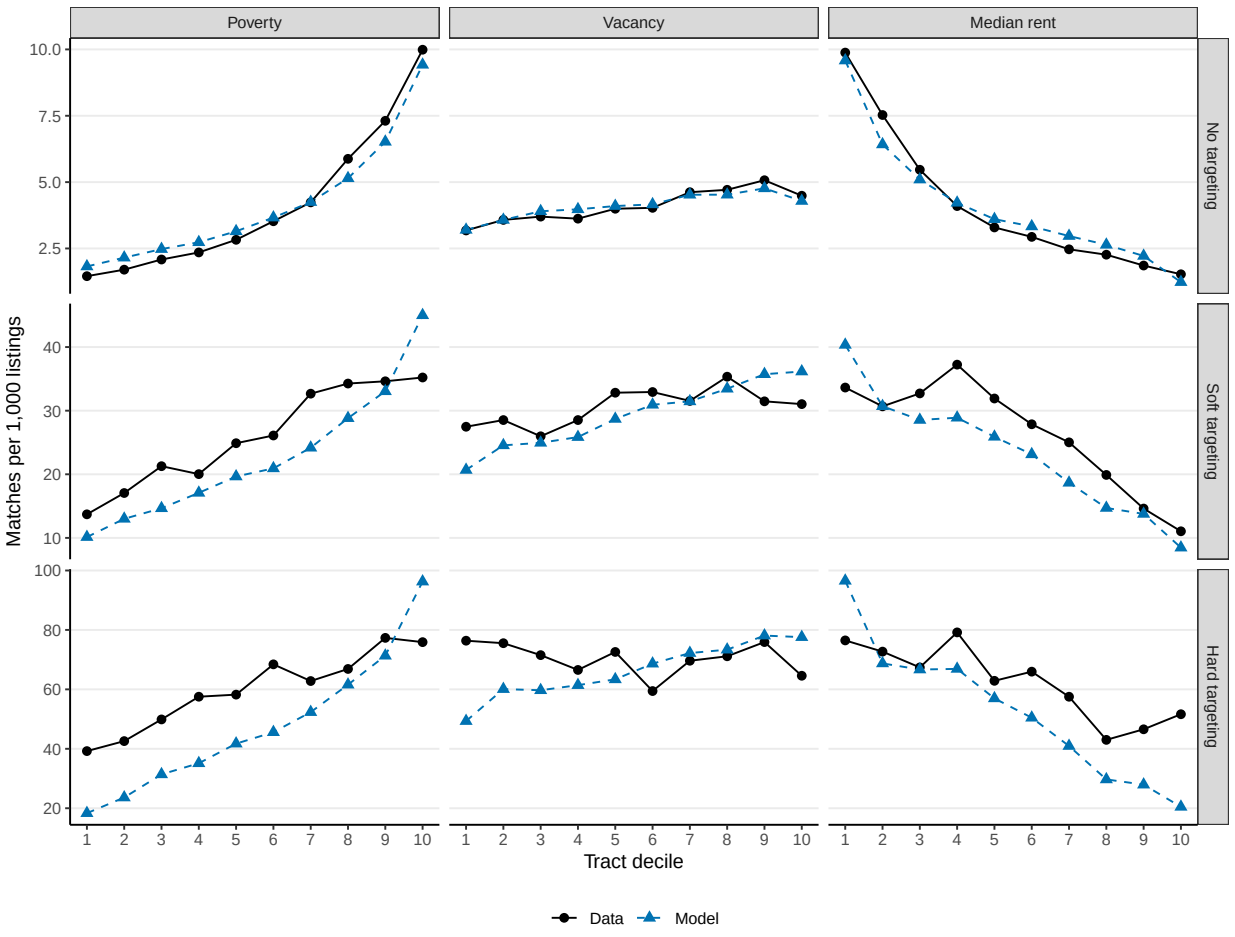
Note. The figure plots the distribution of tenant out-of-pocket rent payment as a share of household income. The left panel shows voucher contracts in non-MTW PHAs, and the right panel shows voucher holders in MTW PHAs. The sample includes all new admissions between 2015 and 2025 in HUD administrative data.

Figure D.11: Stage 2 model fit by targeting status

(a) Match rate by listed rent and targeting status



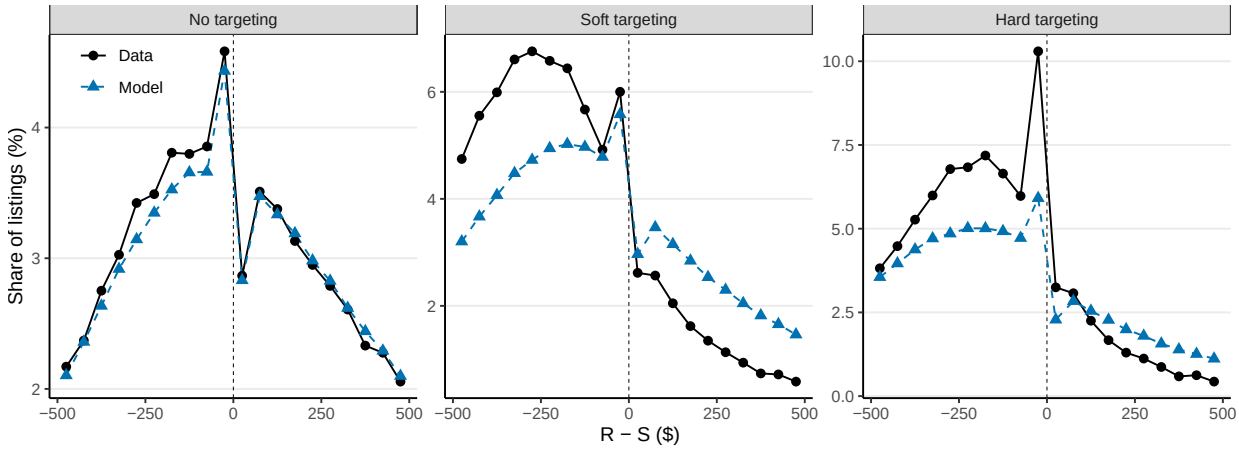
(b) Match rate by neighborhood characteristics by targeting status



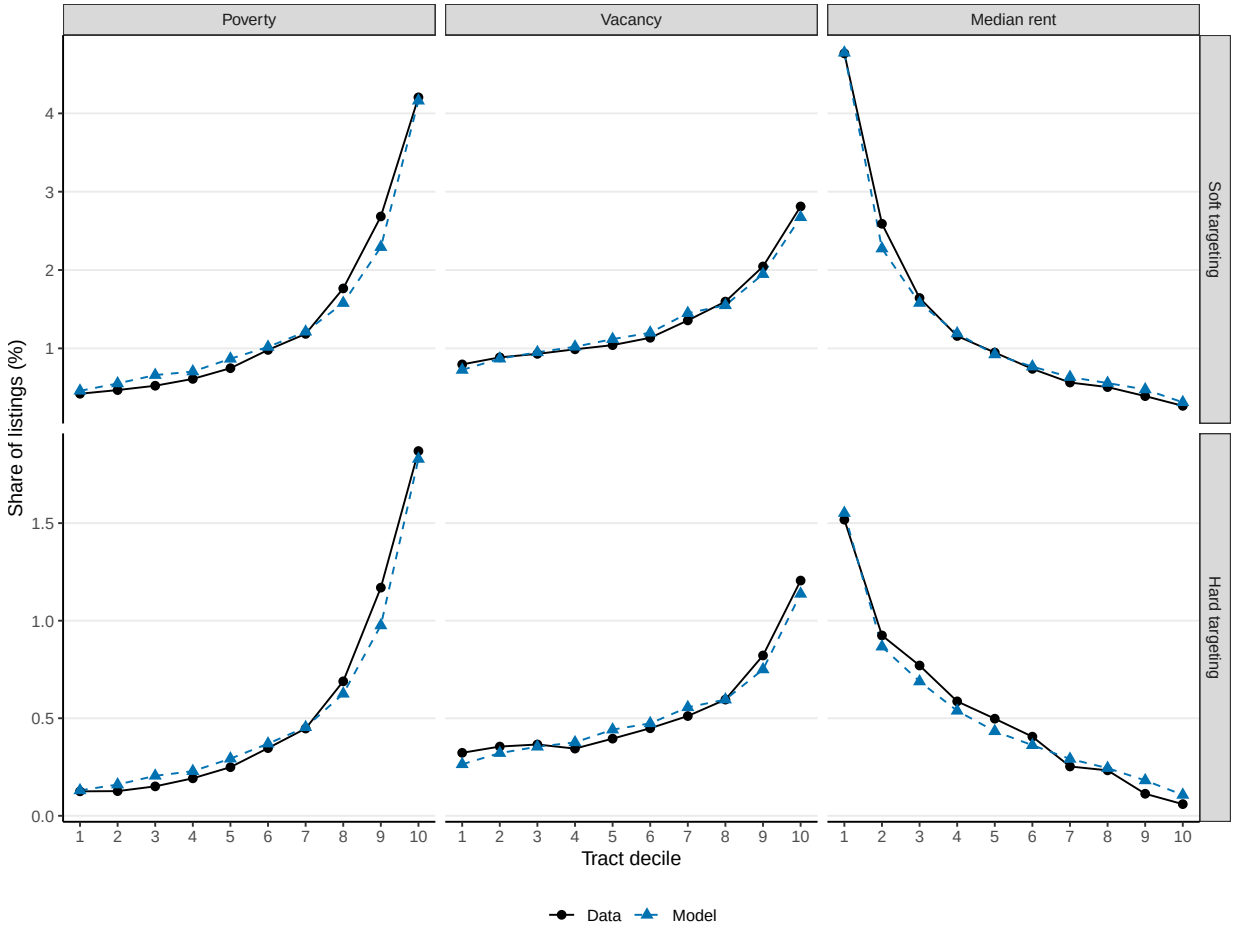
Notes. This figure depicts Stage 2 model fit by targeting status. Panel (a) shows voucher matches per 1,000 listings across \$50 bins of $R - S_j$ over $(-500, 500]$, separately by targeting status, in the data and model. Panel (b) shows voucher matches per 1,000 listings across deciles of tract poverty, rental vacancy, and median rent, separately by targeting status, in the data and model.

Figure D.12: Stage 1 model fit by targeting status

(a) Listed rents relative to the payment standard by targeting status



(b) Targeting by neighborhood characteristics



Notes. This figure depicts Stage 1 model fit by targeting status. Panel (a) shows the distribution of listings across \$50 bins of $R - S_j$ over $(-500, 500]$, separately by targeting status, in the data and model. Panel (b) shows the shares of listings that are soft-targeted and hard-targeted across deciles of tract poverty, rental vacancy, and median rent, in the data and model.

Table D.4: Counterfactual outcomes, in-market admissions

	Voucher matches, in-market (%)	Poverty Q1 (%)	Δ Poverty percentile	Spending ($)	Δ Spending ($)	Cost per percentile ($)
Status quo	16.26	8.28		1,523		
Increased payment standards						
+10%, all tracts	17.21	8.54	−0.31	1,644	121.0	386
+20%, all tracts	17.76	8.74	−0.49	1,741	217.8	445
+10%, Q1 tracts	16.39	9.73	−0.87	1,539	16.6	19
+20%, Q1 tracts	16.48	10.76	−1.49	1,555	31.9	21
+10%, Q1–Q2 tracts	16.57	9.48	−1.62	1,563	40.5	25
+20%, Q1–Q2 tracts	16.78	10.29	−2.69	1,599	76.6	28
Rent-indexed standards						
Indexed	15.14	9.78	−2.06	1,473	−49.6	–
Hold harmless	16.82	8.90	−0.71	1,593	70.6	100
Budget neutral	16.51	9.79	−2.01	1,606	83.5	–
Sign-up bonus						
1 month, all tracts	16.39	8.16	0.29	1,587	64.5	–
2 months, all tracts	16.53	8.02	0.62	1,652	129.5	–
1 month, Q1 tracts	16.26	8.31	−0.02	1,529	6.1	301
1 month, Q1–Q2 tracts	16.27	8.31	−0.05	1,539	16.4	313
Rent-reasonableness enforcement						
$\rho = 0.5$	16.18	8.37	−0.21	1,495	−27.6	–
$\rho = 0$	16.11	8.43	−0.37	1,468	−54.3	–
Budget-neutral enforcement						
$\rho = 0.5$, all tracts	16.47	8.45	−0.33	1,526	3.6	–
$\rho = 0.5$, Q1–Q2 tracts	16.48	9.54	−1.78	1,532	8.9	–
$\rho = 0.5$, indexed	16.71	9.84	−2.11	1,609	86.0	–
$\rho = 0$, all tracts	16.80	8.68	−0.73	1,537	14.6	–
$\rho = 0$, Q1–Q2 tracts	16.77	10.99	−3.71	1,552	29.6	–
$\rho = 0$, indexed	16.96	9.89	−2.20	1,614	91.0	–

Notes. This table reports counterfactual outcomes pooled across the 2019–2025 new voucher admissions matched to observed rental listings, the in-market admissions, under the baseline outside-pool responsiveness, $\zeta = 0.46$, defined in Appendix C.4. Voucher matches, in-market, is the share of all new admissions matched to observed listings. Table 3 reports the outcomes among all new admissions. Poverty Q1 is the share of in-market admissions in tracts at or below the 25th poverty percentile of the metropolitan area, and Δ Poverty percentile is the change in the mean poverty percentile of their tracts. Spending includes the housing assistance payment and any sign-up bonus per in-market admission and month, and Δ Spending is its change. Cost per percentile is Δ Spending divided by the decrease in the mean poverty percentile and is reported when spending rises and the mean percentile falls. Budget-neutral policies hold program-wide spending fixed, so their spending per in-market admission can change, and no cost per percentile is reported for them.

Table D.5: Counterfactual listings

	Soft-targeted (%)	Hard-targeted (%)	At payment standard (%)
Status quo	1.35	0.53	1.42
Increased payment standards			
+10%, all tracts	1.41	0.58	1.32
+20%, all tracts	1.45	0.64	1.16
+10%, Q1 tracts	1.35	0.53	1.42
+20%, Q1 tracts	1.36	0.53	1.40
+10%, Q1–Q2 tracts	1.36	0.53	1.41
+20%, Q1–Q2 tracts	1.37	0.54	1.35
Rent-indexed standards			
Indexed	1.29	0.45	1.76
Hold harmless	1.37	0.54	1.67
Budget neutral	1.34	0.50	1.78
Sign-up bonus			
1 month, all tracts	1.39	0.57	1.44
2 months, all tracts	1.43	0.63	1.46
1 month, Q1 tracts	1.35	0.53	1.42
1 month, Q1–Q2 tracts	1.35	0.53	1.42
Rent-reasonableness enforcement			
$\rho = 0.5$	1.33	0.50	1.42
$\rho = 0$	1.31	0.47	1.42
Budget-neutral enforcement			
$\rho = 0.5$, all tracts	1.34	0.51	1.41
$\rho = 0.5$, Q1–Q2 tracts	1.33	0.50	1.41
$\rho = 0.5$, indexed	1.33	0.49	1.74
$\rho = 0$, all tracts	1.33	0.49	1.36
$\rho = 0$, Q1–Q2 tracts	1.32	0.48	1.28
$\rho = 0$, indexed	1.33	0.48	1.68

Notes. This table reports counterfactual outcomes under the baseline outside-pool responsiveness, $\zeta = 0.46$, defined in Appendix C.4. Soft- and hard-targeted are the shares of observed listings that target voucher holders, and At payment standard is the share of observed listings priced at the payment standard.

Table D.6: Counterfactual outcomes at the bounds of the outside-pool responsiveness

	Poverty Q1 (%)			Δ Poverty percentile			Δ Spending ($)		
ζ	0.17	0.46	0.75	0.17	0.46	0.75	0.17	0.46	0.75
Status quo	7.22	7.22	7.22						
Increased payment standards									
+10%, all tracts	7.33	7.37	7.42	−0.12	−0.19	−0.26	93.7	93.7	93.8
+20%, all tracts	7.41	7.50	7.59	−0.20	−0.33	−0.45	170.0	169.9	169.9
+10%, Q1 tracts	7.65	7.94	8.24	−0.25	−0.43	−0.60	9.5	10.2	11.0
+20%, Q1 tracts	7.96	8.48	9.00	−0.44	−0.75	−1.05	17.7	19.4	21.2
+10%, Q1–Q2 tracts	7.58	7.83	8.07	−0.49	−0.84	−1.19	25.2	26.7	28.2
+20%, Q1–Q2 tracts	7.85	8.27	8.68	−0.85	−1.45	−2.02	46.6	50.2	53.7
Rent-indexed standards									
Indexed	7.56	7.87	8.21	−0.47	−0.89	−1.37	−104.7	−102.1	−99.1
Hold harmless	7.43	7.55	7.67	−0.22	−0.36	−0.49	38.5	38.8	39.2
Budget neutral	7.67	7.97	8.29	−0.57	−1.00	−1.43	0.0	0.0	0.0
Sign-up bonus									
1 month, all tracts	7.20	7.18	7.16	0.07	0.11	0.15	55.9	55.8	55.7
2 months, all tracts	7.16	7.12	7.09	0.16	0.24	0.32	111.8	111.6	111.4
1 month, Q1 tracts	7.24	7.24	7.25	−0.01	−0.01	−0.01	4.6	4.6	4.6
1 month, Q1–Q2 tracts	7.23	7.24	7.24	−0.01	−0.03	−0.04	12.9	12.9	13.0
Rent-reasonableness enforcement									
$\rho = 0.5$	7.24	7.26	7.27	−0.05	−0.08	−0.10	−22.8	−22.7	−22.7
$\rho = 0$	7.26	7.28	7.30	−0.09	−0.14	−0.19	−45.5	−45.4	−45.3
Budget-neutral enforcement									
$\rho = 0.5$, all tracts	7.28	7.30	7.32	−0.09	−0.14	−0.19	0.0	0.0	0.0
$\rho = 0.5$, Q1–Q2 tracts	7.62	7.84	8.04	−0.56	−0.90	−1.20	0.0	0.0	0.0
$\rho = 0.5$, indexed	7.70	8.01	8.33	−0.62	−1.06	−1.50	0.0	0.0	0.0
$\rho = 0$, all tracts	7.35	7.41	7.47	−0.21	−0.33	−0.43	0.0	0.0	0.0
$\rho = 0$, Q1–Q2 tracts	8.15	8.64	9.04	−1.26	−1.97	−2.55	0.0	0.0	0.0
$\rho = 0$, indexed	7.73	8.06	8.37	−0.67	−1.13	−1.57	0.0	0.0	0.0

Notes. This table reports counterfactual outcomes pooled across the 2019–2025 new voucher admissions, including those matched to observed rental listings and those allocated to the outside pool, at the lower bound, the baseline, and the upper bound of the outside-pool responsiveness, $\zeta \in \{0.17, 0.46, 0.75\}$, defined in Appendix C.4. Poverty Q1, Δ Poverty percentile, and Δ Spending are defined as in Table 3.